Spatial and land-use considerations of high renewable penetration into energy systems: Case of PEI's onshore wind and agriculture colocation

by

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Declaration of Committee

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Abstract

Transitioning to renewable energy raises concerns about land use conflicts due to the high land intensity of renewable technologies. This is especially pressing for onshore wind, as seen in cases like Prince Edward Island, where onshore wind and agriculture are both vital to the local economy and highly overlapping in land use. These conflicts are largely driven by poorly modeled land use metrics and a lack of spatial explicitness in capacity expansion energy models. Therefore, in this thesis, we develop a geospatial and optimization energy model connection and incorporate it to conduct land tracking, constraints, site selection and land use colocation into energy system modeling. Our results show that colocating onshore wind and agriculture in PEI decreases emissions and costs by 35% and 22%, respectively, when integrated into the model. This approach enables energy modelers and policymakers to consider land and spatial factors in their modeling and planning practices.

Keywords: land use; geographical information systems; GIS; energy model linking; colocation; onshore wind; Prince Edward Island; cropland

Dedication

To my parents, whose fight and sacrifice built, from nothing, a life of possibility for their children.
Everything I have and everything I am becoming, I owe to your love and your hard work.

To my brother, Mohammad, whose future is quietly one of the reasons I keep trying to build things worth building, and to the long walks and conversations we have shared along the way.

And to the justice-seeking people of Iran and the world, particularly its equity-deserving minorities, who continue to resist totalitarianism in pursuit of a more just and equal world.

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Declaration

I acknowledge the use of ChatGPT [chat.openai.com] and Claude [claude.ai/chat] to assist me in improving the readability of my writing. I used various prompts such as: “Check the grammar and sentence structure of this paragraph.” “Suggest better word choices to make this clearer.” “Improve the flow and readability of this sentence.” I used the AI-generated suggestions to help me identify possible improvements in grammar, word choice, and sentence structure. However, I did not accept the suggestions automatically. I carefully reviewed each one and made changes only when I believed they improved the clarity and quality of my work. All ideas and final decisions in the writing are entirely my own.

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List of Acronyms

IRL	Indigenous Reserve Lands
PCA	Protected and Conserved Areas
ITC	Investment Tax Credit
RPS	Renewable Portfolio Standard

List of Symbols

U	Land use
I	Land intensity
C	Capacity
A	Activity
c	Capacity index
a	Activity index
r	Region
T	Technology
v	Vintage
O	Objective function
L	Land limit
η	Efficiency

Chapter 1.

Introduction

Renewable energy deployment is central to electricity sector decarbonization, but it also increases the importance of spatial constraints in long-term energy system planning. Compared with conventional fuel-based generation, renewable technologies such as wind and solar generally require larger land areas per unit of installed capacity or generated energy, making land availability, land suitability, and land-use conflict increasingly important in long-term energy-system planning [1]. This is especially relevant in jurisdictions pursuing net-zero targets, where replacing fossil-fuel generation with low-carbon technologies may shift planning challenges from fuel supply and emissions control toward questions of space, siting, infrastructure access, and compatibility with existing land uses. Consequently, evaluating renewable energy transitions requires not only estimating future capacities, costs, and emissions, but also representing the spatial feasibility of new infrastructure and its interactions with environmental, agricultural, technical, governmental, and social constraints.

Onshore wind presents a distinct spatial modelling challenge because its estimated land requirements depend strongly on how land occupation is defined. Wind turbines have relatively small physical footprints, but they require substantial spacing between turbines to reduce wake losses and maintain generation performance. This spacing can make onshore wind appear highly land-intensive when the full wind farm area is counted as occupied land; however, the same area may remain available for other uses, including agriculture, grazing, roads, or other compatible infrastructure, if colocation is explicitly represented [1], [2]. Accordingly, the distinction between turbine footprint, turbine spacing, and shared land use constitutes a modelling assumption that can materially affect estimated land requirements and feasible wind capacity. If spacing is treated as unavailable land, wind development may appear to impose strong land-use constraints; if spacing is treated as potentially shareable, the same wind resource may support both electricity production and existing land-based activities. This distinction is central to studying wind development in agricultural regions and motivates the use of spatially explicit methods in this thesis.

The specific problem addressed by this thesis is that capacity-expansion models can represent land too coarsely to capture site-specific wind feasibility and can treat turbine spacing as unavailable land even where compatible land uses may coexist. To address this problem, the thesis links GIS[1]-based onshore wind siting analysis with the Atlantic Canada Energy System model, which is based on TEMOA, a capacity expansion optimization framework [3]. The work first evaluates land-cover and wind-resource conditions across Canada to identify how different spatial constraints affect onshore wind potential, then focuses on Prince Edward Island as a case study where wind development, land scarcity, and cropland availability intersect. By integrating spatially defined wind cells, land-use intensity parameters, and land constraints into the energy-system model, the thesis evaluates how land availability, clean-energy policies, and wind–cropland colocation affect future technology choices, system costs, emissions, and land-use outcomes. In doing so, the research connects national-scale land-cover screening with a regional energy-system planning model and provides a framework for assessing renewable energy pathways where land is not treated as an unlimited or spatially uniform resource.

1.1. Background

This section first introduces the land-use metrics used to interpret energy infrastructure, distinguishing between direct and indirect land use. The chapter then turns to Canada's land-cover context as a separate background on spatial considerations, examining the shares of existing land layers and their relevance to onshore wind siting. In this part, the focus is on how wind production potential spatially overlaps with land-cover layers across provinces. Cropland is then discussed, and the chapter's focus shifts to Prince Edward Island, where the province's strong wind potential, limited land availability, and high cropland coverage make land-aware energy planning particularly relevant. Finally, the TEMOA-ACES model is introduced as the energy-system planning framework used in this thesis and as the basis for integrating spatially defined wind resources, land-use accounting, and policy scenarios into a capacity expansion model.

[1] Geographical Information Systems

1.1.1. Land-Use Metrics for Energy Systems Development

Land-use metrics are reported inconsistently across the energy literature, underscoring the importance of consistent terminology when comparing technologies and interpreting model results. Turkovska et al. identify more than 11 terms and seven meta-metrics used to describe land-use requirements in the energy sector and emphasize that these inconsistencies can lead to substantial variation in reported land-use estimates [4]. They summarize four land-use types: infrastructural, spacing, temporary, and remote land use. In this thesis, infrastructural land use is referred to as footprint, while remote land use is treated as indirect land use, because it represents land requirements outside the generation site, such as upstream fuel extraction and mining. Fig. 1 presents these four land-use types and their visual representation for onshore wind.

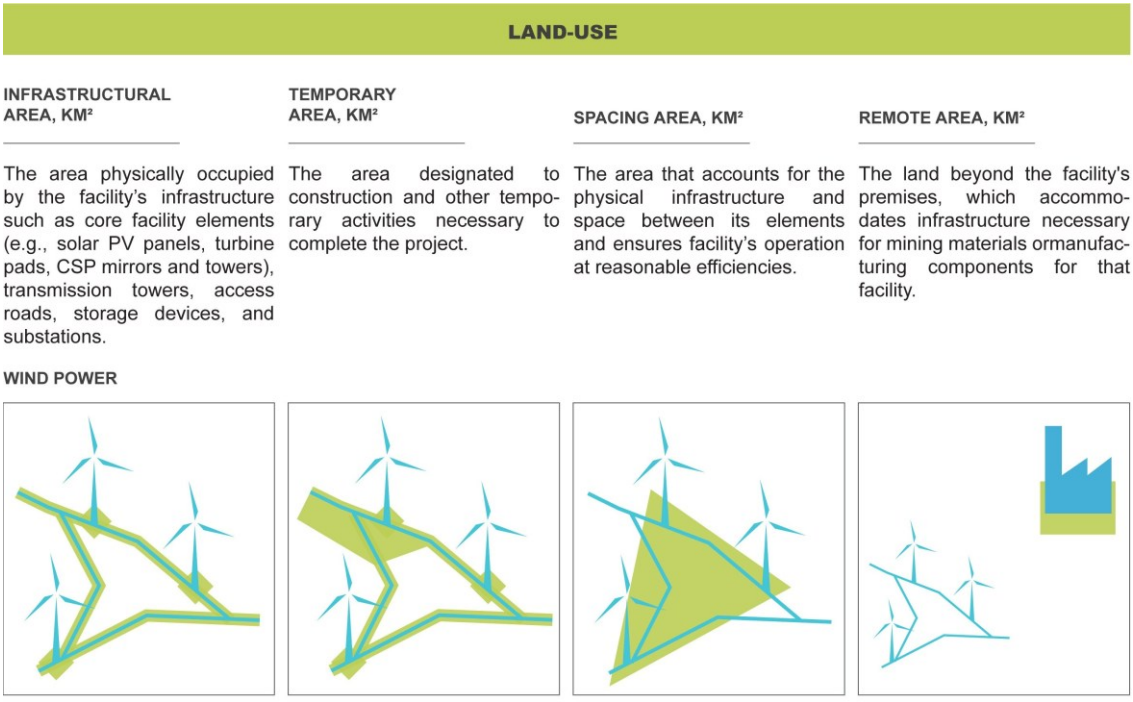


Fig. 1. Classification of land-use types associated with onshore wind power plants, including infrastructural land, temporary land, turbine-spacing area, and remote land requirements. In this thesis, infrastructural land is treated as the direct physical footprint of wind infrastructure, whereas remote land represents indirect upstream land requirements[4] licensed under CC BY.

Land use for energy production is typically described using two broad metrics: direct and indirect land use [2]. Indirect land use refers to land required for upstream

extraction, mining, processing, or fuel-supply activities, and is therefore most relevant to fuel-based energy sources. Direct land use refers to land occupied by power-generation facilities themselves. For onshore wind, direct land use is commonly separated into footprint and spacing: footprint refers to land physically occupied by turbines, access roads, foundations, substations, and other infrastructure, while spacing refers to the land between turbines within a wind farm [1].

These distinctions are central to this thesis because onshore wind can appear highly land-intensive when turbine spacing is counted as fully occupied land, but much less land-intensive when only the physical footprint is considered. The difference is especially important where wind turbines are placed on agricultural land, since the spacing between turbines may remain available for farming. For this reason, the thesis distinguishes between conventional wind development, wind development that excludes cropland, and wind-cropland colocation. The technology-specific land-intensity values used in the model are reported in the Methods chapter under Land-Use Intensity Inputs.

1.1.2. Canadian Land-Cover Context for Onshore Wind Siting

This chapter examines how selected land-cover layers intersect with onshore wind potential across Canada, focusing on both the share of land affected by each layer and the extent to which these layers overlap with high-wind areas. The background is organized into three parts for smoother discussion and understanding.

First, ‘Low-Coverage Exclusion and Buffer Layers’ considers land categories that occupy relatively small provincial land shares while still remaining important for local siting decisions. Second, ‘High-Coverage Constraints and Wind-Resource Overlap’ examines layers that cover large areas in some provinces but do not necessarily coincide with the strongest wind resources. Finally, ‘Cropland as a Constraint and Colocation Opportunity’ focuses on agricultural land as a distinct case because cropland may either restrict wind development or support wind–agriculture colocation depending on the siting assumptions used. This structure provides the spatial context for later methods, results and discussions and helps set the floor for narrowing the focus to Prince Edward Island, where cropland coverage, wind potential, and limited land availability make land-aware wind planning especially relevant.

Canada's large land area and diverse geography create substantial variation in wind resources and land availability. Fig. 2 shows the shares of the land-cover layers considered in this thesis. Data indicates that the constrained layers exclude less than 39 percent of Canada's land from onshore wind production, leaving more than 61 percent available under the selected screening assumptions. However, these constraints are not evenly distributed across provinces, and this heterogeneity motivates a more detailed spatial analysis of where land constraints meaningfully affect wind-development potential.

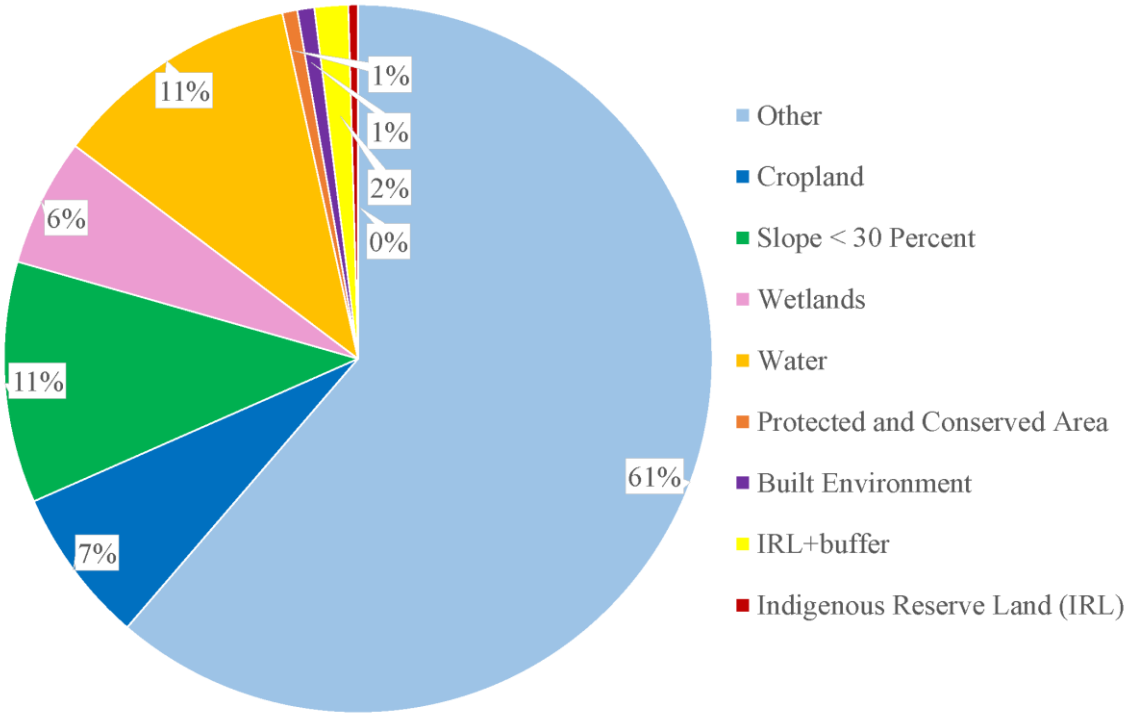


Fig. 2. **Relative shares of the land-cover and spatial-constraint classes considered in the Canada-wide onshore wind siting analysis. The categories include cropland, terrain with slope <30%, wetlands, water bodies, protected and conserved areas, built environment, Indigenous Reserve Lands (IRL), IRL buffers, and land classified as Other/available under the selected screening assumptions. Percentages represent each class as a share of the Canadian land area included in the analysis. [5], [6], [7], [8].**

Low-Coverage Exclusion and Buffer Layers

Land-cover constraints vary by province and influence available land for onshore wind development in different ways. Fig. 3 compares seven land-cover layers and one buffer layer across the provinces, while also showing the remaining land categorized as available for onshore wind development. In the bar chart, each color represents the share of provincial land associated with a given layer. The available land share is slightly conservative because small overlaps between layers are represented in the screening process. Some of the detailed percentages and geographic distributions of these land covers are provided in the Methods Section in Fig. 3.

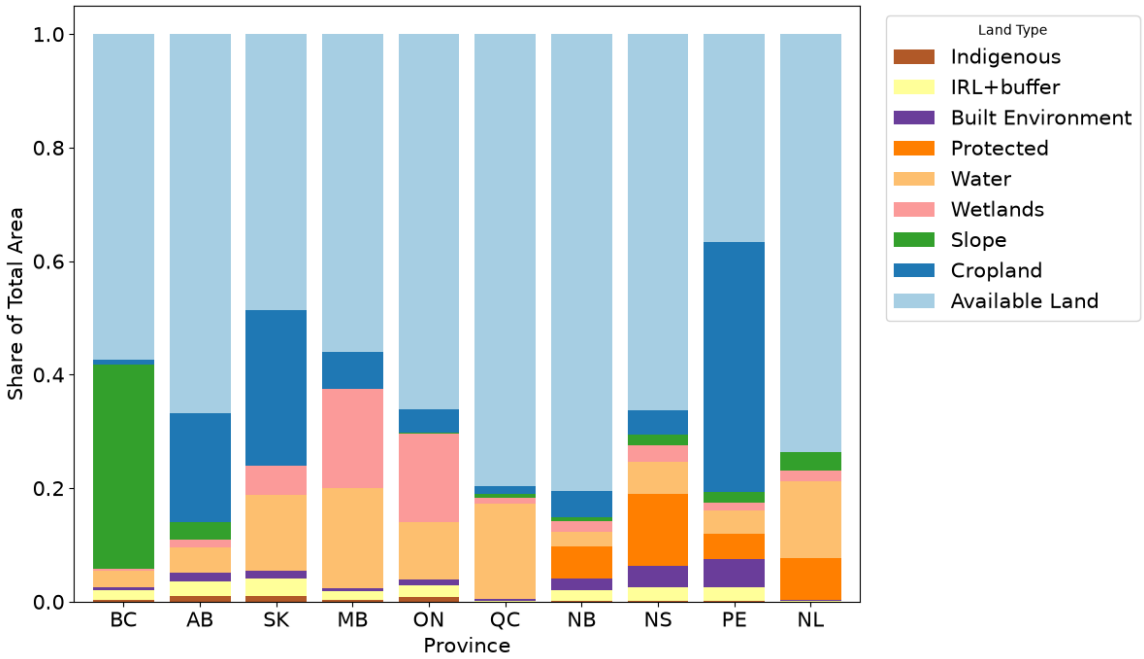


Fig. 3. **Provincial shares of selected land-cover classes, buffer-restricted areas, and remaining available land across ten Canadian provinces used in the onshore wind screening analysis. Each stacked bar represents the fraction of provincial land associated with Indigenous Reserve Lands, IRL buffers, built environment, protected and conserved areas, water bodies, wetlands, slopes < 30 degrees, cropland, and land remaining available after the selected spatial screening assumptions are applied[5], [6], [7], [8].**

The figure shows that protected and conserved areas, the built environment, and Indigenous Reserve Lands occupy comparatively small shares of land in most provinces, even when buffer zones are included. Protected and conserved areas range from 0 to 12.79 percent of provincial land, while built-environment shares range from 0.18 to 4.95

percent. Indigenous Reserve Lands account for 0.02 to 0.97 percent, increasing to 0.09 to 2.3 percent when buffers are included. These layers remain important because they represent environmental, technical, governmental, or social constraints; however, at the provincial scale, their spatial extent is generally not large enough to strongly reduce aggregate onshore wind potential.

High-Coverage Layers with Limited Wind-Resource Overlap

Other land-cover layers occupy larger shares in some provinces. Based on Fig. 3, slope, water bodies, and wetlands cover substantial areas in selected regions. Water bodies account for approximately 16 percent of Quebec and 17 percent of Manitoba. Manitoba also has the largest wetland share, at approximately 17 percent, followed by Ontario at approximately 15 percent. British Columbia has by far the highest share of steep slopes, at approximately 37 percent. However, high land-cover shares do not necessarily translate into large reductions in high-quality onshore wind potential.

Canada's highest wind-speed areas are primarily located in northern regions and along eastern coastlines, as shown in Fig. 4 Provinces with large water or wetland shares, such as Ontario, Quebec, and Manitoba, generally have lower wind speeds than the most wind-rich regions considered in this analysis. Therefore, the exclusion of these layers is not concerning for reducing the highest-quality onshore wind opportunities at the national scale.

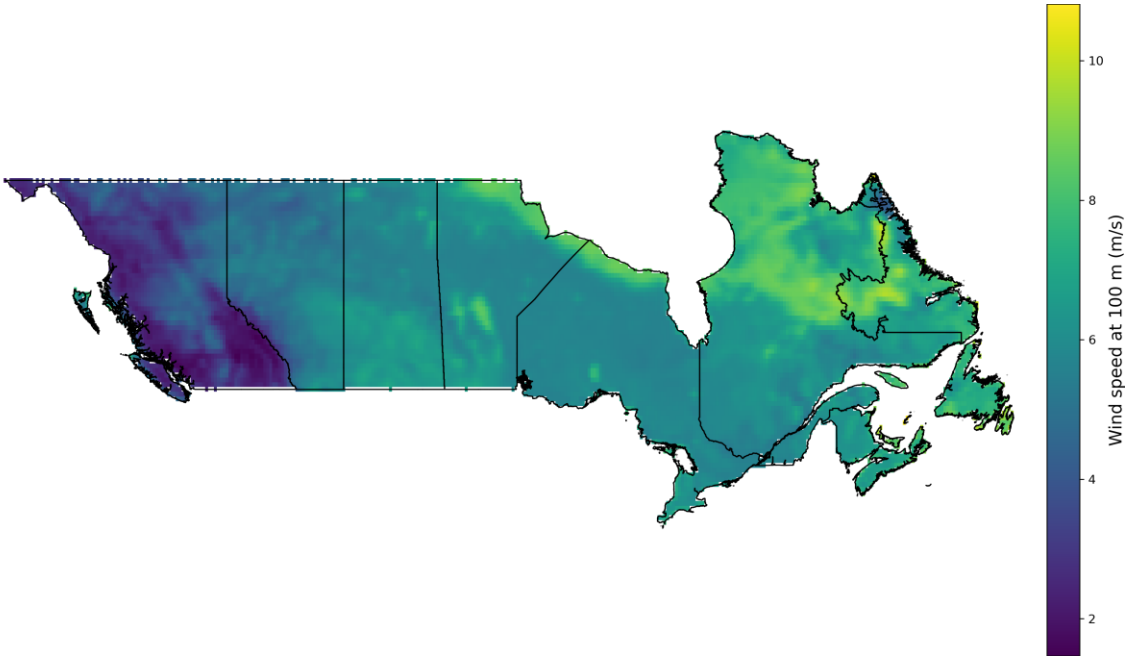


Fig. 4. **Spatial distribution of average wind speed across the ten Canadian provinces included in the national screening analysis, derived from ERA5 data at approximately 30 × 30 km spatial resolution and a reference elevaion of 100 m. Wind speed is reported in m/s and is used to characterize the geographic distribution of onshore wind resource quality before application of land cover constraints [9].**

British Columbia provides a similar example for slope. The province has the lowest wind speeds among the considered provinces and a large share of steep terrain. Fig. 5 shows that the province's higher wind-speed areas are generally located on higher-slope land. However, the high cost of wind development on high-slope land makes these areas less favorable for building wind farms. Given the higher costs and the lower wind speed of BC compared to other provinces, excluding these locations does not materially constrain the wind potential of BC.

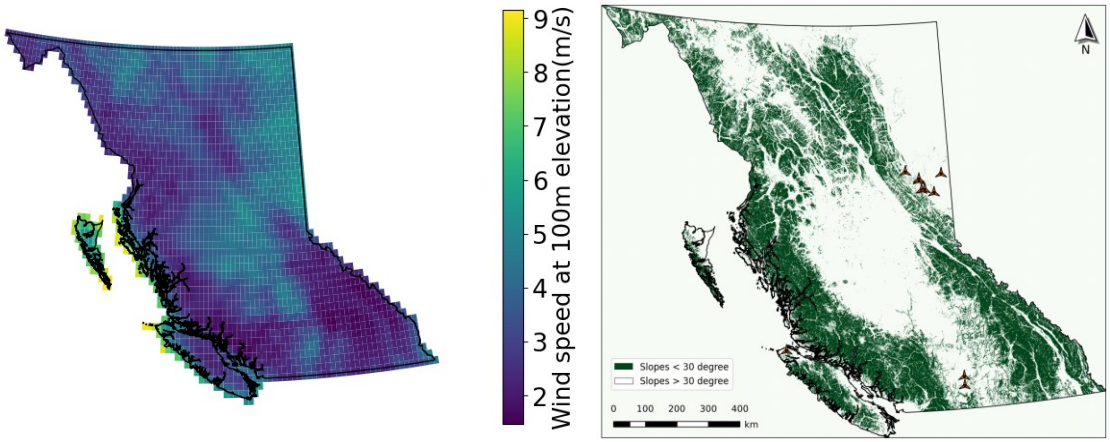


Fig. 5. **Spatial distribution of wind resource and terrain slope in British Columbia. (a) Average wind speed at 100 m elevation, derived from ERA5 at approximately 30 × 30 km resolution and reported in m/s [9]; (b) provincial terrain-slope distribution classified into slope categories used in the GIS siting analysis [6]. The comparison illustrates the spatial relationship between higher-wind locations and terrain conditions relevant to technical suitability for onshore wind development. Sources.**

BC's existing wind farms are located on low-slope areas. BC currently has 10 onshore wind sites and 6 of them overlap with class 5, which includes slopes of 8 to 16 degrees. The highest slopes with onshore wind farms are class 6 (16 to 30 degrees). The current slope profile of BC's wind farms shows that despite excluding the high slope area (class 7 and higher), the province is still generating onshore wind and is not materially impacted by this exclusion.

Cropland as a Constraint and Colocation Opportunity

Among the land-cover categories considered in national wind-siting analysis, cropland requires particular attention because it is not always an absolute exclusion. Unlike water bodies, dense urban areas, or protected lands, agricultural land may remain partially productive when wind turbines are installed, depending on turbine spacing, access roads, and the treatment of the surrounding land area. This makes cropland a context-dependent layer rather than a purely unavailable land category. The distinction is especially important for Prince Edward Island, where agricultural land occupies a large share of the province's limited land base and where future wind development may interact directly with existing agricultural uses.

Cropland is one of the largest land-cover categories in several provinces. Prince Edward Island, Saskatchewan, and Alberta have approximately 44, 27, and 19 percent cropland cover, respectively. If cropland is treated as an exclusion layer, these shares can impose significant constraints on onshore wind development, particularly where cropland overlaps with high-quality wind resources.

Existing wind development suggests that areas rich in cropland are not always avoided in practice. As Fig. 6 shows, many existing onshore wind farms and high-wind locations in Alberta overlap with cropland. Saskatchewan shows a similar pattern, as presented in Fig. 7.

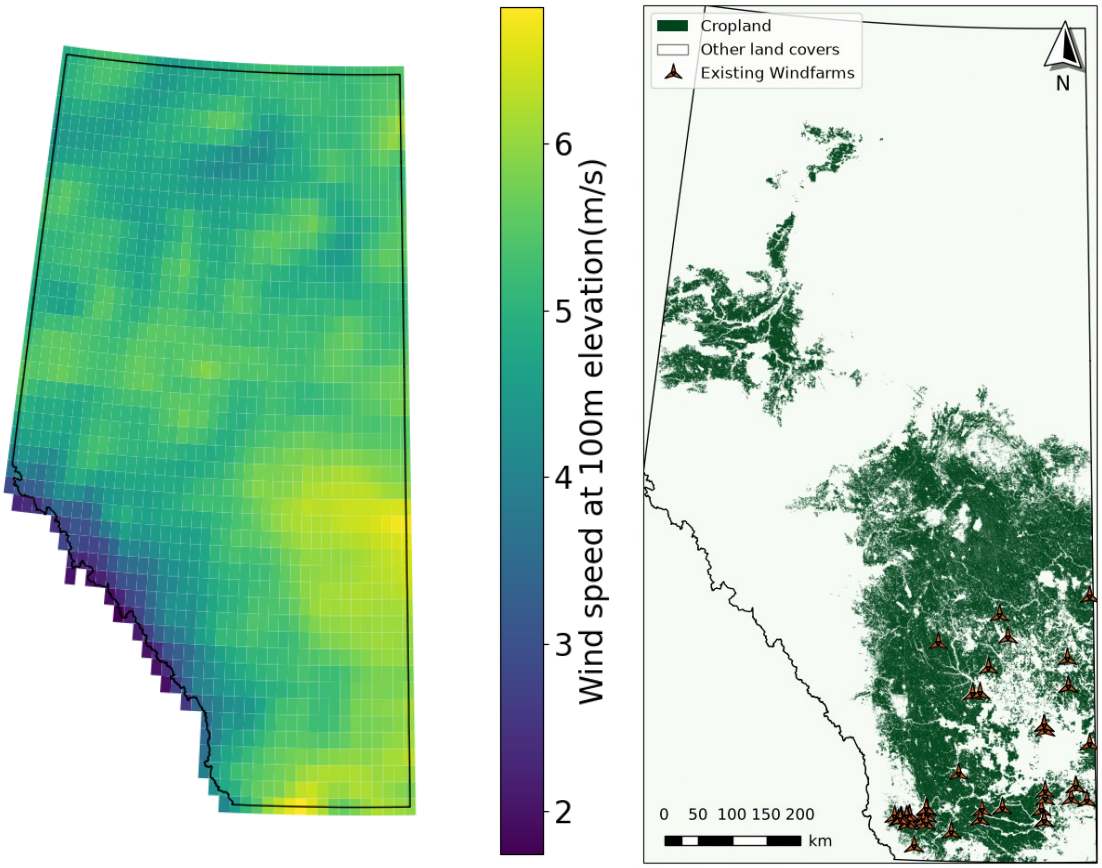


Fig. 6. **Spatial relationship between wind resources, cropland, and existing onshore wind development in Alberta. (a) Average wind speed at 100 m above ground level, reported in m/s [9]; (b) cropland distribution [6] overlaid with the locations of existing onshore wind farms [10]and other land cover classes. The figure is used to assess whether high-quality wind resources and existing wind development occur in proximity to agricultural land, supporting the treatment of cropland as a context-dependent rather than absolute exclusion layer.**

These examples indicate that cropland should not be treated in the same way as hard exclusion layers such as water bodies or dense built environments. Instead, cropland can be represented through multiple siting assumptions: it may be excluded to protect agricultural land, included under conventional wind-siting assumptions, or treated as land available for wind-agriculture colocation. This distinction motivates the PEI-focused colocation scenarios developed later in the thesis.

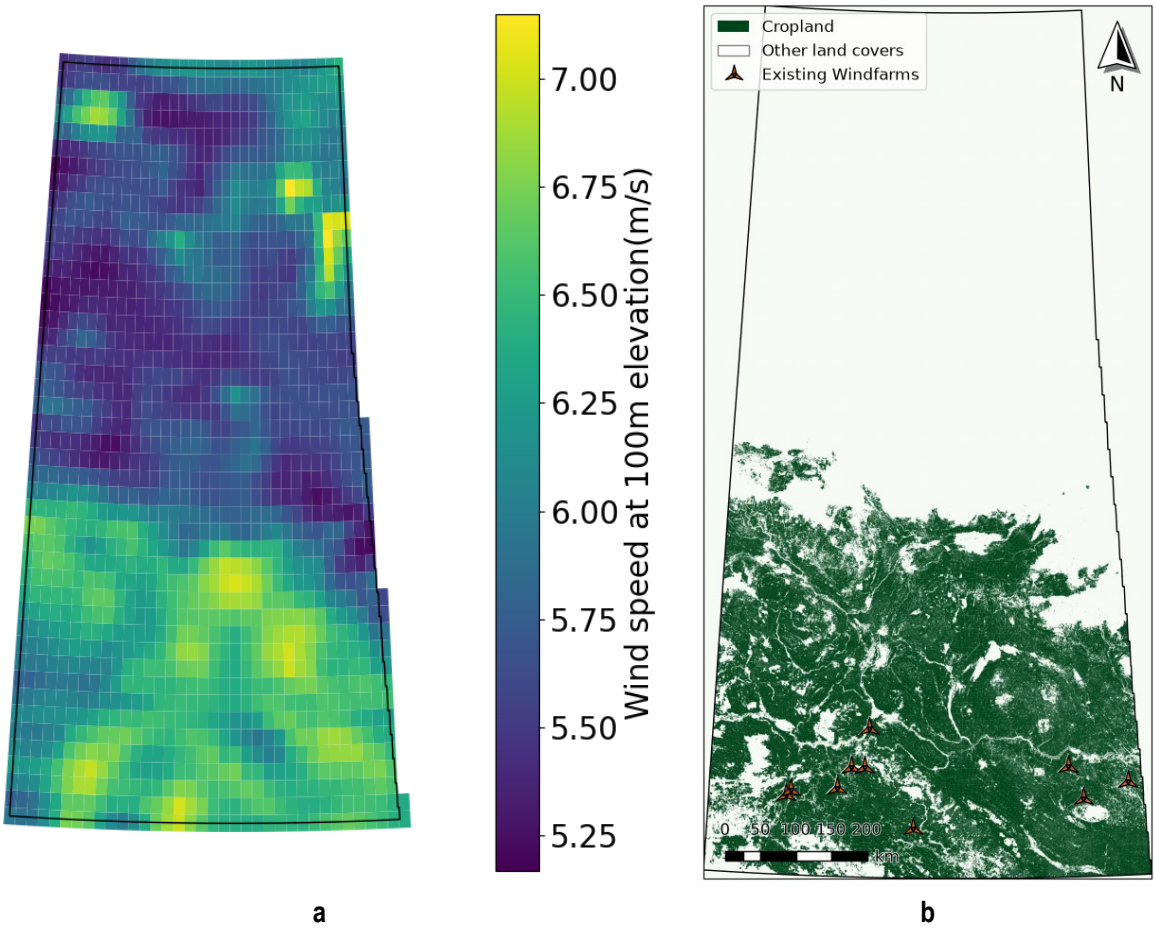


Fig. 7. **Spatial relationship between wind resources, cropland, and existing onshore wind development in Saskatchewan. (a) Average wind speed at 100 m above ground level, reported in m/s [9]; (b) cropland distribution [6] overlaid with existing onshore wind-farm locations [10] and other land-cover classes. The comparison illustrates the proximity of agricultural land to relatively strong wind-resource areas and provides spatial context for evaluating cropland exclusion and potential wind–agriculture colocation.**

As shown on Fig. 6 and Fig. 7, Alberta and Saskatchewan's existing wind farms are located around the cropland areas; although, based on Canada land cover data[5] and existing onshore wind locations[10], no wind farm coordinate sits within a 100*100 cropland cell, meaning that proximity does not guarantee overlap. However, in the larger province scale, these proximities are accounted for as colocations and farmers have contracts with governments and plant developers for compensations and incentives.

PEI shows a related but distinct pattern. Fig. 8 shows that existing onshore wind sites and high-wind locations are spatially associated, but these sites have limited overlap

with cropland. Among the nine wind farms in PEI, only one is located near cropland areas: the Summerside wind farm, which is community-owned by the City of Summerside. Other wind farms located on privately owned land are present or are under construction; however, none of the existing ones are explicitly reported to be on agricultural land. These privately owned lands are acquired through lease contracts by independent power producers[11].

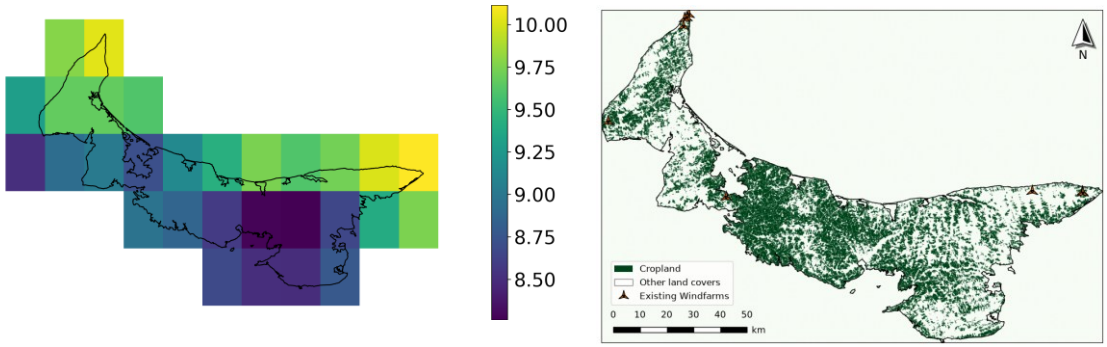


Fig. 8. **Spatial relationship between wind resources, cropland, and existing onshore wind development in Prince Edward Island. (a) Average wind speed at 100 m above ground level, reported in m/s [9]; (b) cropland distribution [6] overlaid with existing onshore wind-farm locations [10] and other land-cover classes. The figure shows the geographic relationship between PEI's strong wind resource, substantial agricultural land base, and existing wind development, providing the spatial basis for the wind–cropland colocation scenarios examined later in the thesis.**

Research has shown wind and agriculture colocation can have both negative and positive effects depending on the region's climate and the crop. A study on corn and soybean in Iowa showed that onshore wind development on these farms increased yields by 0.4–0.8% and reduced the chance of disease and harvest drying costs[12], [13]. However, research on crops such as potato and climates similar to Atlantic Canada is limited and is beyond the scope of this work. Other nexus models such as CLEWs are recommended to investigate such research questions.

On the other hand, wind and agriculture colocation economics for farmers suggests a positive impact on farmers by diversifying their income over volatile, weather-dependent yields[14], [15]. Some reverse effects have also been reported where collocated farms were developed and compromised agricultural production over energy generation; however, this is more prevalent for solar plants[16]. While this research

addresses the economy and engagement of farmers to a limited extent, the field remains largely unexplored.

1.1.3. Prince Edward Island's energy and land system

PEI is a useful focal case because it combines strong wind potential, a limited land base, substantial agricultural land cover, and dependence on electricity imports. These characteristics make the province well-suited for examining how spatial constraints and wind-cropland colocation can affect energy-system planning. Appendix A shows the network diagram of PEI's energy system (a.k.a Reference Energy System (RES)) for a better understanding of the province's energy system, including imports from neighboring provinces and exports to them.

PEI's Energy System

Prince Edward Island's broader energy system continues to rely heavily on fossil fuels, including diesel, which contributes to provincial emissions. As shown in Fig. 9, diesel accounts for 60 percent of the energy mix reported in the PEI dataset [17]. Meeting Canada's net-zero by 2050 target, therefore, requires a shift toward cleaner energy sources, including renewable electricity generation.

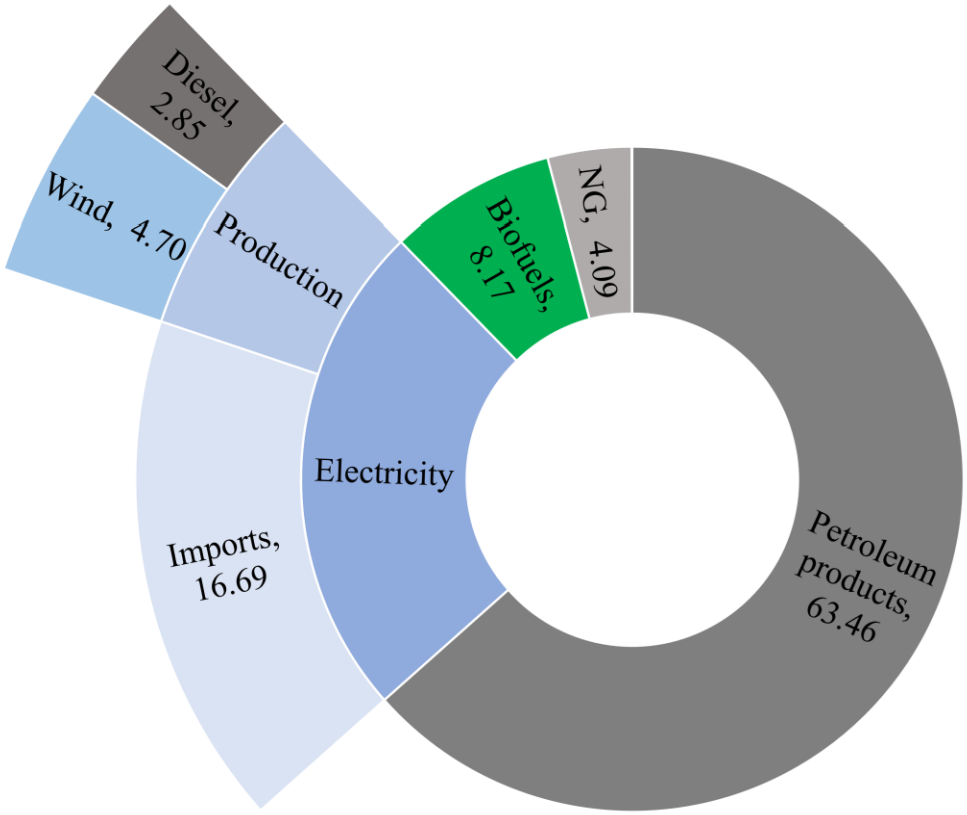


Fig. 9. **Composition of Prince Edward Island's reported energy and electricity supply, showing the contribution of petroleum products, natural gas, biomass, electricity imports, and domestic electricity production, including wind and diesel generation. The figure provides the baseline energy-system context used to motivate future electrification, renewable generation expansion, and reduced dependence on imported and fossil energy [17].**

PEI's electricity demand projection shows a growth from 5.7 to 8.7 PJ by 2050 [18], driven by factors such as population growth, modernization, and electrification (Fig. 10). The electricity system will therefore need additional supply and infrastructure to meet future demand while reducing emissions.

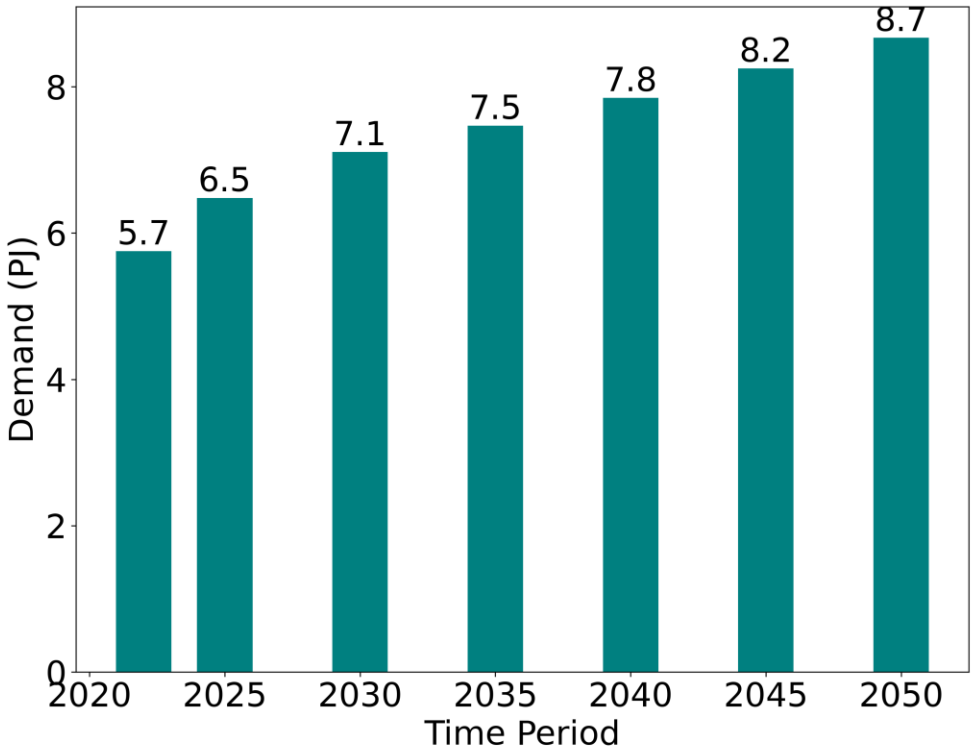
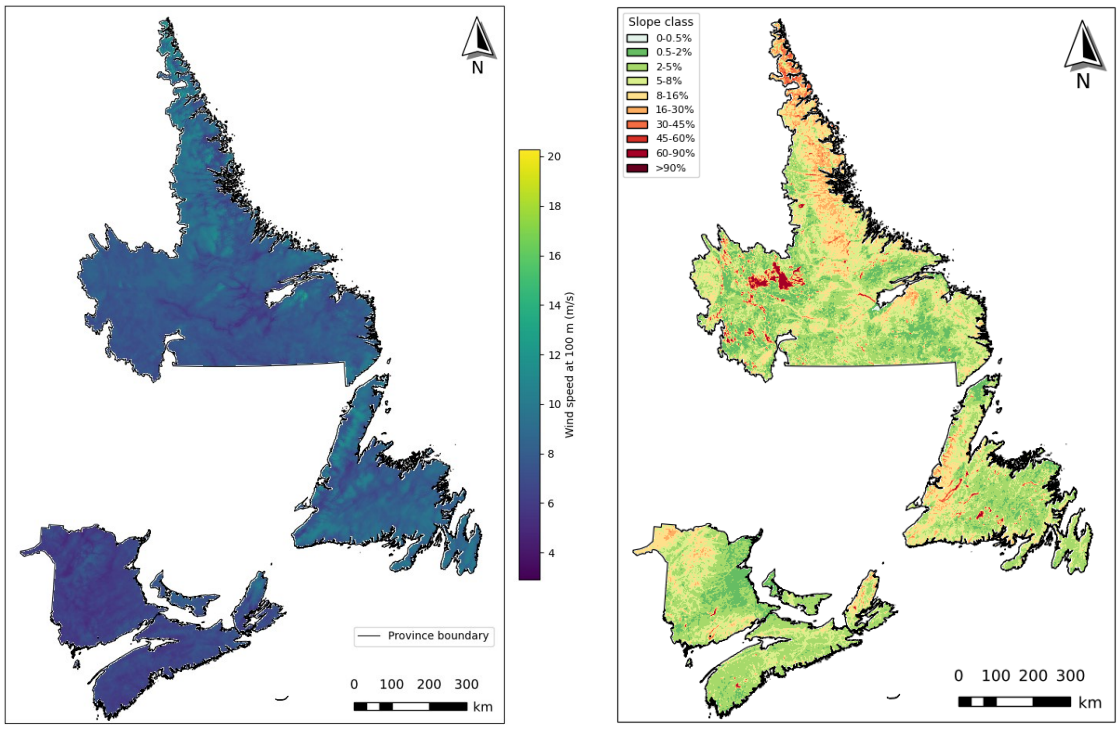


Fig. 10. **Projected electricity demand in Prince Edward Island from 2020 to 2050. Demand increases from approximately 5.7 PJ in 2020 to 8.7 PJ in 2050, providing the exogenous electricity-demand trajectory that motivates future supply and capacity expansion in the energy-system analysis. [18]**

PEI's geography gives the province strong wind potential due to the high wind speed as indicated by the Global Wind Atlas [19] and the relatively flat terrain[6] shown in Fig. 11. Fig. 9 shows that wind already plays an important role in PEI's electricity generation. However, additional low-carbon generation will be required to meet projected demand growth and long-term emission-reduction targets.



a b

Fig. 11. **Regional wind-resource and terrain characteristics across Atlantic Canada. (a) Wind speed distribution at 100 m elevation [19]; (b) slope classification [6] across the Atlantic provinces. Together, the panels illustrate the regional variation in wind-resource quality and topographic suitability that influences the spatial feasibility of onshore wind development, including the comparatively favourable wind and terrain conditions in Prince Edward Island.** **licenced under CC BY.**

PEI also imports a substantial share of its electricity from neighbouring provinces. At the same time, the province's wind potential is higher than that of its neighbours in several candidate locations (Fig. 12). This creates an opportunity to reduce reliance on imports while supporting decarbonization, provided that sufficient land is available for renewable-energy development.

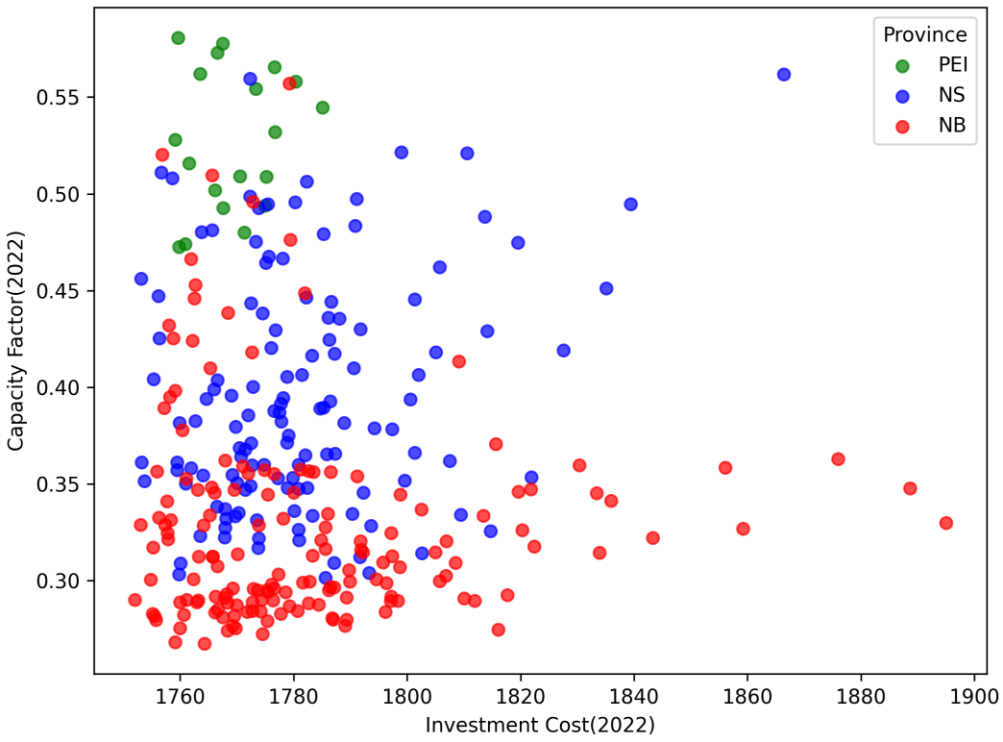


Fig. 12. **Capacity factor and investment cost characteristics of GIS-derived candidate onshore wind development locations in the Maritime provinces. Each point represents a spatial wind cell in Prince Edward Island, Nova Scotia, or New Brunswick, with capacity factor on the vertical axis and estimated investment cost in M$ on the horizontal axis. The distribution illustrates the site-specific trade-off between wind-resource quality and development cost used to differentiate candidate onshore wind technologies in the spatially enhanced TEMOA-ACES model**

Land availability is a central challenge in PEI because cropland accounts for 44.19 percent of the province's land area(Fig. B.1(c)), a larger share than in neighbouring Atlantic provinces. New wind development could therefore compete with agricultural land and potentially affect food production capacity. When cropland is excluded, the available land area for wind development is 83.3 percent, and when cropland is included for colocation, this area is 41.1 percent.

1.1.4. Evolution of Energy System Models Toward Nexus Modelling

Energy system models were initially developed to represent the relationship between energy resources, conversion technologies, and end-use demands. Over time, these models expanded from supply-demand balancing and cost minimization toward

broader representations of climate policy, emissions constraints, water use, land use, and other sustainability dimensions. This evolution is important for the present thesis because land use and spatial constraints are increasingly relevant to the planning of renewable-dominated energy systems.

Energy system models represent the interconnections between energy resources and energy demands, helping engineers and policymakers understand trends, challenges, and opportunities. Broadly, energy system models can be classified into three categories: energy-economy models, which represent economic aspects of the energy transition but often lack technical detail; capacity expansion models, which identify long-term investment pathways but generally do not include full economic feedbacks or operational detail; and operational models, which capture short-term operational challenges but are not designed to evaluate long-term system expansion [20]. This thesis focuses on capacity expansion modelling because it is directly suited to long-term planning under technology, policy, and land-use constraints.

Capacity expansion models are generally implemented as linear programming models, where some operational non-linearities are simplified for computational tractability, or as mixed-integer linear programming models, where selected operational complexities can be represented at the cost of increased computational burden. In either case, the objective is to identify a technically feasible long-term development pathway that satisfies demand and policy constraints, such as emissions reductions, renewable-energy targets, or technology-specific requirements. Policy instruments such as carbon pricing, coal phase-outs, renewable standards, or investment incentives can therefore be represented and evaluated through their effects on modelled energy-system outcomes.

TEMOA Energy System Model

TEMOA(Tools for Energy Model Optimization and Analysis), is an open-source capacity expansion model designed for long-term energy-system planning [3]. It has been identified in the literature as a strong option for energy-system modelling and has been compared with other established energy models [21], [22], [23], [24], [25], [26]. It has also been applied in global research [27] and in regional studies, including applications in the United States [28] and South Africa [29]. Its open-source structure has supported continued model development, including the implementation of Modeling to Generate Alternatives (MGA) methods [30] and improved representation of energy efficiency [28].

TEMOA was originally designed to minimize system cost while meeting demand and satisfying technical and policy constraints. Model costs include direct costs such as investment, fixed and variable costs, as well as indirect costs imposed by emissions. Equation (1) shows the cost elements of the objective function for each period and for the total modeling horizon (C). Each of the following cost values is the discounted summation of that cost group for all technologies present in the model.

$$O = \sum_{p} C_p = \sum_{p} \Big(\Big(C_{invest_p} + C_{invest_p} \Big) * Capaccity + \Big(C_{variable_p} + C_{emissions_p} \Big) * Activity \Big) \quad (1)$$

Emission caps are already a key feature of the model and allow policy targets such as net-zero pathways to be represented. However, land constraints were not included in the original TEMOA formulation, despite the land-use implications of deep decarbonization. Other energy optimization models, including OSeMOSYS, TIMES, and Balmorel, have incorporated land-use considerations [1], [31], [32]. More recently, land integration has been applied to a TEMOA-Italy case study for Pantelleria Island [33], but applications to other regions and to wind-cropland colocation remain limited.

ACES Energy System Model Instance

The Atlantic Canada Energy System (ACES) model provides the capacity expansion modelling framework used to answer the research questions in this thesis. ACES is based on TEMOA and implemented through the ACES model for Atlantic Canada. ACES follows the same general formulation and framework as TEMOA, but is customized for Atlantic Canada through region-specific input data, technology assumptions, and sectoral representation. All the input data used in the ACES model are references from reliable data sources and are cited within each data table. The list of data input sources is included in Appendix C.

This study requires an energy model that can perform scenario-based cost optimization, represent existing infrastructure, model multiple investment periods, include multiple regions and sectors, and support flexible scenario development. To validate the model choice, ACES/TEMOA was compared with OSeMOSYS, PyPSA, MESSAGEix, and TIMES using the criteria summarized in Table 1. The comparison was based on the

ACES curator's model-screening criteria(accessed through personal communication with the ACES developer), which included required capabilities and desirable features.

In Table 1, the color codes indicate the degree to which each model satisfies each criterion: Green cells indicate that the model fully satisfies the criterion in the reviewed framework, yellow cells indicate partial satisfaction or additional implementation effort, red cells indicate a major limitation for this application, and grey cells indicate criteria that were not evaluated in the comparison notes available for this thesis. Rows 7 and after are ‘nice-to-have’ features.

Table 1. Feature comparison for TEMOA and similar models.

	Criterion	TEMOA	OSeMOSYS	PyPSA	MESSAGEix	TIMES
1	Scenario-based cost optimization	Full	Full	Full	Full	Full
2	Existing infrastructure	Full	Full	Full	Full	Full
3	Scenario management	Full	Partial	n/e	Full	Full
4	Open-source license	Full	Full	n/e	Full	Full
5	Multiple regions	Full	Full	n/e	Full	Full
6	Multiple sectors	Full	Partial	n/e	Partial	Partial
7	Flexible temporal resolution	Full	Full	Full	Full	Full
8	Multiple investment periods	Full	Full	Limited	Full	Full
9	Documented codebase	Full	Partial	n/e	Full	Full
10	Commercial and open-source solvers	Full	Full	n/e	Limited	Limited
11	User documentation	Full	Full	n/e	n/e	n/e
12	Active development	Full	Full	n/e	n/e	n/e
13	Python-based platform	Full	Full	n/e	n/e	n/e
14	Established use	Full	Full	n/e	n/e	n/e

The comparison shows that TEMOA and OSeMOSYS are well-suited to the needs of this thesis. In particular, TEMOA scenario-based cost optimization, representation of existing infrastructure, technology vintaging, multiple regions, and multiple sectors. Its structure is also suitable for running multiple scenarios, which is necessary for comparing the ACES model, the spatially enhanced Base model, policy scenarios, and colocation scenarios. It also provides an open-source license, documented model formulation and can use both commercial and open-source optimization solvers, which supports transparency and reproducibility in the academic context. While OSeMOSYS meets most of these criteria, it does not meet the bar for scenario and sector management and the documentation is not as comprehensive as TEMOA. This feature comparison validates the use of TEMOA/ACES as the core energy modeling platform. More importantly, it confirms that the model structure is capable of supporting the additional developments introduced in this work: land-use tracking, land-use constraints, spatially differentiated onshore wind technologies, and scenario analysis under policy and land-scarcity conditions."

For an example energy flow of the ACES model and its technologies and commodities, please check PEI's network diagram as shown in Fig. A. 1 in Appendix A.

In addition to the overall costs, discount rates also influence the model's decision. There are two discount rates used in the TEMOA model: the global and the technology-specific discount rates. In ACES, the global discount rate is equal to 5 percent and reflects the decay of value over the time horizon of the model. The technology-specific discount rates represent the investment value of each particular generation technology and vary between 5.67 and 8 percent. Onshore wind's discount rate is 6.33 percent, which shows its growth compared to other technologies and the global discount rate.

The TEMOA-ACES model is designed based on 7 representative days that represent the supply and demand variabilities over the day, season, and the year. These clustered data cover renewable generation capacities on the supply side and demands on the end-user side. The original ACES model is developed based on the clustered data for a full year and assigns segment fractions to each of the 7 representative days based on their impact in a full year. These clusters are then aggregated for each 5-year period starting from the first year, and follow to the next periods based on NREL learning rates[34]. These learning rates represent how costs change as developed capacity grows over time.

Taken together, these background elements show why land-aware energy-system planning requires both spatial and optimization-based perspectives. Land-use metrics clarify how energy infrastructure occupies land and why onshore wind is especially sensitive to the distinction between physical footprint and spacing. The Canadian land-cover analysis shows that land constraints vary across provinces and that their importance depends not only on their total land share, but also on their overlap with wind-resource potential. Cropland emerges as a particularly important layer because it is not always a strict exclusion; instead, its treatment depends on whether agricultural land is assumed to be unavailable for wind development or potentially compatible with it. Prince Edward Island provides a useful case for examining these issues because its electricity system needs, wind potential, limited land base, and high cropland coverage make the interaction between renewable energy development and land availability especially visible. Finally, TEMOA-ACES provides the energy-system modeling framework through which these spatial and land-use considerations can be evaluated in long-term planning scenarios.

The following literature review builds on this foundation by examining current research findings, identifying how similar questions have been approached, and clarifying the gaps that motivate this thesis.

GIS models

GIS models are models with location data that visualize data on maps to show geospatial relations. Multiple professional software packages are developed for GIS analysis, including ArcGIS and QGIS. However, Python packages such as GeoPandas provide adequate tools to develop Python-based GIS models.

GIS models play a pivotal role in the energy sector; however, integration of geospatial data into energy models is still an emerging field. The energy system encompasses a variety of location-based data, and GIS tools are used to help energy modelers represent these data in their models. Different models integrate spatial elements; however, these instances among capacity expansion models are limited. Improving capacity expansion models with GIS provides strong tools for analysing location-based scenarios.

Python's Atlite package

Atlite is an open-source Python package that converts weather data into power system data. This tool is professionally designed to bridge the gap between energy modeling and spatial modeling by working with land data alongside energy data. Atlite is equipped with multiple features that ease the process of turning weather data, such as wind speed, into energy data, such as capacity factors. Other significant Atlite features are its exclusion command that enables us to filter out land layers and locations based on land use constraints and energy potentials. Atlite is designed to work with large datasets, such as climate reanalysis data, without heavy computation[35], [36].

1.2. Literature Review

In this chapter, existing research on land-use analysis, spatial planning, and renewable energy colocation is reviewed. The review establishes the methodological basis for integrating four areas that are often treated separately: land-use accounting, GIS-based spatial analysis, social and sustainability constraints, and wind–cropland colocation. The review is organized into five subchapters. Section 1.1.4 traces the

evolution of energy-system models from cost- and supply-focused planning tools toward broader nexus modelling frameworks. Section 1.2.1 discusses the growing attention to equity, social considerations, and the Sustainable Development Goals in energy modelling. Section 1.2.2 reviews how land use and spatial representation have been incorporated into energy models, with particular attention to renewable energy and onshore wind. Section 1.2.3 examines colocation of energy infrastructure, including the potential synergies and challenges associated with using the spacing between wind turbines for other applications. Finally, Section 1.2.4 reviews approaches for linking GIS and energy models, showing how spatial data can be integrated into long-term energy-system planning.

Recent energy-system modelling research has increasingly incorporated interactions with other systems and sectors. For example, Vinca et al. combine the MESSAGEix capacity expansion model with the CWatM water-management tool to evaluate how water-energy interactions affect feasible decarbonization pathways [37]. The climate, land, energy, and water systems (CLEWs) framework similarly illustrates the expansion of energy-system modelling toward cross-sectoral nexus analysis [38]. CLEWs-based approaches have been applied in a variety of country contexts, including Vietnam and Canada [39], [40]. Recent scholarly work also includes the integration of biodiversity into energy modeling [41].

1.2.1. Social Dimensions, Equity, and the SDGs in Energy Modelling

With the adoption of the Sustainable Development Goals (SDGs) in 2015, researchers increasingly considered how energy-system models could account for broader sustainability objectives. Niet et al., for example, call for models to incorporate more SDG-related dimensions so that trade-offs can be identified and co-benefits quantified for policy analysis [42]. The 17 SDGs include environmental, economic, and social objectives, each with associated indicators and targets for progress by 2030. Because these goals interact with one another, policies aimed at one goal can produce co-benefits or trade-offs across several others [43]. Energy-system models can therefore provide more coherent policy insights when they represent cross-sectoral interactions rather than treating energy in isolation.

Beyond SDG framing, a growing body of literature has attempted to incorporate equity and justice considerations into energy modelling. Studies that directly combine

equity and optimization remain limited, and many modelling studies that address equity do so indirectly through community engagement or social-science input rather than through formal model constraints. Rubiano et al. argue that techno-economic optimum solutions may overlook justice considerations and lead to inequitable policy outcomes [44]. Belaïd reviews the energy-poverty implications of poorly designed climate policies and emphasizes the importance of addressing SDGs holistically for energy justice [45]. McGookin et al. discuss participatory approaches to energy modelling, including community engagement [46], while Heleno et al. combine sociodemographic and techno-economic models to reduce energy burdens on vulnerable populations in a Detroit case study [47].

This thesis does not conduct a full equity assessment; however, social considerations are relevant to its spatial modelling framework. Indigenous Reserve Lands and associated buffers are included as socially sensitive spatial layers, while cropland is treated as a context-dependent layer because wind development may affect agricultural land users differently depending on whether land is excluded, shared, or colocated. In this way, the thesis connects technical energy-system planning with selected spatial and social dimensions of land-use decision-making.

1.2.2. Land Use and Spatial Representation in Energy Models

Land and energy systems are spatially coupled: energy infrastructure occupies land, while land characteristics constrain the feasible siting of energy technologies. This interaction becomes more important as energy systems transition toward higher shares of renewable generation, because many renewable technologies are more spatially extensive than conventional fossil-fuel generation. Spatially explicit land representation is therefore needed to understand not only how much land is required, but also where that land is located and what other uses it may affect.

As energy systems transition toward a higher share of renewables to achieve net-zero targets, the land use associated with energy production can rise substantially because several renewable technologies have relatively high land intensities [1]. The choice of land-intensity metric plays a critical role in shaping these findings, particularly for onshore wind, where turbine spacing may or may not be counted as occupied land. Many studies account for wind-farm spacing as fully occupied land [1], [31], [32], [48], [49].

However, this assumption can overlook cases where the spacing between turbines remains available for other uses, such as agriculture or solar PV integration [50].

Energy-modelling studies have used several approaches to account for land use. Some studies post-process the land footprint of existing or modelled energy systems [48], [51], [52]. Others impose land limits on the entire energy system to assess system-wide effects [1], [31], [53], while some constrain individual production technologies [49], [54], [55]. Recent work has also introduced cost-based land-use optimization [33] and Modeling to Generate Alternatives for land-related planning questions [32]. Although this literature has advanced land representation in energy models, further work on representing onshore wind's distinctive spacing characteristics and the potential for shared land use with agriculture is needed.

1.2.3. Colocation of Energy Infrastructure, Synergies and Challenges

Colocation is a land-use strategy for reducing effective spatial competition associated with spatially extensive energy infrastructure. Researchers have examined the functional use of spacing within energy production sites by colocating multiple applications, including wind and solar power plants [56], storage facilities, and agricultural activities [57]. Wind energy is particularly relevant for colocation because turbine spacing can occupy a large area while leaving much of that area physically available for other uses. Colocation has been studied in offshore contexts [58], [59] and in onshore applications, including wind-solar integration [60] and wind-agriculture combinations [61]. Krauland et al., for example, examine the colocation of onshore wind and solar PV and its implications for land use [62].

Despite this emerging literature, colocation is not always represented explicitly in energy-system models. Many studies continue to treat turbine spacing as occupied land within wind farms [1], [48], [49], [63], even though spacing may be used for other purposes in practice. Land-intensity values for onshore wind and other energy technologies have been widely discussed in the literature [52], [64], and the selected land-intensity value can substantially change model outcomes. For example, Palmer-Wilson et al. use a wind land-intensity value of 368.3 km^2/GW [1], whereas Chen et al. use 50.505 km^2/GW [32], based on Enevoldsen et al.'s work to improve estimates of occupied land [65]. Brown and Botterud also incorporate wind-solar colocation to account for shared land use [63]. These

studies show that land-use assumptions matter, but a comparison of models' reaction to turbine spacing and colocation is still needed. In addition, wind-agriculture colocation and alternative colocation options need to be studied.

The high apparent land intensity of onshore wind therefore creates both a modelling challenge and a land-allocationopportunity. If spacing is treated as fully occupied land, onshore wind can appear highly land-intensive and may become less attractive under land constraints. If spacing can be shared with agriculture or other compatible uses, the effective exclusive land requirement of wind development will be lower. This thesis builds on this distinction by comparing conventional wind siting, cropland exclusion, and wind-cropland colocation within a spatially enhanced energy-system modelling framework.

1.2.4. Energy and GIS Model Linking

Energy system optimization models and Geographic Information Systems (GIS) provide complementary capabilities for spatially explicit energy planning. Capacity expansion models represent existing energy systems and optimize future technology investments under demand, cost, and policy constraints. GIS models provide spatial information on resources, land cover, infrastructure, and siting restrictions. Linking these tools can therefore improve the spatial relevance of energy-system planning by allowing geographically explicit resource and land data to influence long-term investment decisions.

Model linking for spatial explicitness has gained attention in recent energy-systems literature. Javanmardi et al. review mutual land-energy system interactions and different model-linkage approaches, concluding that recursive soft-linking can effectively integrate spatial models into energy-system models [66]. Such linkages are useful when spatial information is too detailed to represent fully inside the optimization model but remains essential for defining feasible technologies, constraints, or scenarios.

One recurring application of spatial model linkage is transmission planning. Brown et al. co-optimize capacity planning and dispatch to study the value of inter-regional transmission in decarbonizing the U.S. electricity system [63]. Bennett et al. use the GIS-

energy tool SimWIND to examine wind-farm siting, transmission development, and environmental justice implications [67].

Another major application is renewable-energy siting and land-use assessment. Mosso et al. integrate land use into the TEMOA-Italy capacity expansion model by linking it to a GIS model and adding land prices to the TEMOA objective function [33]. Strachan et al. similarly implement a soft-linked GIS and MARKAL model for hydrogen infrastructure development [68].

For renewable technologies, GIS-energy model linkage is particularly valuable because resource quality, land availability, environmental constraints, and social acceptance vary spatially. Radu et al. study renewable siting using a combination of site selection and a PyPSA-based capacity expansion model [69]. Grimsrud et al. focus specifically on onshore wind, assessing biodiversity and social-acceptance dimensions through a linked GIS and TIMES model [41]. These studies show the value of linking spatial analysis with energy-system optimization, but there remains a need for approaches that explicitly combine wind siting, land-use accounting, and wind-cropland colocation within capacity expansion modelling.

Overall, the literature shows that energy-system models have evolved from tools focused primarily on supply-demand balance, cost minimization, and emissions reduction toward broader frameworks that increasingly consider interactions with water, land, social outcomes, and sustainability goals. Existing studies have made important contributions by incorporating land-use constraints, evaluating renewable energy land requirements, exploring colocation opportunities, and linking GIS-based spatial analysis with energy models. However, several gaps remain. Many energy models still represent land at an aggregated level or quantify land use through post-processing rather than embedding spatially differentiated land availability directly in the optimization formulation. For onshore wind, studies often treat turbine spacing as occupied land, even though this area may remain available for other uses under certain siting or colocation assumptions. At the same time, GIS models can identify spatially feasible wind locations, but their outputs are not always integrated into capacity expansion models in a way that affects investment decisions, technology selection, system costs, emissions, and land-use outcomes. These limitations motivate the approach developed in this thesis: linking GIS-based wind siting with a land-aware TEMOA-ACES model to evaluate land constraints, policy scenarios,

and wind–cropland colocation. The following chapters build on this review by identifying the specific research gap, defining the research questions and objectives, and outlining the thesis contributions.

1.2.5. Summary of Research Gaps

The literature reviewed above shows that energy-system models have become increasingly capable of representing decarbonization pathways, technology investment, emissions constraints, and interactions with other resource systems. However, land use is still often represented at an aggregate regional or technology level, or calculated through post-processing, rather than incorporated as a spatially explicit constraint that directly influences investment decisions. This limitation is particularly important for renewable electricity planning, where technologies such as wind and solar can require substantially larger land areas than conventional generation technologies and where site suitability depends on the overlap between resource quality, land cover, infrastructure access, and social or environmental constraints [1], [31].

Onshore wind creates a specific modelling challenge because its land-use implications depend on the definition of land occupation. The physical footprint of turbines and associated infrastructure is relatively small, but the spacing between turbines can be large and is frequently included in land-intensity estimates. Many modelling studies therefore treat the full wind farm area as occupied land, even though the space between turbines may remain available for agriculture, grazing, roads, or other compatible uses under certain siting assumptions [1], [49], [63]. This creates uncertainty in how onshore wind should be represented in land-constrained energy-system models, especially in agricultural regions where wind development may either compete with cropland or coexist with it.

At the same time, GIS-based studies can identify spatially feasible renewable energy locations and quantify overlaps between wind resources and land-cover layers, but these outputs are not always integrated into long-term capacity expansion models in a way that affects technology selection, capacity additions, system cost, emissions, and land-use outcomes. Recent work has begun linking GIS and energy-system models, including applications that integrate land use into optimization frameworks [33], [66]. Nevertheless, there remains a need for approaches that connect spatially resolved wind-

siting information with land tracking, land constraints, and policy scenarios in a regional capacity expansion model.

This thesis addresses this gap by developing a spatially enhanced and land-aware version of the TEMOA-ACES model for Atlantic Canada. The work links GIS-derived onshore wind locations with the energy-system model, incorporates direct and indirect land-use accounting, applies land constraints, and evaluates how policy and land-use assumptions affect long-term electricity-system outcomes. The thesis further applies this framework to Prince Edward Island, where wind potential, limited land availability, and high cropland coverage make the interaction between renewable energy development and agricultural land particularly important.

1.3. Research Questions and Objectives

The main research question guiding this thesis is:

How can spatial land-cover information and land-use accounting be integrated into energy system optimization to evaluate land-aware clean electricity pathways for Atlantic Canada, with particular attention to onshore wind development and wind-cropland colocation in Prince Edward Island?

This overarching question is addressed through the following research questions:

1. How do selected Canadian land-cover layers overlap with onshore wind potential, and which layers are most relevant for wind-siting constraints or opportunities at the provincial scale?

2. How does replacing generalized onshore wind technologies in ACES with GIS-derived, spatially explicit wind locations affect modelled capacity expansion, activity, cost, emissions, and land-use outcomes?

3. How do clean-energy policy instruments, specifically the Investment Tax Credit (ITC) and Renewable Portfolio Standard (RPS), affect the spatial and land-use implications of electricity-system development in Atlantic Canada?

4. How do different treatments of cropland, including conventional siting, cropland exclusion, and wind-cropland colocation, affect Prince Edward Island's optimal electricity mix under alternative land-availability constraints?

To answer these research questions, the thesis meets the following objectives:

- Develop a GIS-based onshore wind siting framework that combines wind-speed data, land-cover layers, slope, cropland, Indigenous Reserve Lands and buffers, protected areas, wetlands, water bodies, built environments, and grid-connection distance.

- Quantify how selected land-cover layers intersect with onshore wind potential across Canadian provinces, with particular attention to layers that are spatially extensive, socially or environmentally sensitive, or relevant to wind-agriculture colocation.

- Integrate land-use tracking into TEMOA-ACES by representing capacity-based direct land use and activity-based indirect land use through technology-specific land-intensity parameters.

- Develop land constraints within TEMOA-ACES so that land availability can influence the optimization process rather than being evaluated only after model results are produced.

- Replace generalized onshore wind technologies in ACES with GIS-derived potential wind cells characterized by available land, capacity factor, maximum capacity, and grid-connection cost.

- Design and run policy scenarios for Atlantic Canada to evaluate the individual and combined effects of the ITC and RPS on costs, technology choices, emissions, and land-use outcomes.

- Design and run Prince Edward Island scenarios to compare conventional wind siting, cropland exclusion, and wind-cropland colocation under a range of land limits.

- Validate the modelling framework through input-data checks, model-feature comparison, result comparison, and edge-case testing of land-intensity assumptions.

1.4. Contributions

This thesis makes four main contributions. First, it makes a conceptual contribution by clarifying how land use should be interpreted in energy-system planning, particularly for onshore wind. Second, it makes a methodological contribution by integrating land-use tracking, land constraints, and GIS-derived wind-resource locations into the TEMOA-ACES capacity expansion model. Third, it makes an applied case-study contribution by using this framework to assess Canadian land-cover constraints, Atlantic Canada clean-energy policy scenarios, and PEI wind–cropland colocation. Fourth, it makes a policy-analysis contribution by evaluating the individual and combined effects of ITC and RPS policies under a spatially enhanced, land-aware energy-system model.

Conceptual contribution: The thesis clarifies how different land-use definitions affect onshore wind modelling. By distinguishing between footprint, spacing, and indirect land use (Section 1.2.2), and by comparing conventional, exclusion-based (spacing scenario in 2.2.2), and colocation-based treatments of cropland, the work demonstrates how land-intensity assumptions can alter the interpretation of wind development in energy-system planning (Section 3.2.3).

Methodological contribution: The thesis extends TEMOA-ACES by adding land-use tracking and land-limiting functionality. This enables the model to estimate both direct and indirect land use and to represent land availability as a constraint within the optimization problem. The work also develops a soft-linking workflow between GIS and TEMOA-ACES, where spatially defined wind locations are translated into energy-system technologies with site-specific capacity factors, land availability, and grid-connection costs. This formulation enables explicit evaluation of land-constrained and colocation scenarios within the capacity-expansion optimization.

Applied contribution: The thesis provides a Canada-wide assessment of selected land-cover layers and their overlap with onshore wind potential, then applies the spatially enhanced modelling framework to optimize Atlantic Canada and Prince Edward

Island's energy system development. This quantifies how land availability, wind resource quality, clean-energy policies, and cropland assumptions influence technology selection, electricity generation, imports, emissions, system costs, and land-use outcomes in the case-study regions.

Policy and planning contribution: The thesis provides a framework for evaluating renewable energy pathways in which land is treated as a finite and spatially differentiated planning constraint. The Atlantic Canada policy scenarios show how ITC and RPS instruments can affect land-use and spatial outcomes, while the Prince Edward Island colocation scenarios show how wind-agriculture interactions may influence land-constrained electricity planning. These results provide a quantitative basis for planning decisions concerning renewable energy siting, agricultural land protection, and long-term decarbonization pathways.

1.5. Thesis Organization

This thesis is organized into four chapters. Chapter 2 presents the methodology. It describes the development of the land tracking and land constraint features in TEMOA-ACES, the GIS-based onshore wind siting model and the linkage between GIS outputs and the energy-system model. This chapter also explains the scenario design of the ACES-to-Base model comparison, the Atlantic Canada policy scenarios, and the Prince Edward Island colocation scenarios. The chapter is then concluded with a validation process, approached from three angles: input and output data, and the model.

Chapter 3 presents the results. It first reports the Canada-wide land-cover and wind-potential analysis, then compares the original ACES model with the spatially enhanced Base model. It then presents the Atlantic Canada policy scenario results and the Prince Edward Island land-constrained colocation results, focusing on changes in capacity, activity, costs, emissions, imports, and land-use outcomes.

Chapter 4 discusses the implications of the results. It interprets how spatially explicit wind representation changes energy-system outcomes, how policy instruments interact with land-use constraints, and how cropland treatment affects the feasibility and interpretation of onshore wind development in Prince Edward Island. The chapter concludes the thesis by summarizing the main findings, answering the research questions,

and outlining the broader contributions of the work. It also discusses model limitations, data limitations, and implications for energy planning and future research, including improved social and equity representation, more detailed agricultural impacts, stronger observational validation of wind resources, and further development of spatially linked energy-system models.

Chapter 2. Methods

Our methods for this work consist of three pieces: model development for 1) land tracking, 2) land constraints, and 3) spatial analysis. In this chapter, we first summarize each of these pieces and elaborate on how they connect to answer our research questions.

The remaining part of the chapter covers our scenario design. We eventually validate the model and data at the end of this Chapter.

2.1. Model Development

As mentioned above, the model development phase consists of three phases: one is spatial processes and GIS modeling, and the other two involve the TEMOA [3] capacity expansion optimization model, which is developed for our case study under the ACES model (as explained in Section 1.1). The following subchapters summarize these three modeling steps and two models.

We develop a GIS framework, integrate land into an energy model, and connect these two models to exchange data. Fig. 13 presents the workflow of our approach and model connection. The workflow starts with two separate models (yellow box): the GIS model and the ACES energy model. Each of the models receives different data (green box) as inputs, according to its model structure and targets. On the energy model side , land intensity data are integrated as inputs, and on the GIS side, land layers, wind speed data and the grid connection nodes are the inputs. The GIS model provides potential onshore wind locations, also referred to as cells. Each cell is represented by three values as intermediate results (blue box): its available land, available wind, and distance to the nearest grid node. These cells' data are then integrated into the energy model in a unidirectional approach to facilitate reaching an optimal energy system solution. The energy model can then process its land intensity inputs independently, allowing us to track land use for the electricity system. However, a land constraint can be applied to the energy model to observe model behaviour under land scarcity. This is the last step in the sequence and its outputs include the costs, capacities, emissions and land uses of the energy system, among other parameters. The following chapters will elaborate on each of these steps, starting from the top, the GIS model development and model connection,

followed by land intensity integration and constraints. Among the parts explained, the pieces highlighted in grey were developed in this work and integrated into the existing ACES model workflow.

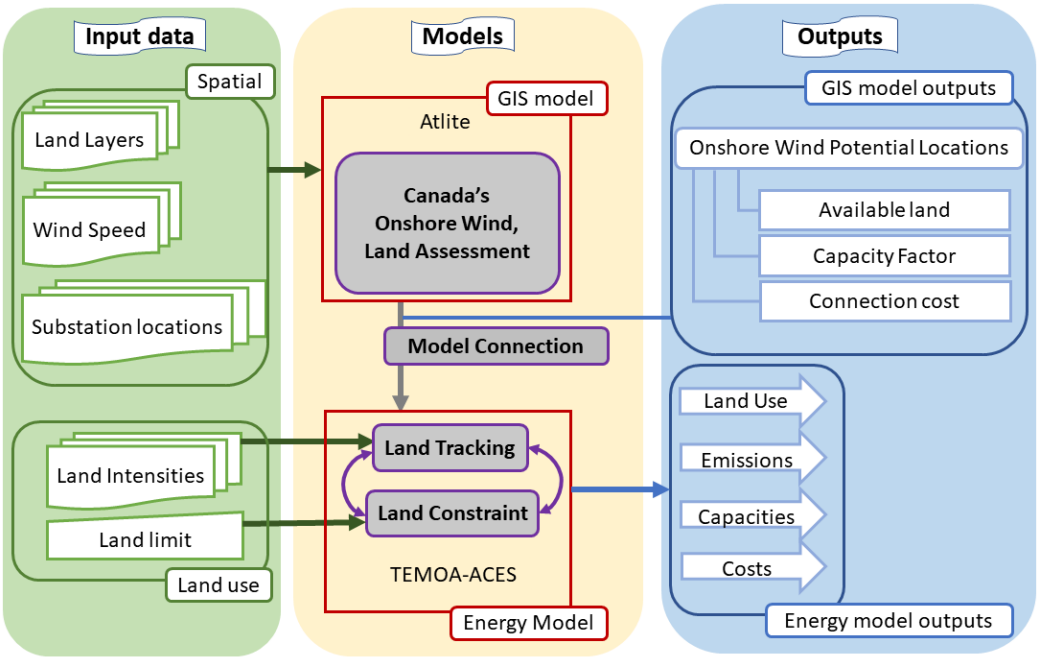


Fig. 13. **Integrated GIS–TEMOA-ACES modelling workflow developed in this study. The GIS component combines wind resource data, land cover constraints, and distance to grid infrastructure to identify candidate onshore wind cells and characterize each cell by available land, capacity factor, and grid connection-associated cost. These spatially differentiated wind technologies are transferred unidirectionally to TEMOA-ACES, where technology-specific land intensity parameters enable land use tracking and land constraints can be imposed within the capacity expansion optimization. The resulting model outputs include technology capacity, activity, system cost, emissions, and land use. Grey elements indicate components developed as part of this thesis.**

2.1.1. Spatial Land Modeling of Onshore Wind

This step is the first step in our unidirectional model connection, so it happens first in the chain as described in Fig. 13 (GIS model). This geospatial model is coded based on GIS fundamentals in a Python setting for easier integration with the Atlite package. GIS tools such as QGIS are now equipped with built-in Python features; however, these features have limited capabilities compared to Python. This work is highly reliant on the

Atlite package, which can only be implemented in an independent Python environment. Therefore, the spatial part of this work is coded in Python. This decision has some minor compromises compared to using a professional GIS tool; however, due to the focus on the energy sector, the minor trade-offs are negligible.

Wind speed data extracted from ERA5 are collected in the EPSG:4326 coordinate reference system(CRS); however, for visualizations, all data and maps are exported to the official provincial projections based on their Universal Transverse Mercator(UTM) system for consistency. These projection values are as follows:

Table 2. Projections used for visualizing provincial maps

Province	CRS
PEI	EPSG:26920
SK	EPSG:26913
AB	EPSG:26912
MN	EPSG:26914
BC	EPSG:3005
ON	EPSG:3161
QC	EPSG:6622

The spatial model is designed to find the location and qualities of potential onshore wind farms under different constraints. The constraints include two input data categories: wind potential and land cover.

The wind potential data provides information on the onshore wind capacity each location can produce, using wind speed data and calculating capacity factors. These data are gathered in the form of hourly historical wind speed time series at 100 m elevation for 2019(the reference year of our energy model) at a 30*30 km resolution [9] and are extrapolated to 110 m elevation, which is the hub height of our wind turbine. This database is chosen since it offers a high temporal resolution, which matches our energy model's representative day structure. The original ACES model's onshore wind resolution is based on the Pan-Canadian Wind Integration Study (PCWIS) [70] and provides only four technology zones, which is a low spatial resolution. This work improves this technology resolution and increases the number of onshore wind zones to tens for PEI and hundreds for NS, NB and NL. Therefore, even though the 30*30 km resolution is relatively low in the geospatial context, it provides a ten- to hundred-fold improvement in the technology resolution of our model.

A cutout is then generated from these data to match their format with energy system models using the Atlite package[35]. These data are filtered for our target location based on the Canada boundary file to minimize the data capacity burden[5]. These boundaries are available in both national and provincial scales. Now we know the wind speeds for our desired location and resolution. Then we extrapolate the speeds to 110 m height using Atlite's extrapolation feature and the logarithmic law described by [71]. We finally apply production array losses, including turbulence and wake effects [70] to the speed data to be able to calculate capacities.

To derive capacity factors from speed data, we first classify them into ten NREL wind classes based on their speed range[34]. Eventually, capacity factors are calculated for each sample cell based on a chosen wind turbine model that is consistent with the ACES model's choice of turbine[72]. This turbine is NREL ATB's moderate case reference turbine and is pre-defined in Atlite [73]. Atlite's 'cutout.wind' feature derives capacity factors for 2030 from wind speeds based on this turbine model, which operates according to the reference power curve. Electrical losses are applied to capacity factors at this point [70]. Eventually, capacity factors for previous time periods, starting from 2022, are calculated from the 2030 estimates by applying a wind learning rate factor as a function of wind class and year [34]. Now we have the capacity factors for all cells and time periods.

Land layers provide information on the areas where wind farm development is not allowed, desirable, or feasible. These layers comprise various data types and their characteristics, which are elaborated in Table 3. Hard exclusions are inevitable, while soft exclusions are flexible. Binary layers can be included or excluded, while continuous and incremental layers can be positioned somewhere in between. The last column shows the category of each exclusion land layer in terms of its type of impact. These land layers have a 100*100 m resolution; however, it is aggregated and downscaled to 30*30 km to match the wind resolution.

Table 3. Data layers and corresponding references.

Exclusion Land Layers	Exclusion type	Data type	Impact type
Urban Environment [8]	Hard	Binary	Technical
Protected and Conserved Areas [7]	Hard	Binary	Governmental
Wetlands [8]	Hard	Binary	Environmental
Water Bodies [8]	Hard	Binary	Technical
Cropland [8]	Soft	Binary	Environmental
Indigenous Reserve Lands [5]	Soft	Binary	Social
Buffers to the above layers [74]	Soft	Continuous	Social
Slope [6]	Soft	Incremental	Technical

Lastly, the distance of each identified location(cell) to the closest substation [10] is estimated for grid connection based on a straight-line distance. This distance impacts the investment cost [75] for developing each of the potential sites, based on their estimated grid connection costs, while the original investment costs are derived from NREL ATB's moderate case [34].

Each 30*30 km cell represents a potential technology for the optimization model, which in our case is the TEMOA-ACES model. Cells are defined with three key parameters: their wind class(average capacity factor), land availability(area eligible for wind development) and Investment cost(costs of development and connection to the grid). The number of cells varies for each province based on their area and in cases other than PEI, they are filtered based on wind and land availability to keep the required processing power manageable. The threshold for this filtering depends on the model and dataset sizes in each case.

The technologies(onshore wind only) are chosen for the optimization based on their land availability and wind class. Land availabilities are defined as percentages and wind classes are integers between 1 and 10, where 1 is the strongest. For the three Maritime provinces, we ran each provincial model separately and with the full list of available technologies, without applying any primary filters. So our availabilities were over 10 percent and our wind classes included all 10 classes.

We then picked the optimal technologies from each of the 4 individual runs and ran an Atlantic scenario including these technologies. We chose provincial optimals over

regional optimals rather than applying the same threshold on all provincial technologies, since local supply is more favorable than importing from other provinces in our case.

For Newfoundland and Labrador, the primary number of cells without filtering was 966, which required unavailable computing power. So we applied a primary filter of wind classes lower than 8 and available lands higher than 80 percent and narrowed down the number of technologies to 419. This threshold was chosen since any number of technologies higher than this would result in the model collapsing or not converging. Then the NL provincial model was run and the optimal technologies were added to the maritime *technology list to provide an Atlantic model dataset. Table 4 shows sensitivity analysis results on NL's number of technologies based on wind class and land availability and indicates that the number of cells is sensitive to available land and lower than 80 percent availability gives us significantly fewer cells. However, cells are not significantly sensitive to wind classes.*

Table 4. NL's number of technology cells under different wind class and land availability filters

Land availability \ Wind Class	8	9	10
70%	522	589	597
80%	419	447	450
90%	128	142	142

2.1.2. Integrating Land Use into the TEMOA model

Land integration into the TEMOA model includes two steps: land tracking [48], [52], [76] and land limiting [1], [31], [32]. For the land tracking part, we first explain our land intensity data and then present the equations for model development. For the land-limiting chapter, we show the equations applied to the objective function.

Developing Land Tracking Function for the TEMOA model

As discussed earlier, the land tracking feature is defined in two parts: one is a capacity-based function for direct land use and the other is an activity-based function for indirect land use. These two equations allow us to calculate the land use of the optimized energy system, both for individual technologies and for all technologies. These land use estimates are available as model results based on different regions, vintages, and periods. Eqs. (2) and (3) show the indirect and direct functions, respectively.

$$\mathrm{U_{c,r,T,v}} = \sum_{\mathrm{r}} \sum_{\mathrm{T}} \sum_{\mathrm{v}} \left(\mathrm{I_{c,r,T,v}} \times \sum_{\mathrm{t}} \mathrm{C_{r,t,v}} \right) \quad (2)$$

$$U_{a,r,T,v} = \sum_{r} \sum_{T} \sum_{v} \left(I_{a,r,T,v} \times \sum_{t} A_{r,t,v} \right) \quad (3)$$

In Eqs. (2) and (3), the main variables are as follows: ‘U’ is land use, respectively for capacity-based and activity-based land uses. ‘I’ is land intensity. ‘C’ and ‘A’ are capacity and activity and are both a main variable and an index (‘c’, and ‘a’), and other indices are: ‘r’ for region, ‘T’ for technology group, ‘v’ for vintage, and ‘t’ for technology.

Vintage in our case is the year when the technology was built and is different than ‘period’, which is each 5-year period based on which the model runs. Therefore, vintage values can be lower or higher than the starting year of the model, while period values can only be equal to or higher.

The starting year of the model was decided based on the original ACES model and is 2019. However, no technologies are allowed to be built before 2022, and some before 2025 depending on the local policies and particular scenario assumptions. These choices minimized the time and resources required to update the model data within the context of this study and are validated to be representative and valid for the current energy system in 2026.

It is noteworthy that we apply a greenfield approach in our land modeling as opposed to a brownfield approach [77], [78], meaning that in our case, land is taken once and not used again after decommissioning. Therefore, a land that is used for an energy generation technology remains occupied during its operation and after.

Methods for Incorporating Land Limits into the TEMOA model

Limiting land use in the ACES model allows us to run scenarios in which future land use is constrained for energy system development. While land tracking functions provide a clear picture of existing and future land use, it also prepares the foundation for applying a cap to this parameter. As Equation (4) shows, the land use constraint applies a limit on the summation of both capacity-based and activity-based land uses.

$$\min \quad O$$

$$\text{s.t.} \quad \sum_{r,T,p} U_{c,r,T,p} + U_{a,r,T,p} \leq L_{r,T,p} \tag{4}$$

In Eq. (4), the main variables ‘O’ and ‘L’ are respectively TEMOA’s cost objective function and land limit, where O is a variable and L is an exogenously defined parameter. The index ‘p’ is the period andll other variables and indices are the same as in Eqs. (2) and (3).

This constraint limits the overall land use of technologies; however, it does not include the cost of land in the model’s objective function. As mentioned earlier in the literature review, a few papers have incorporated the cost of land in energy models’ objective function to directly impact the optimal minimum cost. However, in our case, the cost of land is included in the objective function as a portion of the investment cost based on NREL data values[34]. Due to the calculation limitations of the scope of this work, these original costs were left untouched and only the overall land use in km^2 was capped using the land limit integrated as explained above.

Land-Use Intensity Inputs

Building on the land-use definitions introduced in the background chapter, land use in TEMOA was represented using two intensity parameters: capacity-based land intensity for direct land occupation and activity-based land intensity for indirect land requirements. Capacity-based intensities are reported in km2/GW, while activity-based intensities are reported in km2/PJ. These parameters were assigned by technology and used in the land-tracking equations described below.

Our intensity metric for diesel encompasses both direct and indirect land use in unit of km^2/GW. The land intensity of Biomass is limited to the generation(direct) land use per capacity and the indirect fuel-based land use per activity is not included. The reason for this is that the fuels are forestry residues and byproducts and are not dedicated energy crops used for energy generation purposes only. Table 5 and Table 6 summarize these data, where the former shows the per-capacity and the latter shows the per-activity land intensity.

Table 5. Capacity-based(direct) land-use intensity inputs used in this study.

Technology	Land Area Impact factors per capacity (km2/GW)	Reference
COAL	4	[1]
DIESEL	5.14	[79]
NG, NGACT	1.8	[1], [80], [81]
NGACC-CCS	2.2	[1]
BECCS	5	[1]
BIO, BIO-COGEN	4	[1]
NUC-SMR	4.15	[79]
GEO	38.8	[1], [52]
HYDRO-LIMDSP	84.6	[1]
SOLPV-CEN	34.4	[52]
SOLPV-RES	0	[1]
WIND-ONSHORE	368.3	[1], [52]

Table 6. Activity-based(indirect) land-use intensity inputs used in this study.

Technology	Land Area Impact factors per fuel (km^2/PJ)	Reference
Imported natural gas	0.074	[81]
Imported wood	0.000	[1]
Imported coal and coke	0.075	[82]

While we incorporate all the above intensities into our land use tracking feature, the intensity for onshore wind is particularly important. Our important data for this work are land use intensities of onshore wind under different colocation scenarios. The land intensity included in Table 5 accounts for the spacing scenario. We also use 6.5 km^2/GW [83] for onshore wind without spacing. This intensity is applicable to the colocation of wind and agriculture, explained further in the scenario development section.

2.2. Scenario Development

In this section, first we summarize the methodology implemented on the GIS model of Canada. Later, we elaborate on the scenarios developed and studied under the energy optimization model.

2.2.1. Canada's Land and Energy

In this section, we present the highest-impact land layers in their target provinces as shown in Table 7. To assess how these land layers are impacting the onshore wind

potential, we compare scenarios where these layers are included or excluded from site selection and calculate the average capacity factors of the cells impacted after this exclusion. We then compare these average capacity factors with the average capacity factor of all the province's cells to estimate whether this exclusion is impacting the onshore wind potential and if the impact is concerning.

Table 7. Studies land layers and provinces

Province	Land layer	Included	Excluded
BC	Slope (9 classes)	Average capacity factors	
MN	Wetland		
	Water bodies		
QC	Water bodies		
ON	Wetland		
AB	Cropland		
	IRL+buffer		
SK	Cropland		
NS	PCA		
PEI	Cropland		

2.2.2. Spatially enhanced TEMOA model

In this section, we discuss the spatially enhanced TEMOA model(coded as 'SE') under three categories of scenarios. Fig. 14 shows a visual of these scenarios. First, we compare the TEMOA model with our spatially enhanced model (red box). Then we implement scenarios on the spatially enhanced model to assess policies and their land use implications in Atlantic Canada (green box). Lastly, we design and implement scenarios to study the colocation of onshore wind and cropland in PEI (yellow box).

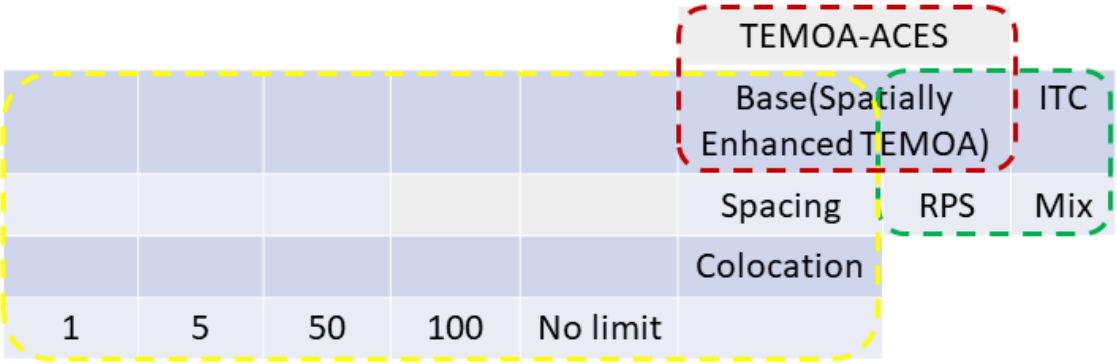


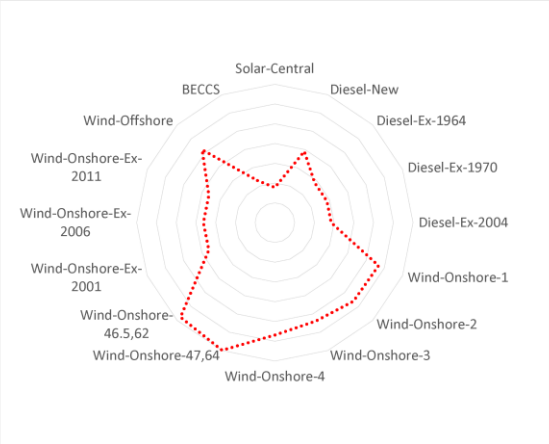
Fig. 14. **Scenario architecture used to evaluate spatial enhancement, clean energy policy, wind–cropland colocation in TEMOA-ACES, and land constraints. The original TEMOA-ACES configuration is compared with the spatially enhanced Base model; policy cases apply the Investment Tax Credit (ITC), Renewable Portfolio Standard (RPS), or their combined Mix scenario; and PEI land use cases compare conventional/spacing and colocation assumptions under land limits of 1, 5, 50, and 100 km² and a no-limit case. The colour-coded groups distinguish the principal scenario families evaluated in the thesis.**

ACES vs. Base Changes Under TEMOA Model Developments

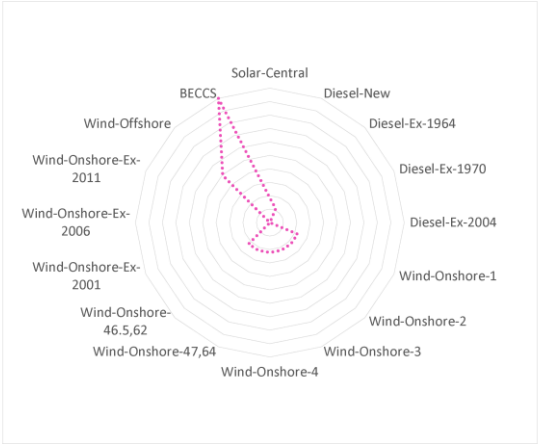
In this section we analyze the spatially enhanced TEMOA model against the previously developed TEMOA model for Atlantic Canada's electricity sector(ACES). These scenarios do not include any policies, land limits or colocation. The cropland layer is included in site selection for both scenarios and colocation of wind and other technologies with cropland is allowed. However, colocation intensities as in 'colocation scenarios' shown in the 'yellow-coded' scenarios is not considered. We will elaborate on this piece in the next sections.

As explained above, the only difference between the ACES and Base models is their onshore wind technologies. In the ACES model, four onshore wind technologies are defined based on the Pan-Canadian Wind Integration Study (PCWIS) [70] and are named "Wind-Onshore-1 to 4". However, as an example, the Base model of PEI includes the 17 potential sites following the mentioned procedure and only two of them are picked in our scenarios as optimal technologies. Fig. 15 shows these optimal onshore wind sites and the other optimal and existing technologies for the ACES and Base models. The other

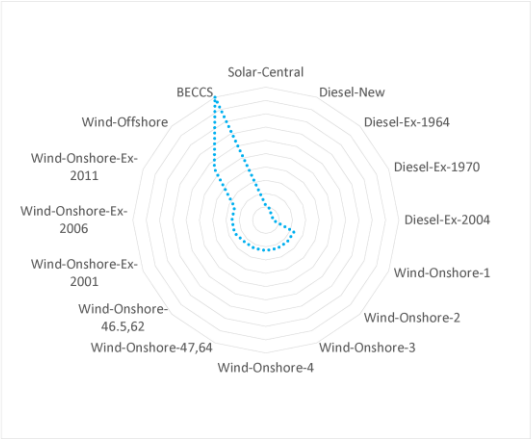
three provinces have more potential sites; however, due to visualization limits, they are not presented here.



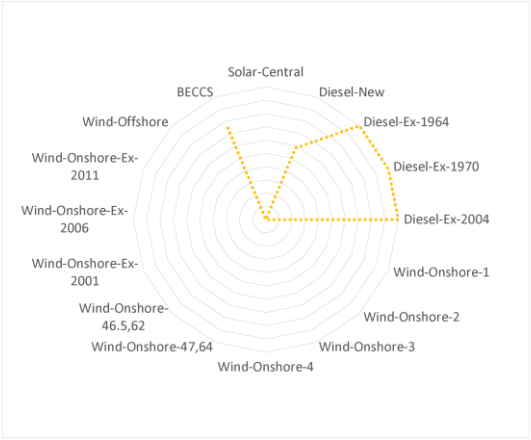
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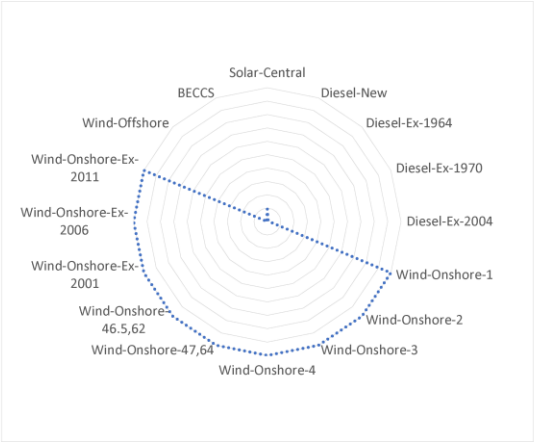
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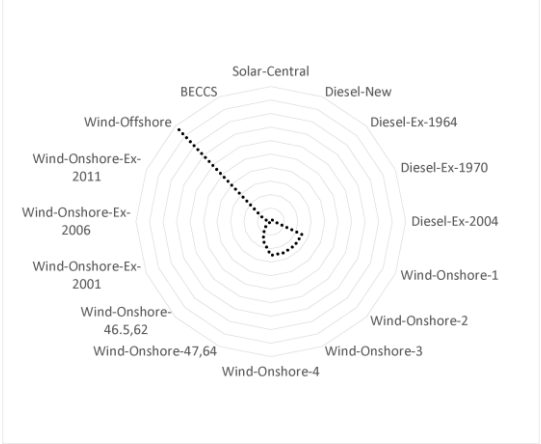
c



d



e



f

Fig. 15. Radar chart comparison of existing and candidate electricity generation technologies represented in the original ACES and spatially enhanced Base scenarios. Panels compare technologies according to (a) capacity factor or conversion efficiency, (b) investment cost(M$), (c) fixed operation and maintenance cost(M$), (d) variable cost(M$), (e) direct land-use intensity(KM^2/GW), and (f) maximum available capacity(GW). The comparison illustrates how the GIS-derived onshore wind technologies differ from the generalized ACES wind technologies in the technical, economic, land use, and resource potential parameters that enter the optimization.

The parameters compared in Fig. 15 have a significant impact on the objective function, which represents model costs. To find the relation between parameters from (1) and the objective function, we break down "Activity". Equation (5) is the inequality that connects Activity to Capacity and Capacity Factor. Fuel use, which is a component of fixed cost, is then connected to Activity and Efficiency(eta) by Equation (6). Emissions are eventually found by Fuel use in Equation (7), contributing to the calculation of emission costs in the objective function. However, according to the objective function in Equation (1), emission costs are minimized along with the total cost. It is also noteworthy that emission costs are equal to the carbon tax in the ACES model and are eliminated due to federal policies, so this part of the equation is absent in our calculations and Equations (5) to (7) are only presenting examples for clarification.

$$Activity <= Capacity * Capacity\ to\ Activity * Capacity\ Factor \quad (5)$$

$$Fuel\ Use = Activity/\eta \quad (6)$$

$$Emissions = Fuel\ Use * Emission\ Factor \quad (7)$$

Knowing the relations between parameters, we can now look back at Fig. 15.

Radar chart comparison of existing and candidate electricity generation technologies represented in the original ACES and spatially enhanced Base scenarios. Panels compare technologies according to (a) capacity factor or conversion efficiency, (b) investment cost(M$), (c) fixed operation and maintenance cost(M$), (d) variable cost(M$), (e) direct land-use intensity(KM^2/GW), and (f) maximum available capacity(GW). The comparison illustrates how the GIS-derived onshore wind technologies differ from the

generalized ACES wind technologies in the technical, economic, land use, and resource potential parameters that enter the optimization. to compare ACES and Base models' technologies. As mentioned before, the only difference is in the newly built onshore wind technologies and their capacity factors, investment costs, and maximum capacities. Base technologies have higher capacity factors compared to ACES, which is favorable, but have higher investment costs and lower maximum capacities, which makes them less favorable.

Capacity factor changes of spatial enhancement

The demand and all the supply-side data remain unchanged except for the onshore wind capacity factors. Based on Fig. 15(a), the original ACES model's onshore wind technologies have capacity factors ranging between 0.53 and 0.56. However, the spatially enhanced ACES model optimizes onshore wind technologies with capacity factors close to 0.7.

Atlantic Canada's Policy Scenarios

In this section, we study the impacts of two clean energy policies on our case study through our enhanced model. Investment Tax Credit (ITC) and Renewable Portfolio Standard (RPS) are two of the most significant policies in play in Atlantic Canada. Studying these policies with our enhanced model provided valuable insights into our case study from two angles. First, to see how our case study is impacted by these policies in terms of spatial and land considerations. Second, to analyze how enhancing our model contributes to the land use and spatial implications of these policies. The following paragraphs summarize these policies and their implementation details in our scenarios.

Investment Tax Credit Policy (ITC)

Investment Tax Credit [84] is an incentive that encourages clean energy investors by offering tax credits. This policy was designed and introduced by the government of Canada in 2023 to incorporate more clean energy production into the energy system. ITC operates by refunding a percentage of the investment costs of clean energy technologies. These percentages are different based on investment year and are listed per our modeling periods in Table 8. Technologies included in our modeling of the ITC policy are: Wind(onshore and offshore), Solar(central and residential), Battery, Nuclear(only SMR) and Geothermal.

Table 8. Investment Tax Credit percentages applied in each modeling period [84].

Period	2022-2025	2025-2030	2030-2035
Percent decrease	80%	70%	73%

Renewable Portfolio Standard

The Renewable Portfolio Standard [85] is another significant policy in Atlantic Canada that promotes renewable energy production. This policy operates at the provincial level and encourages utility providers to commit to a certain percentage of renewable energy in their electricity mix. Our model implements this policy in NS and NB, and by the percentages for each model period presented in Table 9. Technologies included in our modeling of the ITC policy are: Wind(onshore and offshore), Solar(central and residential), Biomass, and Geothermal.

Table 9. Renewable Portfolio Standard shares applied in each modeling scenario [85].

	2022	2025	2030	2035	2040	2045	2050
NS	0.4	0.4	0.8	0.8	0.8	0.8	0.8
NB	0.4	0.4	0.4	0.4	0.4	0.4	0.4

A 2×2 policy scenario matrix was developed to evaluate the individual and combined effects of the Investment Tax Credit (ITC) and Renewable Portfolio Standard (RPS) within the spatially enhanced TEMOA-ACES model. The four scenarios include a no-policy Base scenario, an ITC-only scenario, an RPS-only scenario, and a combined-policy Mix scenario. This factorial structure allows the analysis to distinguish between the marginal effect of each policy and the interaction effect that emerges when both policies are implemented simultaneously. The single-policy scenarios are used to isolate the effects of each instrument because the ITC and RPS operate through different mechanisms. The ITC primarily affects investment decisions by reducing the effective capital cost of eligible clean energy technologies, thereby influencing capacity expansion. In contrast, the RPS constrains the generation mix by requiring a specified share of electricity production to come from renewable sources, thereby affecting technology activity and dispatch. Although the two policies target overlapping sets of renewable technologies, they do not affect the optimization problem in the same way. Evaluating them both separately and jointly is therefore necessary to identify trade-offs, synergies,

and policy interactions. The Mix scenario is used to assess whether the policies reinforce one another, producing amplified effects in capacity, activity, cost, emissions, or land-use outcomes, or whether they generate countervailing effects, where the influence of one policy partially offsets or modifies the effect of the other.

Table 10. Scenario matrix for ITC and RPS scenarios.

RPS \ ITC	×	✓
×	Base	ITC
✓	RPS	Mix

PEI's Colocation Scenarios

In this section, we study the possibility, implications and impacts of colocating onshore wind and cropland by integrating land uses into energy modeling. As explained in Chapter 1, colocation in this context means building onshore wind power plants on agricultural land, while both operate simultaneously. As mentioned above, this approach contributes to lowering direct land use for onshore wind development and reducing risks to agriculture due to energy system development. However, the identified gaps summarized a lack of suitable modeling tools to be able to model and study energy colocation.

Our developed models (Section 2.1), features and workflow (Fig. 13) provide a tool to study energy colocations. The GIS model allows us to identify potential locations overlapping with the cropland layer. The energy optimization model's land tracking feature helps us to track land use, whether the cells overlap or not. Finally, our TEMOA's land constraint feature supports a methodology to study the energy system under land constraints. This is necessary since space becomes more limiting as we introduce more renewables to the energy system.

Existing approaches to onshore wind placement often allow colocation without modeling it as colocation, meaning that the cropland layer(and other land layers) is not taken into account. The literature review in Section 1.2 discussed some of these examples. We call this scenario 'conventional' in our scenario design. Some rare cases exclude the cropland layer from the site selection process and treat it as a hard constraint (cite); however, this is not a realistic approach. We study it to put the impact into perspective and name it the 'spacing' scenario. Lastly, thanks to our model development,

we are now able to model ‘colocation’ scenarios. These three scenario groups are listed in Table 11.

Table 11. Scenario matrix for the conventional, colocation and spacing scenarios.

Land limit (km^2)	No limit	100	50	5	1
Conventional	ACES-SE	ACES-SE-L100	ACES-SE-L50	ACES-SE-L5	ACES-SE-L1
Spacing	ACES-SE-SP	ACES-SE-SP-L100	ACES-SE-SP-L50	ACES-SE-SP-L5	ACES-SE-SP-L1
Colocation	ACES-SE-CL	ACES-SE-CL-L100	ACES-SE-CL-L50	ACES-SE-CL-L5	ACES-SE-CL-L1

Two major spatial data layers are involved in the process of identifying colocated sites using the GIS model: the wind speed layer and the cropland layer. First, we identify the cropland and wind overlaps, where the cropland layer is static and the wind layer is dynamic. This means that outputs from the cropland layer do not change, while the wind layer outputs are dependent on capacity factors and distance to the grid, impacting the energy modeling results. These identified cells show the locations where wind farms can be colocated with cropland. After identifying wind and cropland overlaps from GIS, we add these cells as technologies to the energy optimization model (ACES) to find the optimal cells for wind development.

Land use for these technologies varies based on their overlap with cropland. In the conventional scenario, overlaps were not identified; therefore, potential wind technologies were included in the energy model with their direct land intensity, including the spacing, which is equal to 368.3 km^2/GW, as listed in Section 2.1.1. However, in the colocation scenario, where technologies are grouped into colocated and not colocated, we use a direct land intensity excluding the spacing, which is 1 km^2/GW. This approach allows us to distinguish between the land occupied solely for wind generation and the land used for cropland while housing the wind power plant.

Table 12 lists the scenarios and their assumptions while grouping the scenarios into three categories as explained above. These three categories are color-coded in red, green and yellow similar to Fig. 14. We will elaborate on each scenario category in the following sections, namely: ‘ACES vs. Base’, ‘Atlantic Canada’s Policy’ and ‘PEI’s Colocation’ scenarios.

Table 12. Overall scenario table for this work

		Spatially Enhanced	ITC	RPS	Spacing included	Co location	Land limit	Region
TEMOA-ACES		-	-	-		-	-	Atlantic
Base	ACES-SE	✓	-	-	✓	-	-	Atlantic
ITC	ACES-SE-PITC	✓	✓			-	-	Atlantic
RPS	ACES-SE-PRPS	✓		✓		-	-	Atlantic
Mix	ACES-SE-PMix	✓	✓	✓		-	-	Atlantic
Base	ACES-SE-L100	✓	-		✓	-	100	PEI
	ACES-SE-L50	✓	-		✓	-	50	PEI
	ACES-SE-L5	✓	-		✓	-	5	PEI
	ACES-SE-L1	✓	-		✓	-	1	PEI
Spacing	ACES-SE-SP	✓	-		-	-	-	PEI
	ACES-SE-SP-I100	✓	-		-	-	100	PEI
	ACES-SE-SP-I50	✓	-		-	-	50	PEI
	ACES-SE-SP-I5	✓	-		-	-	5	PEI
	ACES-SE-SP1H	✓	-		-	-	1	PEI
Colocated	ACES-SE-CL	✓	-		✓	✓	-	PEI
	ACES-SE-CL-L100	✓	-		✓	✓	100	PEI
	ACES-SE-CL-L50	✓	-		✓	✓	50	PEI
	ACES-SE-CL-L5	✓	-		✓	✓	5	PEI
	ACES-SE-CL-L1	✓	-		✓	✓	1	PEI

Limiting land use in this approach provides a way to observe the model's results under land constraints. While the land tracking allows us to see the actual land use compared to the spacing, it does not automatically reflect the real-world situation. Applying land limits represents an energy system that has spatial constraints and the optimization model in these scenarios favors potential wind locations with lower land use, which are colocated with cropland. Therefore, each of the scenario groups listed in Table 12 is studied under an incremental range of land limits, from 100 to 1 km^2.

2.3. Model Validation

Model validation is necessary to ensure that the developed modeling framework is internally consistent, grounded in reliable input data, and capable of producing outputs that respond logically to changes in model assumptions. Validation in this study focuses on four complementary checks: first, confirming that the selected energy model structure is appropriate for the research questions; second, validating key spatial and land-use input data; third, comparing model outputs with existing results and external data sources where possible; and fourth, testing model behavior under edge cases to verify that the newly developed land-tracking and land-limiting features operate as expected.

2.3.1. Input Data Validation

The second validation step focused on the input data used to connect land-use modeling with energy system optimization. Since the central contribution of this thesis is the integration of spatial land constraints into ACES, the credibility of the land-use inputs is critical. Wind speed data are important as well, due to their role in our technology resolution improvement. Input validation was performed for three categories of data: land exclusion layers, wind resource data, and onshore wind land intensity values.

Land-cover validation using existing wind farms

The spatial exclusion layers were first checked against the locations of existing Canadian wind farms. Existing wind farm locations were overlaid with the land layers used in the GIS model, including protected and conserved areas, wetlands, water bodies, Indigenous

Reserve Lands and their buffer layers, cropland, built environment and slope. This comparison provides a practical test of whether the exclusion rules are broadly consistent with observed wind farm development.

The overlay results showed very limited overlap between existing wind farms and hard exclusion layers. No major overlap was found with wetlands or water bodies. A small number of overlaps were identified with protected and conserved areas (one in SK) and Indigenous Reserve Lands (2 in ON and 9 more in ON, AB and NS when the buffer was applied). However, these overlaps do not necessarily indicate an error in the GIS layers or model logic, and may rather reflect differences in spatial resolution, boundary definitions, permitting dates, or local project agreements. The results support the treatment of wetlands and water bodies as hard exclusions, while also supporting the treatment of Indigenous Reserve Lands and their buffers as soft or socially sensitive exclusions rather than universally unavailable land.

Slope was also evaluated against existing wind farm locations. The validation notes show that wind development is concentrated in lower and moderate slope classes, particularly classes 2, 3, 4, and 5. Only one existing wind farm was identified in slope class 7 (>30 degrees) in QC, and no wind farms were identified above class 7. This pattern supports the use of slope as an incremental constraint: steep slopes do not make development impossible in every case, but they reduce technical suitability and increase development difficulty and costs.

Wind resource validation

ERA5 was used as the wind-speed source because it is a well-established and widely used reanalysis dataset for climate, weather, and energy-system applications. ERA5 combines historical observations with numerical weather modelling through data assimilation, producing globally complete and physically consistent estimates of atmospheric variables; Copernicus describes climate reanalyses as among the most-used datasets in the geophysical sciences, and ERA5 specifically provides hourly estimates of wind speed and other variables from 1940 onwards [86]. This temporal resolution is appropriate for the present modelling framework because the TEMOA uses hourly historical wind-speed time series for the 2019 reference year [87]. ERA5 is also the standard input dataset in Atlite, the package used in this work for converting weather data into wind power and capacity-factor inputs; Atlite identifies ERA5 as its standard reanalysis

source and notes that ERA5 enables hourly time series for energy system modelling applications [35].

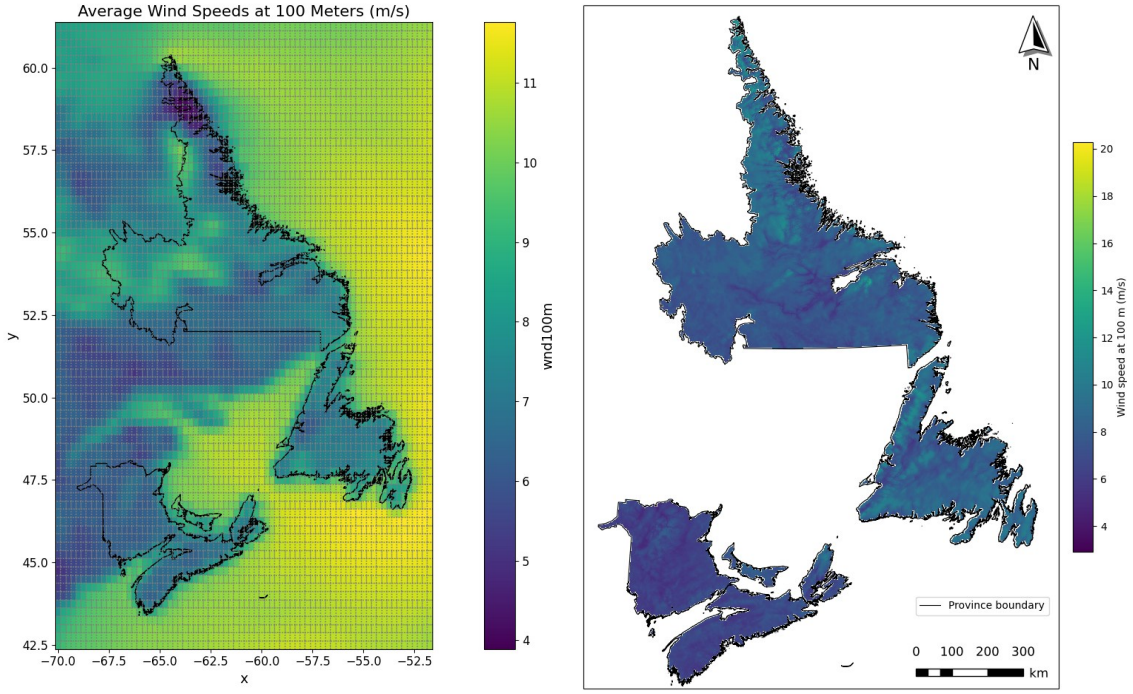


Fig. 16. Comparison of Atlantic Canada wind speed(m/s) patterns from two wind resource datasets used for spatial plausibility assessment. (a) ERA5-derived average wind speed at 100 m above ground level and approximately 30 × 30 km resolution [9]; (b) higher resolution wind speed distribution from the Global Wind Atlas [88]. The comparison is used to assess whether the large-scale spatial pattern of the ERA5 resource is consistent with the Global Wind Atlas.

The comparison with the Global Wind Atlas(Fig. 16) was therefore used as a spatial plausibility check for the ERA5-derived wind resource rather than as a fully independent observational validation. This distinction is important because the Global Wind Atlas is itself partly derived from ERA5: the Global Wind Atlas version 3 and version 4 datasets are produced by dynamically downscaling ERA5 reanalysis data to approximately 250 m resolution using the WRF mesoscale model, DTU's wind-atlas methodology and the WAsP model. Agreement between the ERA5 map and the Global Wind Atlas therefore supports the reasonableness of the ERA5 spatial wind-speed pattern used in this study, especially at the regional screening scale, but it should not be interpreted as validation against direct site measurements. A stronger independent validation would require a meteorological mast, lidar/sodar, weather station, or operational

wind-farm production data; nevertheless, for regional energy-system planning and spatial screening, ERA5 is a defensible source because it is transparent, globally available, temporally complete, compatible with TEMOA's temporal structure, and consistent with the large-scale wind-resource information used in the widely applied Global Wind Atlas.

Land intensity validation

Land intensity inputs were validated through comparison with literature values and project-level estimates for existing wind farms. As discussed earlier, land intensity values for onshore wind vary substantially across the literature because studies differ in whether they include only the physical footprint of turbines and infrastructure or also include spacing between turbines. In this thesis, spacing-inclusive land intensity is used when wind development occupies the full wind farm area, while spacing-exclusive land intensity is used for colocation scenarios where agricultural production and wind generation are assumed to occur on the same land base.

To test the plausibility of the selected land intensity values, several Canadian wind farm examples were evaluated. The CODERS dataset [10] reports permitted onshore wind areas by grid cell. Using the onshore wind area associated with Prince Edward Island and the province's total onshore wind capacity, the implied land intensity is calculated as:

$$\text{Average land intensity for PEI onshore wind} = \frac{4545\ km^2}{0.2336\ GW} = 19{,}456.34\ km^2/GW$$

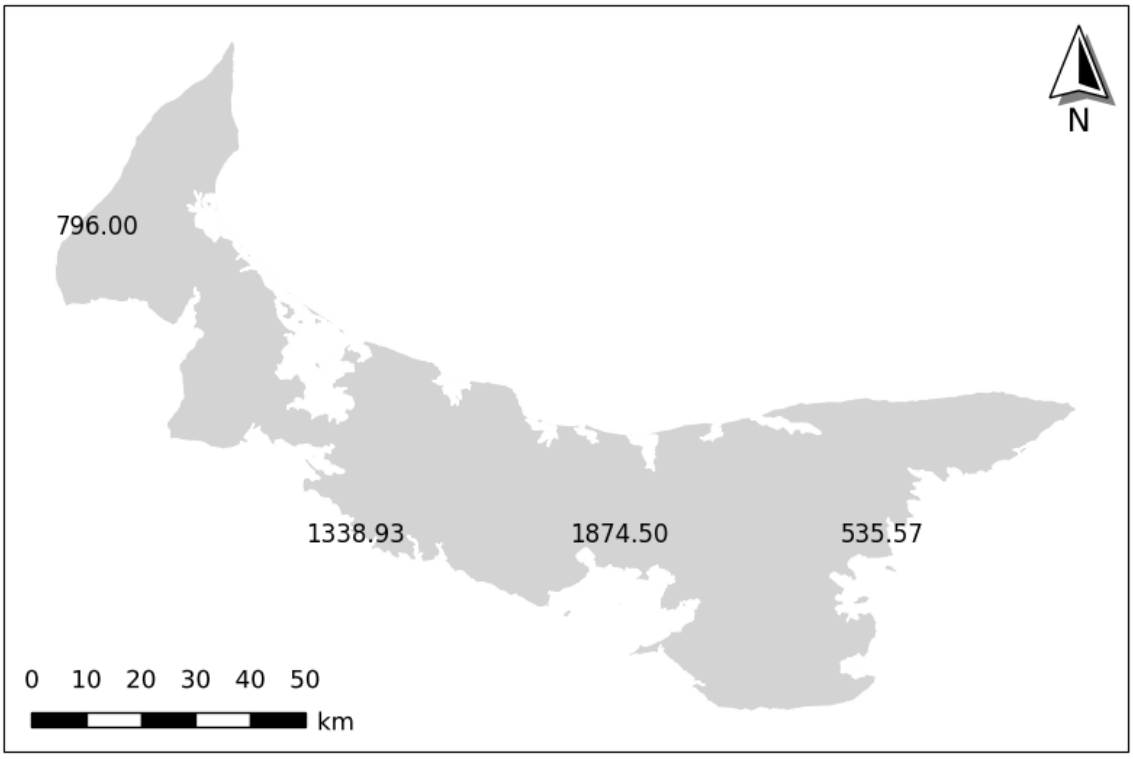


Fig. 17. Spatial distribution of permitted or potentially available onshore wind area in Prince Edward Island from the CODERS dataset [10]. Numeric labels report the wind development area associated with individual spatial cells in km² yielding a province-wide total of approximately 4,545 km² used as an upper-bound land intensity validation case rather than an estimate of physical turbine or windfarm occupation[10]

This estimate is extremely high and should not be interpreted as the actual physical land occupation of wind infrastructure. Rather, it represents a broad permitted or available area and is therefore useful as an upper-bound edge case.

A second validation case was performed for the West Cape wind farm in Prince Edward Island using OpenStreetMap as shown in Fig. 18. Based on the mapped wind farm boundary, the estimated area is approximately 12.9 km^2, and the installed capacity is 0.099 GW. The resulting land intensity is:

$$\frac{12.9\ km^2}{0.099\ GW} = 130.30\ \ km^2/GW$$

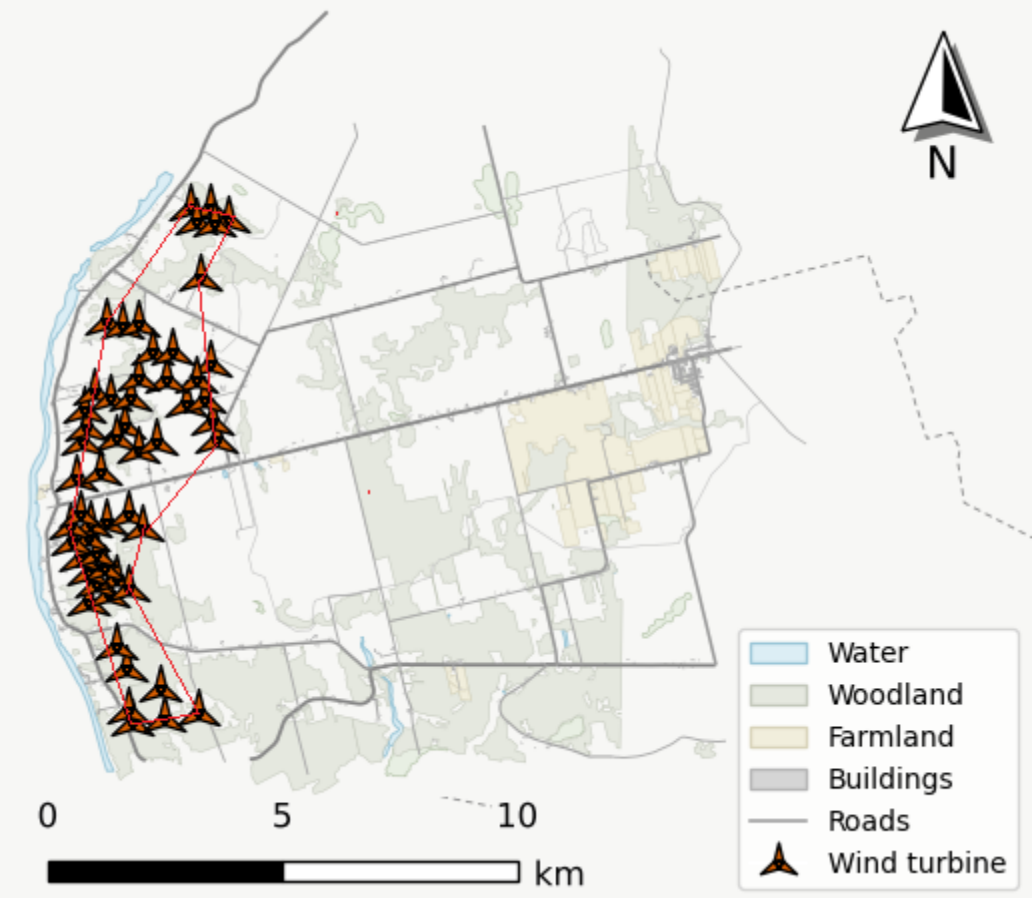


Fig. 18. **Mapped boundary used to estimate the land area of the West Cape onshore wind farm in Prince Edward Island. The approximate wind farm area delineated from OpenStreetMap is 12.9 km²; combined with the installed capacity of 0.099 GW, this produces a project-level, spacing-inclusive land intensity estimate of approximately 130.3 km²/GW for comparison with the literature-based wind land intensity assumptions used in the model [88] licensed under ODbL.**

This estimate was then cross-checked against public descriptions of the West Cape wind farm area [89], which report an area of approximately 25 km^2. Using the same installed capacity, this produces a land intensity of approximately:

$$\frac{25\ km^2}{0.099\ GW} = 252.53\ \ km^2/GW$$

A third validation case was performed for the Summerside wind farm in Prince Edward Island. This project consists of three turbines with a total installed capacity of 12 MW or 0.012 GW, calculated from OpenStreetMap data. The area of the triangle enclosing the three turbines(Fig. 19) was estimated as 0.19 km^2, producing a land intensity of:

$$\frac{0.19\ km^2}{0.012\ GW} = 15.83\ km^2/GW$$

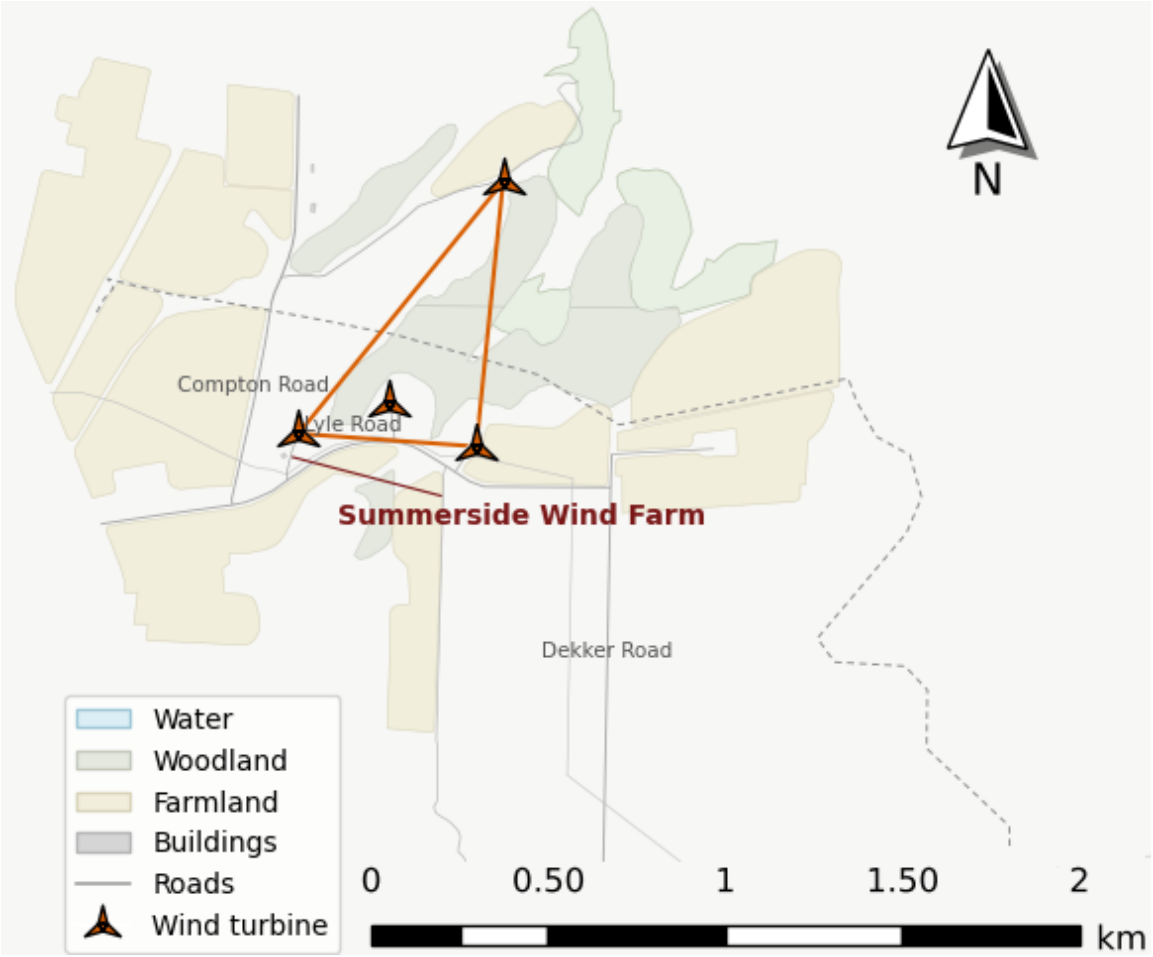


Fig. 19. **Spatial extent used to estimate land occupation for the Summerside onshore wind farm in Prince Edward Island. The mapped triangular area enclosing the three turbines is approximately 0.19 km² and the total installed capacity is approximately 0.012 GW, corresponding to an estimated project-level land intensity of 15.83 km²/GW. This value is used as a lower spacing-inclusive project estimate in the land intensity edge case validation [88] licensed under ODbL.**

These checks confirm that the selected land intensity assumptions cover the expected range of definitions used in the literature and in project-level interpretations. It also supports the use of edge-case testing in Section 1.3.4, where the model is tested under a range of land intensity assumptions from 1 km2/GW to 20,000 km2/GW.

2.3.2. Validation through Result Comparison

The third validation step compared selected model outputs with external references and previous model applications. Since this thesis modifies ACES rather than developing an entirely new capacity expansion model, a key validation requirement was to ensure that the modified model remains consistent with the original ACES framework. Through personal communication with the ACES developer [90], the base ACES structure and results were confirmed to be consistent with the model's previous applications and the outputs used in Atlantic Canada energy system analysis. The results were also consistent with the provincial and regional reports published by the Atlantic Canadian government organizations. The ACES model has also been used alongside the Canada-wide TEMOA model in the multi-model comparison initiative run by the Energy Modeling Hub [91], a nationwide organization known for their contribution to accurate and transparent energy modeling. This comparison proved the ACES model results to be in accordance with other Canadian energy system models widely used for energy planning. This provides qualitative validation that the original model is functioning as intended before the spatial and land-use extensions are added.

The spatially enhanced Base model was then compared with the original ACES representation. This comparison validates whether changes in results are explainable by changes in technology representation. The Base model includes onshore wind technologies with higher resolution technology presentation as explained in Chapter 2, therefore presents different capacity factors, higher investment costs, due to grid connection costs, and lower maximum capacities because they are constrained by site-specific land availability. These differences enter the optimization through investment cost, activity, capacity factors, fuel use, emissions, and total system cost. Therefore, changes between ACES and the Base model can be traced to the spatially explicit technology definitions rather than unexplained model behavior. These changes will be summarized in detail in Chapter 3.

2.3.3. Validating Outputs through Edge Cases

The final validation step tests whether the model behaves logically under extreme input assumptions. This is particularly important because the main methodological contribution of this thesis is the integration of land intensity and land constraints into the

energy optimization model. The edge-case validation, therefore, tests whether the model responds as expected when the land intensity of onshore wind is varied across a wide range.

As shown in Table 13, the lowest project-level land intensity estimate with spacing is 15.83 km^2/GW, based on the Summerside wind farm estimate. The literature values used in this thesis include higher spacing-inclusive values, including 368.3 km^2/GW and 439.7 km^2/GW. The CODERS-based estimate for Prince Edward Island is much higher, at 19,456.34 km^2/GW, and is interpreted as a broad permitted-area estimate rather than a physical land occupation estimate. For the upper edge case, this value is rounded to 20,000 km^2/GW.

Table 13. Different land intensity values used in edge case validation

Land intensity (km^2/GW)	Basis	Role in validation
1	Spacing-exclusive footprint	Lowest footprint-only/colocation edge case.
6.5	Spacing-exclusive footprint	Literature-based footprint-only value for wind without full spacing.
15.83	Project-level spacing-inclusive estimate	Lower spacing-inclusive estimate from the Summerside validation case.
368.3	Literature spacing-inclusive estimate	Primary spacing-inclusive onshore wind land intensity used in the base land accounting.
20,000	Rounded permitted-area estimate	Upper edge case based on the CODERS-derived PEI estimate.

For spacing-exclusive wind development, where only the physical footprint is considered, the literature values used in this thesis include 1 km^2/GW and 6.5 km^2/GW. These values are used to represent cases where wind generation is colocated with agriculture and turbine spacing does not remove the full surrounding land area from agricultural use. Table 13 summarizes the land intensity values used in the edge-case validation.

Five scenarios were then run using different onshore wind land intensity values: 1, 6.5, 15.83, 368.3, and 20,000 km^2/GW. The purpose of these scenarios is not to define the most realistic land intensity value, but to test whether the model responds correctly as

onshore wind becomes more land-intensive. Fig. 20 shows accumulated activity by 2050 for each scenario.

The results show that onshore wind activity declines as the land intensity of onshore wind increases. This trend is consistent with the expected behavior of the land-constrained optimization model. When the onshore wind land intensity is low, at 1 to 6.5 km^2/GW, the model strongly favors new onshore wind. In these scenarios, new onshore wind production reaches approximately 46.33 PJ, and net imports are negative, meaning that exports to New Brunswick exceed imports.

When the onshore wind land intensity increases to 15.83 km^2/GW, new onshore wind production decreases to approximately 28.82 PJ. Although onshore wind remains an important part of the activity mix, the province becomes more dependent on imports. Net imports increase to approximately 13.22 PJ, compared with approximately −2.77 PJ in the lower land intensity scenarios. This indicates that the model substitutes away from onshore wind when its land requirement increases and compensates through imports and other generation options.

When the onshore wind land intensity increases further to 368.3 km^2/GW, new onshore wind activity falls to approximately 1.2 PJ. At this level, onshore wind contributes only a small share of the activity mix. Net imports increase to approximately 25.36 PJ, and the model also adds offshore wind despite its higher cost. This behavior is consistent with expectations because offshore wind has no direct land-use intensity in the model and therefore becomes more attractive under severe land constraints, even when its investment cost is higher.

In the upper edge case, where the onshore wind land intensity is set to 20,000 km^2/GW, new onshore wind activity falls to zero. Net imports and offshore wind remain major contributors to the activity mix, and a small amount of new utility-scale solar is added. The model also adjusts storage and curtailment to satisfy demand under the restrictive land-use condition. This result confirms that when onshore wind becomes effectively unavailable from a land-use perspective, the optimization model shifts to alternative supply and balancing options.

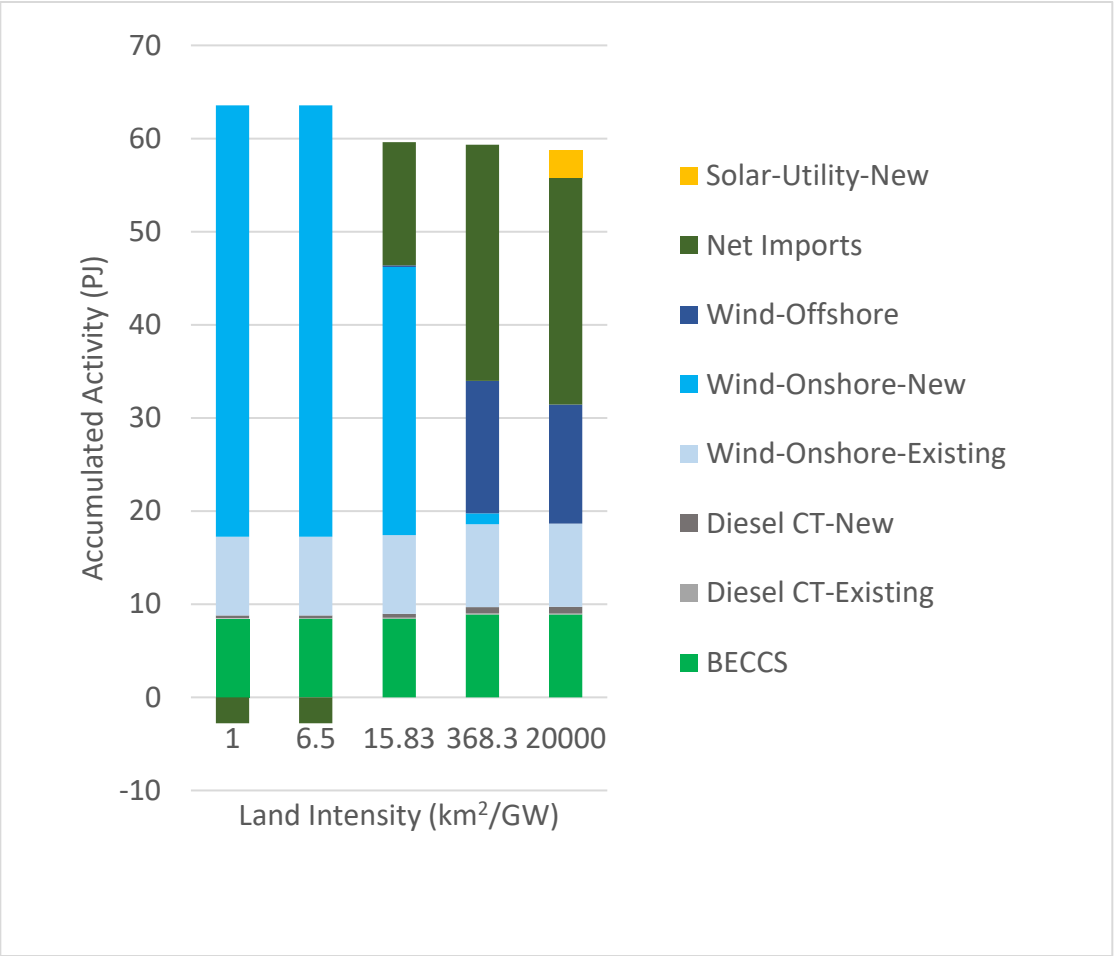


Fig. 20. **Accumulated Prince Edward Island electricity system activity by 2050 under alternative onshore wind land intensity assumptions used for edge case validation. Onshore wind land intensity is varied from 1 and 6.5 km²/GW for spacing-exclusive cases to 15.83 and 368.3 km²/GW for spacing-inclusive cases and 20,000 km²/GW as an upper permitted-area edge case. Stacked bars show the resulting activity contributions from existing and new onshore wind, offshore wind, net imports, utility-scale solar, diesel combustion turbines, and BECCS. The figure tests whether increasing wind land intensity causes the optimization to substitute away from land-intensive onshore wind as intended.**

These edge-case results validate the implemented land-use features. The land-tracking function correctly calculates land occupation based on the assigned land intensity values, and the land constraint correctly changes the feasible and optimal technology mix when land-intensive technologies become less attractive. The decline in onshore wind activity as land intensity increases confirms that the model responds to the land parameter in the intended direction. Therefore, the edge-case validation supports both the technical

implementation of the land-use constraint and the interpretation of the scenario results presented in this thesis.

Chapter 3. Results

In this chapter, we first present the results from wind potential and land cover synergies and trade-offs in Canada. We then show the results of our spatially enhanced energy model for Atlantic Canada. Afterwards, we summarize policy scenarios for Prince Edward Island in the Atlantic context and following this, examine the impacts of land constraints and colocation of onshore wind and cropland in the province.

3.1. Canada's Onshore Wind Deployment under Competing Land Uses

In this section, we present an analysis of the most impactful land layers for Canada's onshore wind development. First, we analyze the layers with relatively low shares, followed by high share layers that overlap with low wind speed locations. Lastly, we discuss the cropland layer, which covers a relatively high share and overlaps with high wind locations for three provinces. The land-cover layers considered in this analysis differ in whether they act as hard or soft constraints on onshore wind development. The built environment is treated as a hard constraint because wind development is not permitted on these lands within the model. This exclusion has the greatest effect in Prince Edward Island, where the cells affected by the built-environment layer have an average capacity factor of 0.53, compared with the provincial average of 0.42 across all cells. This indicates that the built environment overlaps with relatively high-quality wind locations in PEI and therefore removes some attractive sites from the development pool. However, because the built environment is defined as a hard exclusion, these areas remain unavailable for onshore wind development regardless of their wind potential.

In contrast to the built-environment layer, Protected and Conserved Areas (PCA) and Indigenous Reserve Lands (IRL) have limited effects on onshore wind site selection at the provincial scale. PCA has its highest land-cover share in Nova Scotia; however, the cells affected by this layer have an average capacity factor of 0.42, which is lower than the provincial average of 0.51 across all cells. This suggests that most cells excluded by the PCA layer do not correspond to the highest-quality wind locations in Nova Scotia. Therefore, although PCA remains an important exclusion layer from an environmental and

governmental perspective, its removal does not substantially constrain the province's aggregate onshore wind potential.

Buffered Indigenous Reserve Lands (IRL) have their largest spatial footprint in Alberta, where they cover 2.61% of the province's land area. However, the cells affected by excluding this layer have average capacity factors that are comparable to the provincial average. This indicates that buffered IRL areas do not disproportionately overlap with Alberta's highest-quality wind locations. As a result, excluding these areas from site selection does not substantially reduce aggregate onshore wind development potential, although the layer remains important from a social and local siting perspective.

3.1.1. Wetlands and water bodies

High shares of wetlands or water bodies do not necessarily translate into major constraints on onshore wind development; their effect depends on how these layers overlap with provincial wind potential. Several Canadian provinces contain substantial wetland areas, water bodies, or both, but the implications for wind site selection vary by province. This subsection first examines the effect of wetlands in Ontario and Manitoba, then evaluates the effect of water bodies in Manitoba and Quebec. Together, these cases show whether large land-cover shares remove high-quality wind locations or primarily affect cells with lower capacity factors.

In Ontario and Manitoba, wetlands affect a large share of spatial cells, but these cells are generally associated with lower-than-average wind potential. In Ontario, wetland-affected cells represent 86% of the analyzed cells, but their average capacity factor is 0.27, compared with 0.33 when wetlands are included in the candidate cell pool. A similar pattern is observed in Manitoba, where wetlands affect 72% of cells and the average capacity factor of excluded wetland cells is 0.29, compared with 0.32 for the included case. These results indicate that wetlands reduce the amount of land available for onshore wind development, but they do not substantially remove the highest-quality wind locations in either province.

Water bodies show a similar pattern in Quebec and Manitoba: they affect a large share of cells, but the affected cells generally have lower wind potential than the provincial average. In Quebec, water bodies intersect 99% of the analyzed cells, yet the average

capacity factor of the excluded water-body-affected cells is 0.36, compared with 0.54 when these cells are included in the candidate pool. In Manitoba, water bodies affect 72% of cells, and the average capacity factor of excluded cells is 0.29, compared with 0.32 in the included case. These results suggest that although water bodies reduce the amount of technically available land, their exclusion does not remove the strongest wind-resource areas in either province.

Overall, the wetland and water-body results show that high land-cover shares do not necessarily translate into strong wind-development constraints. Ontario and Manitoba were selected for wetlands, and Quebec and Manitoba for water bodies, because these provinces have the highest coverage of those layers. However, the affected cells generally have below-average capacity factors, meaning that their exclusion reduces gross land availability but does not substantially remove the strongest wind-resource areas. Cropland is examined next because it presents a different pattern: in some provinces, high cropland coverage overlaps with higher-quality wind locations.

Cropland differs from the wetland and water-body layers because, in the provinces where cropland coverage is most relevant, it overlaps with relatively strong wind-resource locations. Here we compare the average capacity factors when cropland is included in the candidate cell pool with the average capacity factors of cells affected by cropland exclusion. In Alberta, the average capacity factor increases from 0.166 for the included case to 0.22 for the cropland-excluded cells. This indicates that the cells affected by cropland exclusion have higher wind potential than the provincial average, so removing cropland from the candidate pool eliminates some of Alberta's more attractive onshore wind locations.

Saskatchewan follows the same pattern. The average capacity factor is 0.27 when cropland is included, while the cells affected by cropland exclusion have an average capacity factor of 0.31. These values are higher than those observed in Alberta, indicating that Saskatchewan's cropland-overlapping wind cells are not only above the provincial average, but also represent stronger wind resources in absolute terms. Therefore, treating cropland as unavailable land in Saskatchewan would remove relatively high-quality wind sites from consideration. Prince Edward Island also follows this pattern, but with substantially higher capacity factors. The average capacity factor is 0.49 when cropland is included, while the cells affected by cropland exclusion have an average capacity factor

of 0.52. This shows that PEI's cropland-overlapping cells coincide with very strong onshore wind resources.

Because higher-capacity-factor sites require less installed capacity to generate the same amount of electricity, developing wind in high-wind locations such as PEI can reduce the land required per unit of electricity generation relative to lower-quality sites. Therefore, excluding cropland in PEI may protect agricultural land in a narrow siting sense, but it can also remove some of the most land-efficient wind development opportunities.

Overall, the cropland results contrast with the wetland and water-body findings. Wetlands and water bodies have high coverage in some provinces but mainly affect lower-capacity-factor cells, while cropland in Alberta, Saskatchewan, and PEI overlaps with above-average wind resources. This makes cropland a more consequential layer for wind-energy planning and supports treating it separately from hard exclusion layers in the spatial analysis.

3.2. Spatially enhanced TEMOA-ACES model

This section presents the results from the spatially enhanced TEMOA-ACES model. It first compares the original ACES model with the spatially enhanced Base model to assess how GIS-derived onshore wind technologies affect costs, capacity factors, capacity expansion, and land-use estimates. It then applies the enhanced model to PEI's future electricity and land-use profile, clean-energy policy scenarios, and wind–cropland colocation under land limits. Together, these analyses show how spatial wind representation, policy design, and land constraints shape energy-system outcomes.

3.2.1. Impact of spatial enhancement on TEMOA-ACES outputs

This section evaluates how the spatial enhancement of onshore wind representation affects TEMOA-ACES outputs. The comparison is made between the original ACES model and the spatially enhanced Base model, where generalized onshore wind technologies are replaced with GIS-derived wind cells characterized by capacity factor, available land, maximum capacity, and grid-connection cost. This comparison isolates the effect of spatially enhancing the onshore wind technologies before introducing policy scenarios, land limits, or colocation assumptions.

Cost representation in TEMOA-ACES

Including spatially enhanced onshore wind data changes the cost representation of the model. Compared with the original ACES model, the spatially enhanced Base model reduces total system costs by 5.9%. This cost reduction occurs even though the enhanced model includes additional transmission-related costs, which account for 2%. After accounting for these added costs, generation capacity costs decline by 6.1%. This result indicates that the original ACES representation overestimates the cost of meeting demand because it does not fully capture the availability of higher-quality, spatially specific onshore wind sites.

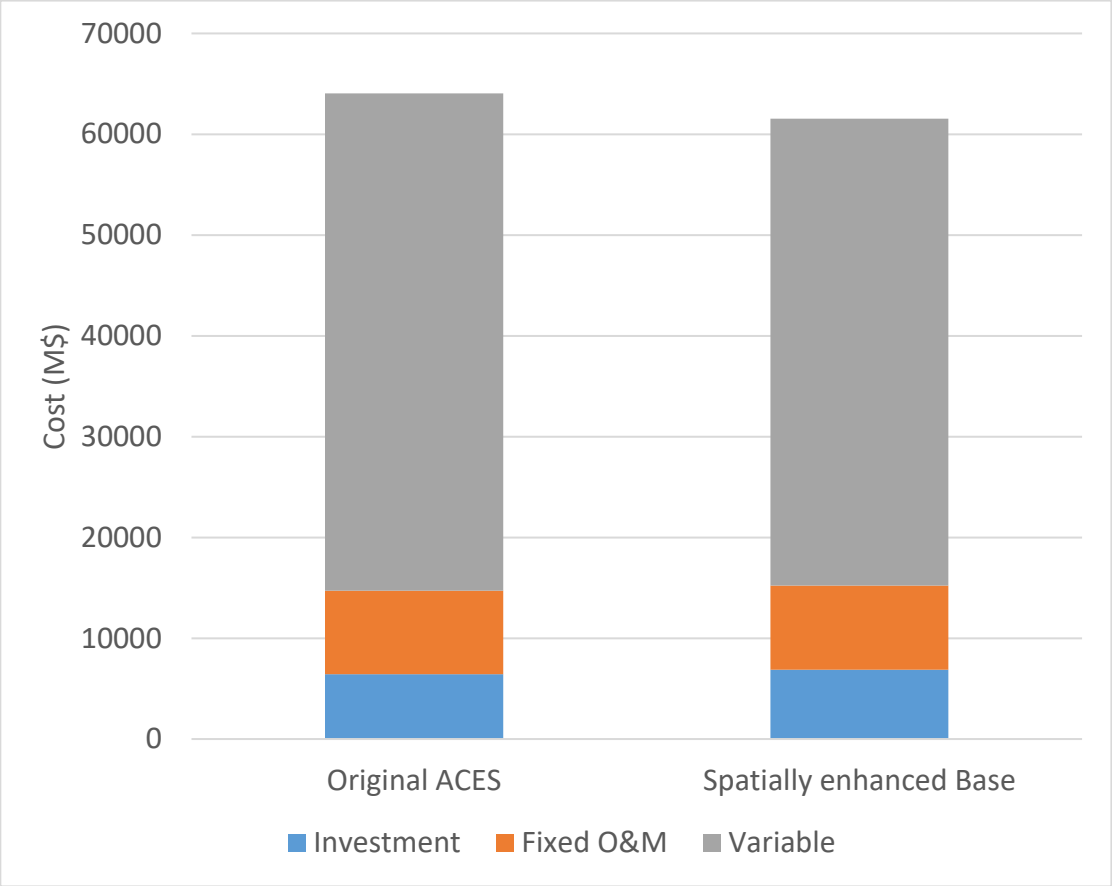


Fig. 21. **Comparison of total energy system cost components between the original ACES model and the spatially enhanced Base model. Stacked bars report investment, fixed operation and maintenance (O&M), and variable costs in million dollars (M$). The comparison isolates the cost effect of replacing generalized ACES onshore wind technologies with GIS-derived spatially differentiated wind sites, including site-specific wind resource and grid connection characteristics.**

Wind potentials and capacity in the spatially enhanced model

The spatial enhancement also reduces the total installed capacity required to meet the same electricity demand. The average capacity factor of newly built onshore wind increases from 0.47 in the original ACES model to 0.60 in the spatially enhanced Base model. Because higher-capacity-factor technologies generate more electricity per unit of installed capacity, the model can satisfy demand with 1.4 GW less total installed capacity. This shows that improving the spatial resolution of onshore wind representation changes not only the cost of the system, but also the amount of capacity required to deliver the same level of electricity service.

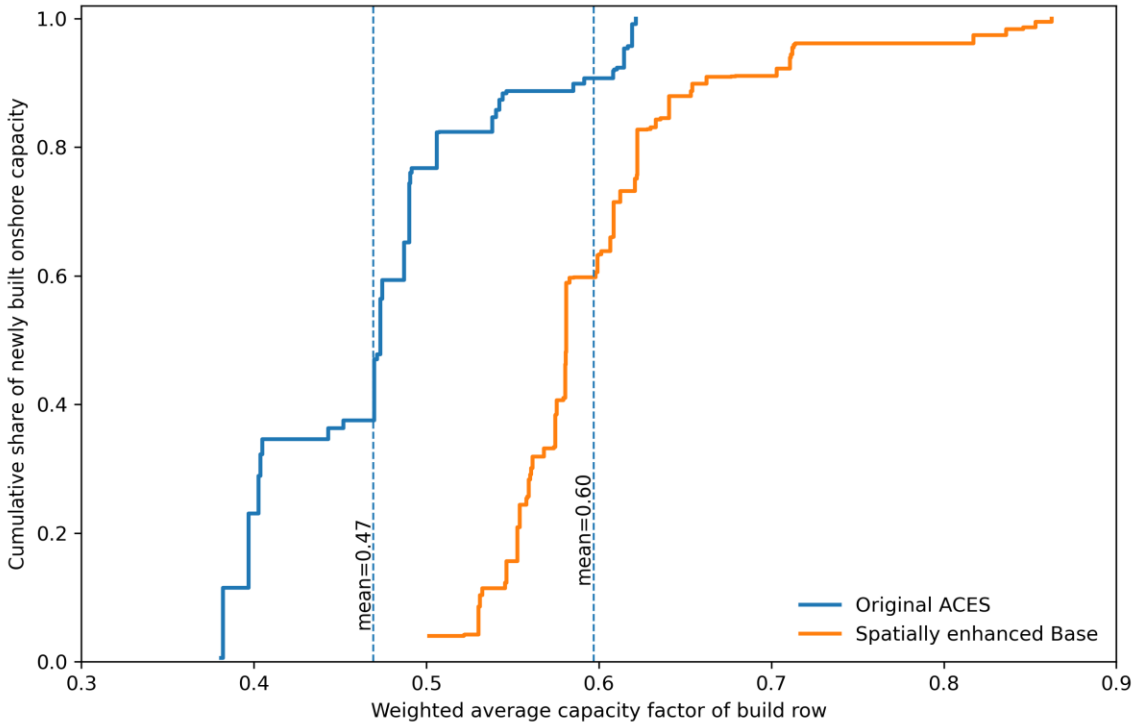


Fig. 22. **Cumulative distribution of newly built onshore wind capacity across the capacity factors of technologies selected in the original ACES and spatially enhanced Base models. The horizontal axis represents the capacity-weighted average capacity factor of newly built wind technologies, while the vertical axis represents the cumulative share of newly installed wind capacity. Dashed reference lines indicate mean capacity factors of approximately 0.47 for the original ACES representation and 0.60 for the spatially enhanced Base model, illustrating the shift toward higher quality wind resources after spatial disaggregation.**

At the same time, the reduction in total capacity does not mean that onshore wind becomes less important. On the contrary, higher-capacity-factor wind cells make onshore wind more favorable within the optimization model. As a result, the model shifts toward greater use of onshore wind, even while the overall capacity requirement decreases. This distinction is important because spatial enhancement improves the quality of available wind options: the model builds less capacity overall, but relies more strongly on high-performing onshore wind sites.

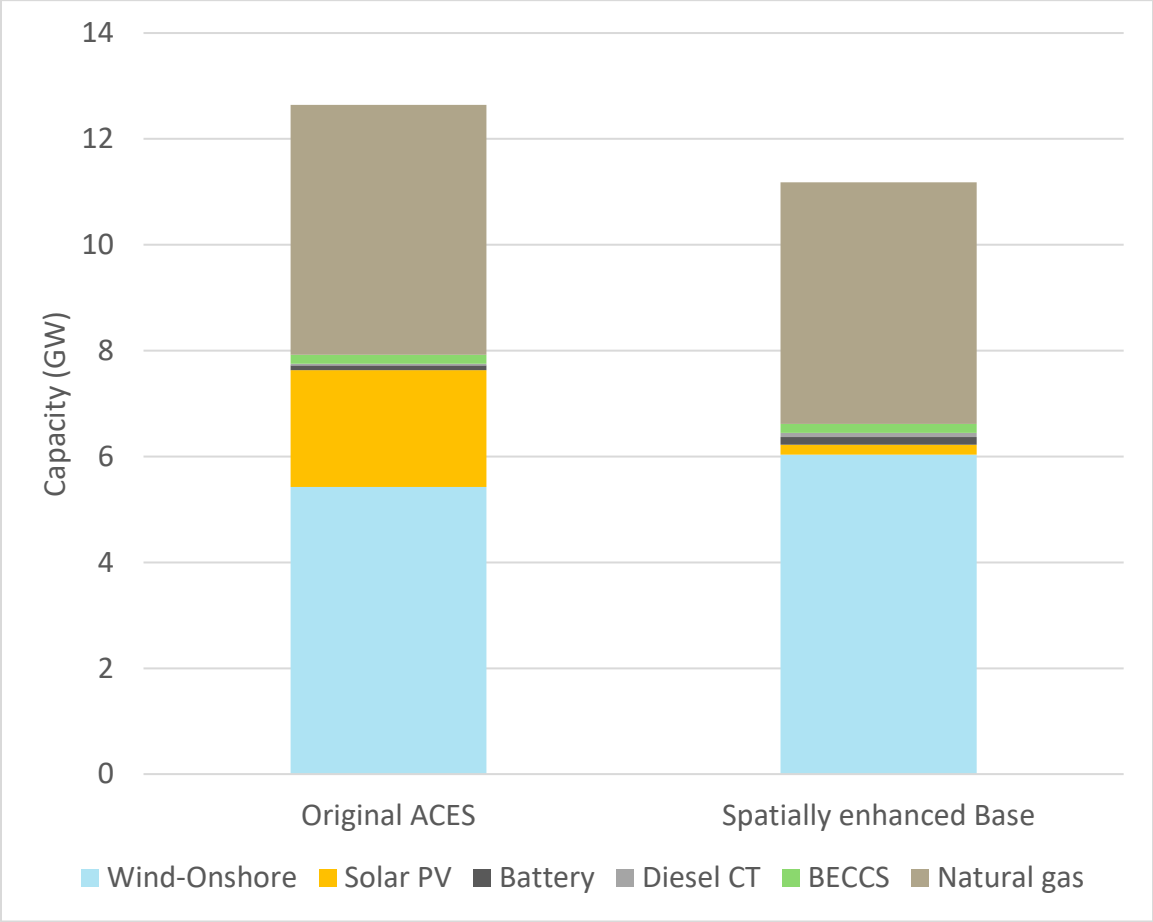


Fig. 23. **Atlantic Canada installed electricity generation capacity by technology in 2050 for the original ACES and spatially enhanced Base scenarios. Stacked bars report total capacity in GW for onshore wind, solar PV, battery storage, diesel combustion turbines, BECCS, and natural gas. The comparison shows how spatially differentiated wind resource representation changes the optimized technology portfolio while electricity demand and non-wind technology assumptions are held constant between the two scenarios.**

Land use changes due to spatial improvements

The land-use results show that spatial enhancement alone does not necessarily reduce land occupation. Although higher-capacity-factor sites can reduce the capacity needed per unit of electricity generation, the model's increased reliance on onshore wind leads to higher total land use because onshore wind is the dominant contributor to land occupation in the model. This result shows that spatial enhancement improves the representation of wind potential, costs, and capacity factors, but it is not equivalent to modeling a land-constrained system. To represent land scarcity directly, land use must be tracked and constrained within the optimization.

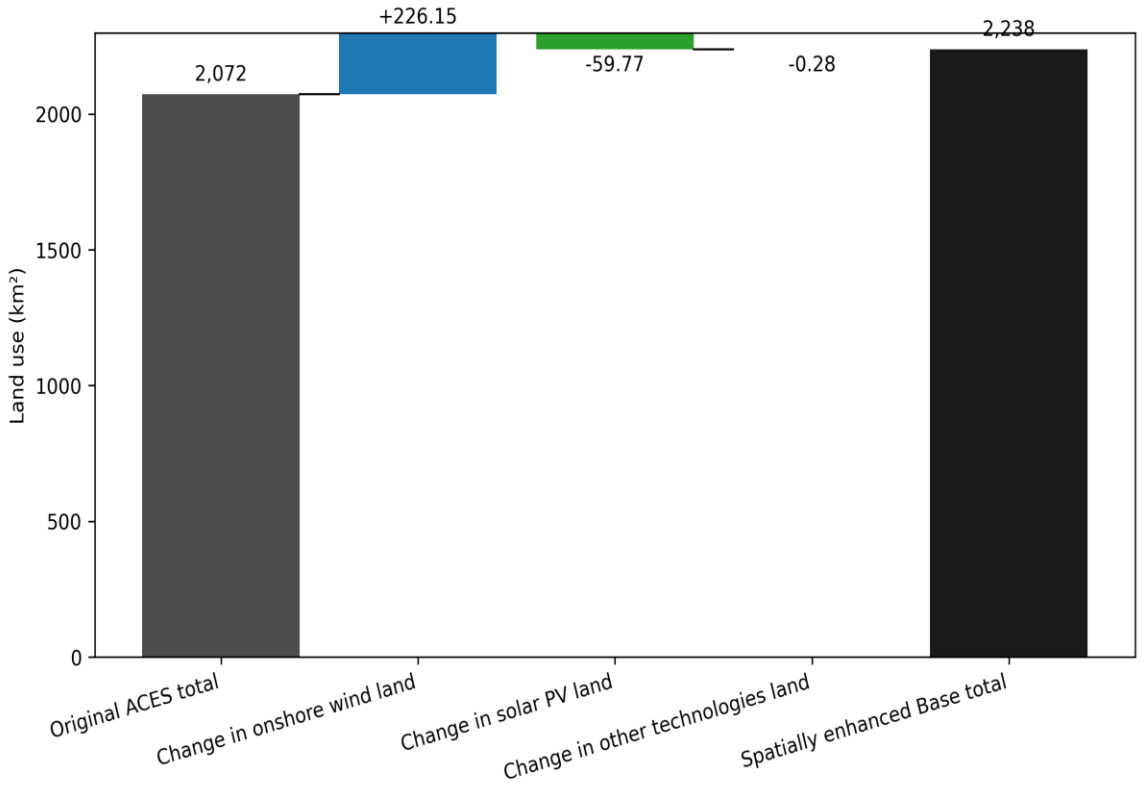


Fig. 24. **Change in Atlantic Canada electricity system land use in 2050 resulting from spatial enhancement of onshore wind representation. The waterfall chart begins with total land use in the original ACES model, identifies the contributions from changes in onshore wind land use, solar PV land use, and other technologies, and ends with the total land use of the spatially enhanced Base model. Land use is reported in km² and increases from approximately 2,072 km² to 2,238 km² despite improved wind resource quality, primarily because the enhanced model selects more onshore wind generation.**

Overall, the ACES–Base comparison shows that spatially enhancing onshore wind representation changes the model in three important ways: it reduces system costs, lowers the total capacity required to meet demand, and increases the attractiveness of high-quality onshore wind sites. However, because the model responds by relying more heavily on onshore wind, total land use can increase unless land limits are explicitly included. This finding motivates the later land-limit and colocation scenarios, where land availability becomes an active constraint.

3.2.2. Future Land Use Profile of PEI's Energy System

PEI's future electricity system requires both cleaner electricity and greater total generation capacity. In this model, demand growth is incorporated exogenously, meaning that the model does not determine future demand but instead optimizes the energy system required to meet the projected demand. Fig. 25 shows the Base scenario capacity mix for PEI through 2050. Total installed capacity increases from 0.487 GW in 2022 to 1.080 GW in 2050, indicating that the province's electricity system must expand substantially over the modelled period.

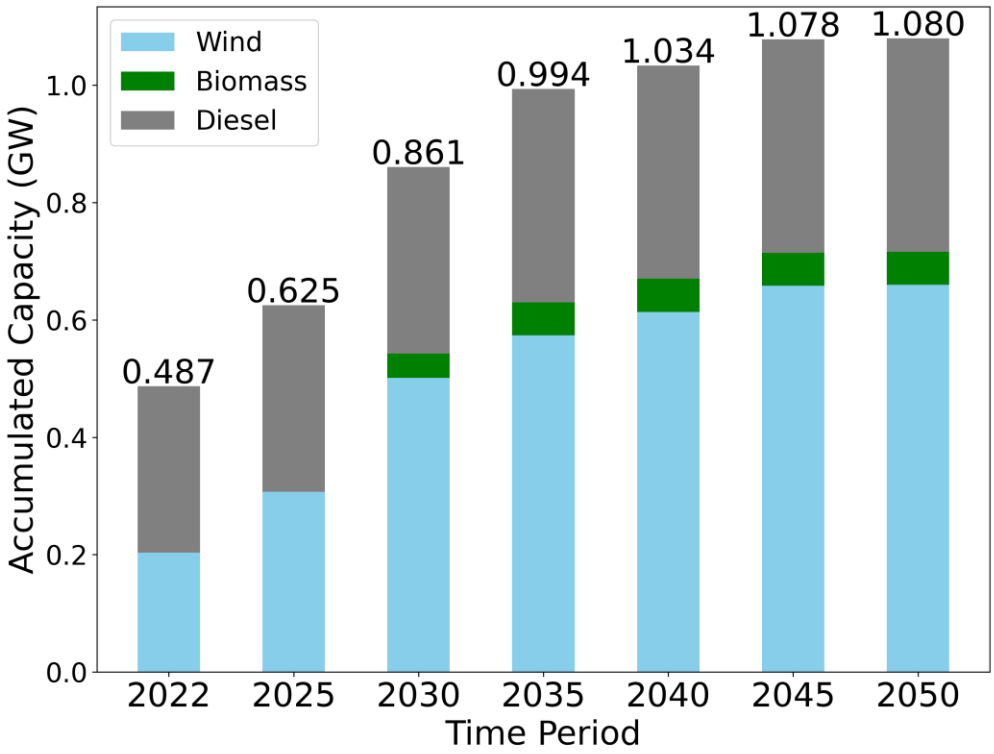


Fig. 25. **Evolution of Prince Edward Island installed electricity generation capacity from 2022 to 2050 in the spatially enhanced Base scenario. Stacked bars report accumulated capacity in GW for onshore wind, diesel generation, and biomass technologies. Total installed capacity increases from approximately 0.487 GW in 2022 to 1.080 GW in 2050, with onshore wind providing the majority of new capacity.**

Onshore wind is the main source of capacity growth in the Base scenario. As shown in Fig. 25, wind capacity increases strongly by 2050, while diesel capacity remains in the system and biomass-based capacity is added after 2030. This result shows that the future electricity system is not simply replacing fossil capacity with one renewable technology; rather, it expands through a combination of additional wind, retained diesel capacity, and new biomass-based generation. However, onshore wind remains the dominant renewable technology in the future capacity mix.

The increase in wind generation contributes to a substantial reduction in emissions. Fig. 26(a) shows that emissions decline from 20.57 MtCO2_22e in 2022 to 6.78 MtCO2_22e in 2050, representing a reduction of approximately 67%. The sharpest

decrease occurs between 2025 and 2030, after which emissions remain much lower than the initial level. However, emissions do not fall to zero under the Base scenario, indicating that additional policy or technology changes would be required to fully eliminate residual emissions.

A significant increase in land use accompanies the reduction in emissions. As shown in Fig. 26(b), accumulated land use increases from 75.0 km2 in 2020 to 243.0 km2 by 2050. This increase is mainly driven by the expansion of onshore wind, which has a much higher direct land intensity than compact generation technologies such as diesel. Therefore, while the Base scenario reduces emissions, it also shifts part of the planning challenge toward land availability and land-use compatibility.

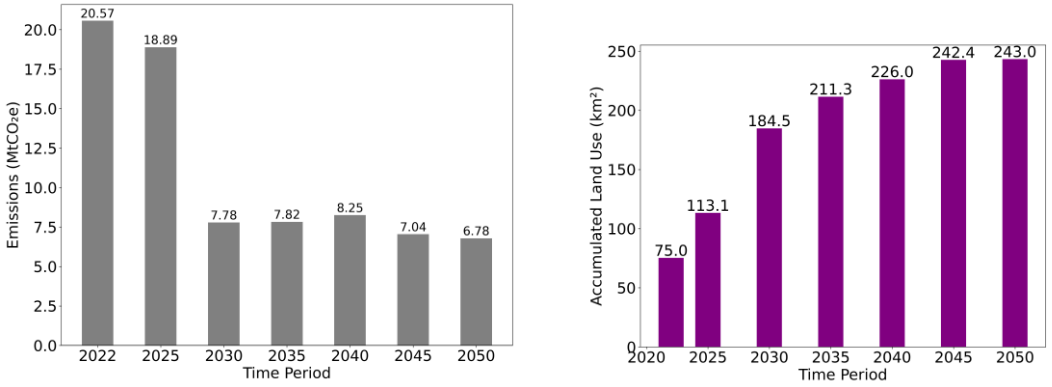


Fig. 26. Evolution of Prince Edward Island electricity system emissions and accumulated land use over the model horizon in the Base scenario. (a) Modelled CO_2-equivalent emissions by period, declining from approximately 20.57 in 2022 to 6.78 in 2050; (b) accumulated electricity system land occupation, increasing from approximately 75 km² in the initial period to 243 km² by 2050. Together, the panels illustrate the simultaneous reduction in emissions and increase in land requirements associated with the transition toward greater onshore wind deployment.

PEI's electricity system also shows a clear mismatch between capacity share and land-use share. Fig. 27 compares the present and future capacity mix with the corresponding land-use distribution. In the present system, diesel accounts for a substantial share of installed capacity, while onshore wind accounts for the majority of land use. Diesel occupies only a negligible share of total land because its direct land intensity is much lower than that of onshore wind. This pattern remains in the future

system: even though diesel and biomass-based technologies remain part of the capacity mix, onshore wind continues to dominate the land-use profile.

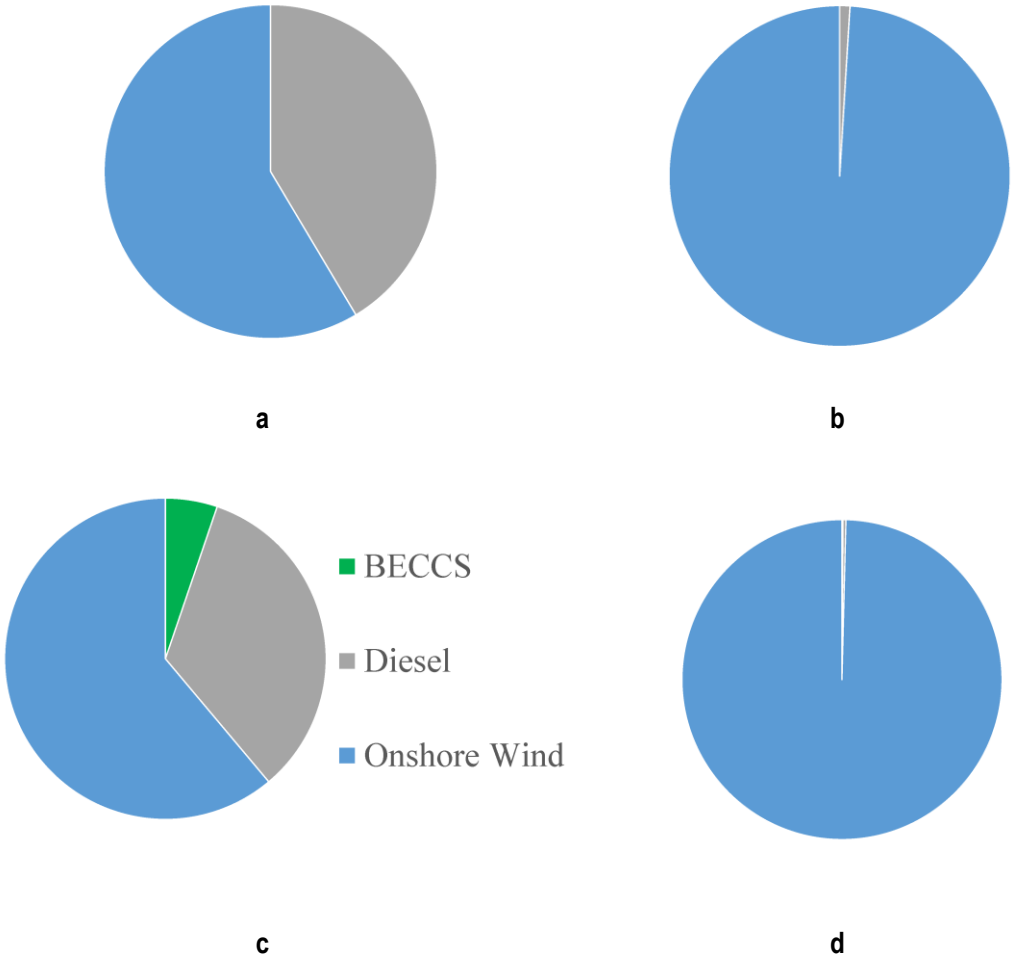


Fig. 27. Comparison of technology shares in Prince Edward Island's installed capacity and electricity system land occupation for the existing and 2050 Base systems. (a) Existing installed capacity share; (b) existing land use share; (c) installed capacity share in 2050; and (d) land use share in 2050. Technologies include onshore wind, diesel generation, and BECCS. The comparison demonstrates that capacity share and land use share are not equivalent: onshore wind accounts for a disproportionately large share of land occupation because of its higher direct land intensity.

Overall, the Base scenario shows that PEI's electricity transition is both an energy-system planning issue and a land-use planning issue. Expanding onshore wind helps meet future demand and reduce emissions, but it also substantially increases the land area associated with electricity generation. This finding establishes the importance of the

following policy and colocation analyses. Policy scenarios are needed to examine how clean-energy instruments affect technology choices and land-use outcomes, while colocation scenarios are needed to evaluate whether onshore wind development can be integrated with cropland in a way that reduces land-use conflict.

How Different Policies Affect Atlantic Canada's Future Energy and Land Use

This subsection compares how the Investment Tax Credit (ITC), Renewable Portfolio Standard (RPS), and combined-policy Mix scenarios affect the electricity system in Atlantic Canada, with particular attention to implications for PEI. The comparison focuses on capacity expansion, electricity generation, total system cost, emissions, and land-use projections.

The policy scenarios produce different 2050 capacity mixes, as shown in Fig. 28. The ITC scenario increases onshore wind capacity relative to the Base scenario, reaching 3.71 GW in 2050 compared with 3.49 GW in the Base scenario. However, the largest wind expansion occurs under the scenarios that include the RPS. Onshore wind reaches 4.82 GW in the Mix scenario and 4.93 GW in the RPS scenario. This indicates that the renewable production requirement has a stronger effect on wind deployment than the investment incentive alone, due to the high investment cost of wind development.

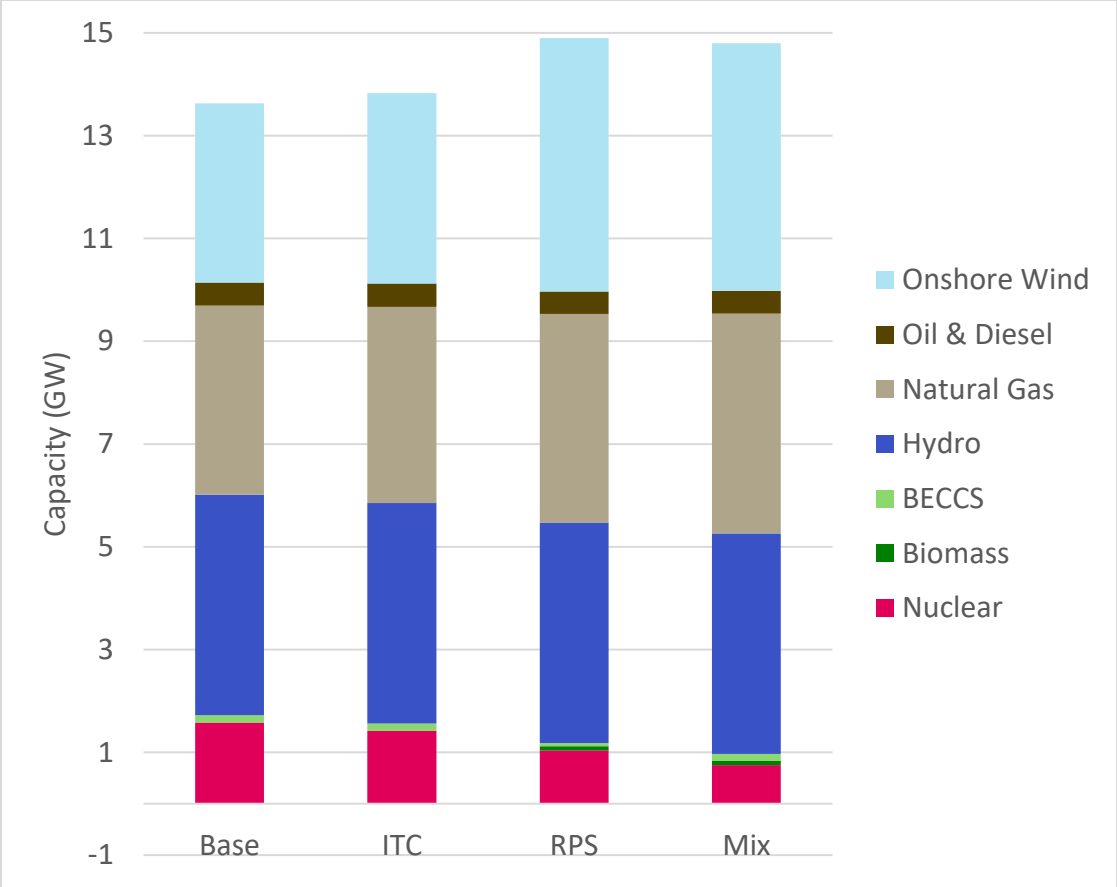


Fig. 28. **Atlantic Canada installed electricity generation capacity in 2050 under the Base, Investment Tax Credit (ITC), Renewable Portfolio Standard (RPS), and combined-policy Mix scenarios. Stacked bars report capacity in GW for onshore wind, oil and diesel, natural gas, hydro, BECCS, biomass, and nuclear generation. The figure compares how an investment cost incentive and a renewable generation requirement alter the optimized technology portfolio, with the RPS-containing scenarios producing the greatest onshore wind expansion.**

Fig. 28 also shows that the ITC maintains a larger role for nuclear generation than the RPS-driven scenarios. Nuclear capacity is 1.42 GW under ITC, compared with 0.75 GW under Mix and 1.04 GW under RPS. This is consistent with the design of the policy instruments: the ITC lowers investment costs for eligible clean technologies, while the RPS shifts the system toward technologies that satisfy the renewable production requirement. As a result, the RPS and Mix scenarios rely more strongly on wind, while ITC allows a more balanced clean-capacity portfolio.

The activity results show the same policy contrast more clearly. As shown in Fig. 29, wind activity reaches 63.90 PJ in the Base scenario and 67.34 PJ under ITC, but

increases to 88.89 PJ under Mix and 90.91 PJ under RPS. Total electricity activity remains broadly comparable across scenarios, but the share supplied by onshore wind is much higher when the RPS is present. This confirms that the RPS is primarily an activity-oriented policy because it changes the production mix, whereas the ITC primarily affects investment decisions.

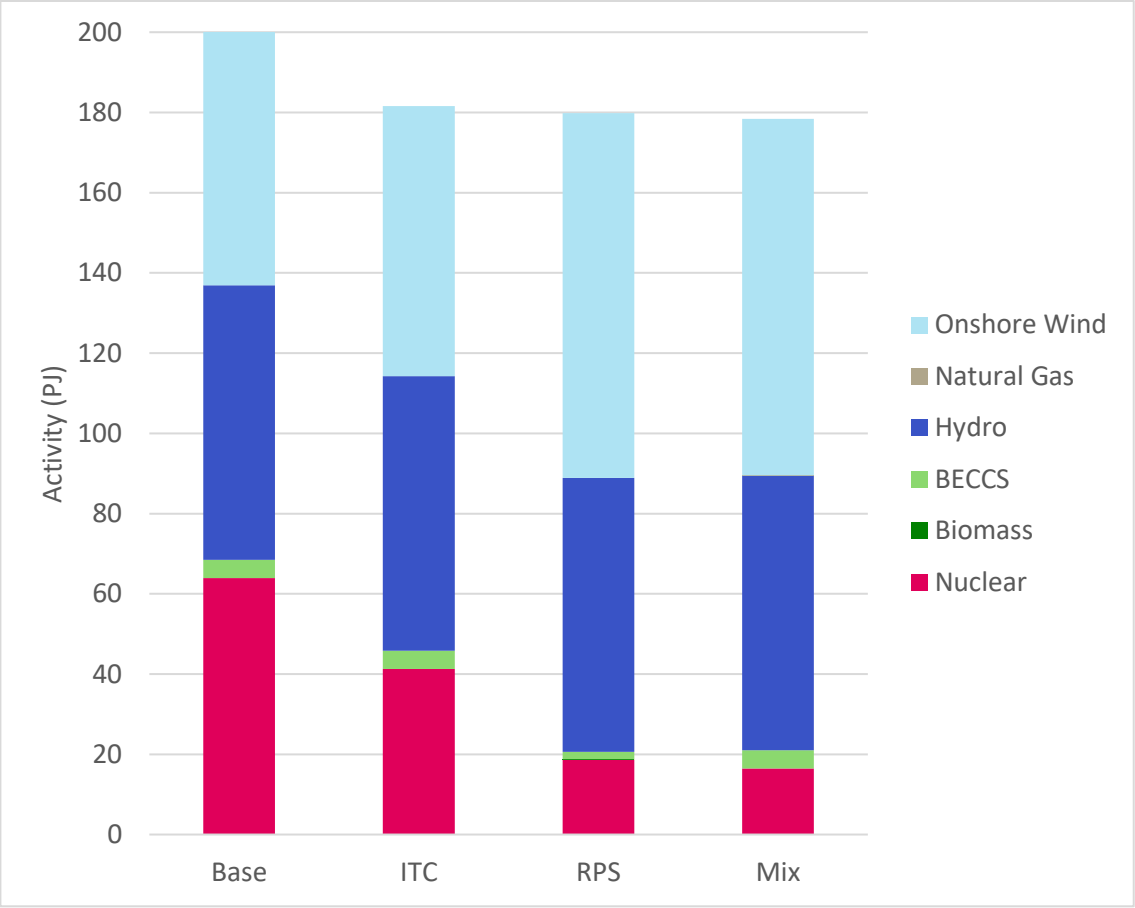


Fig. 29. **Atlantic Canada electricity generation activity in 2050 under the Base, ITC, RPS, and combined Mix policy scenarios. Stacked bars report modelled activity in PJ by generation technology, including onshore wind, natural gas, hydro, BECCS, biomass, and nuclear generation. The comparison shows the effect of policy design on electricity production rather than installed capacity, with RPS-containing scenarios producing the highest onshore wind activity.**

Because land use in the model is strongly driven by onshore wind, the RPS and Mix scenarios create greater land-use pressure than the Base and ITC scenarios. The RPS scenario produces the highest wind capacity and activity, while the Mix scenario remains close behind. This means that policies designed to increase renewable electricity

production can also increase the spatial footprint of the electricity system, especially when onshore wind is the main compliance technology.

Natural gas and diesel are present in capacity, but not in activity, meaning that their main purpose in the system is as backup to intermittent renewables. Natural gas accounts for a large share of the capacity in the system, and increases under all the studied policy scenarios. While this does not impose significant land use and emissions on the model, it results in some cost increase due to capacity buildout and fixed costs. However, this cost is lower than renewable capacity buildout costs and takes a lower share of the overall costs.

The cost results indicate that the ITC has the strongest cost-reduction effect among the policy scenarios. As shown in Fig. 30, total discounted system cost is 64,643 under ITC, compared with 65,247 in the Base scenario, 65,511 in the Mix scenario, and 66,961 in the RPS scenario. Relative to the Base scenario, ITC reduces total cost by approximately 0.9%, while Mix increases cost by approximately 0.4% and RPS increases cost by approximately 2.6%. This suggests that reducing investment costs can improve system economics, whereas requiring additional renewable activity can increase total system cost when it pushes the model toward a more wind-intensive portfolio.

The cost breakdown in Fig. 30separates generation-technology costs from network, storage, fuel/import, and other model entries. Network and transport-related costs dominate the total cost across all scenarios, but generation-technology costs are highest under RPS and Mix because these scenarios develop more onshore wind and associated clean-generation capacity. This breakdown is useful because it prevents the policy comparison from being interpreted only through total system cost, which includes several non-generation components.

The small cost variations between the policy scenarios also indicate that the main concern in these cases is not cost, but other policy implications, such as land use. Capacity expansion models are large-scale models that assist decision-makers in long-term planning; however, they are not able to predict an accurate estimate of costs and require other tools for such estimates. Therefore, the cost findings in this study are useful, along with other results such as land uses, to derive land-aware long-term perspectives and decisions.

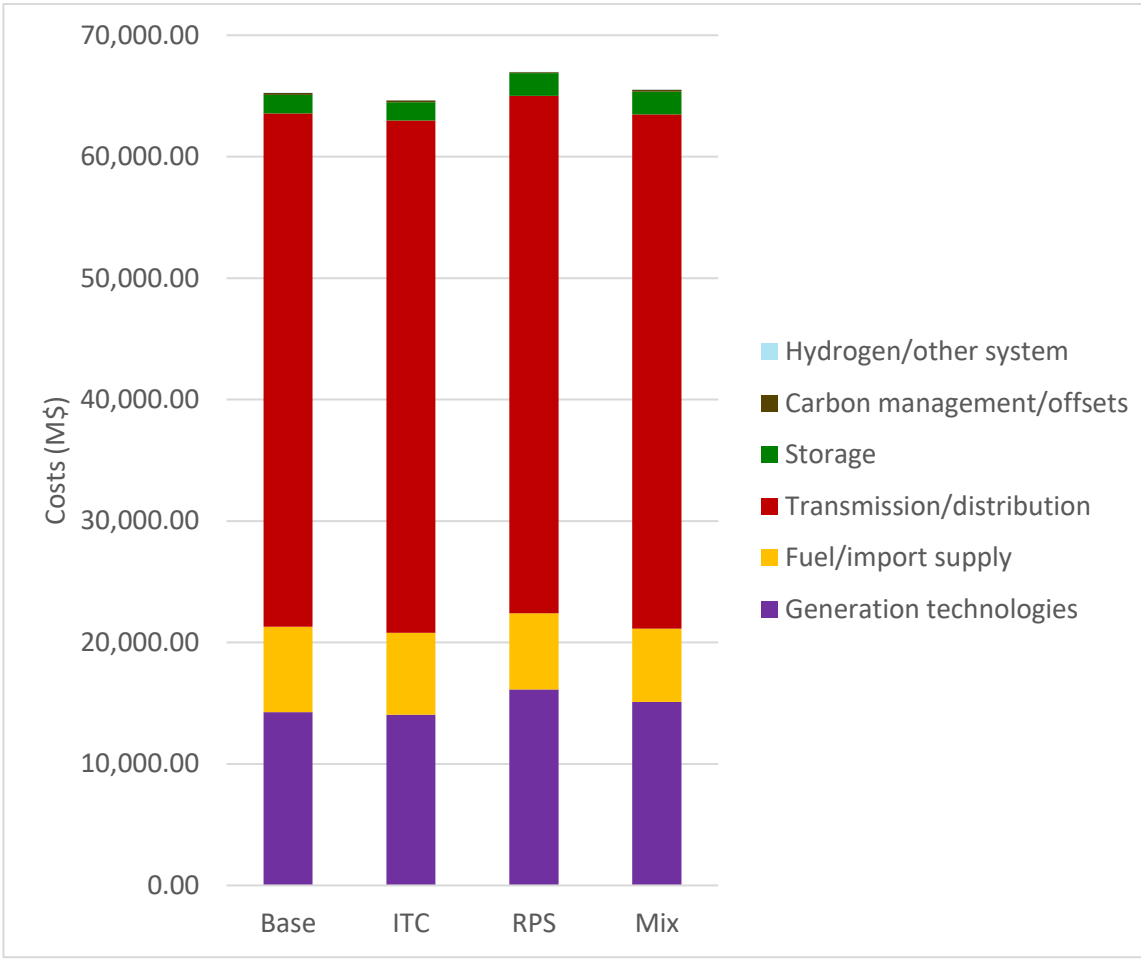


Fig. 30. **Atlantic Canada energy system cost composition under the Base, ITC, RPS, and combined Mix policy scenarios. Stacked bars report costs in million dollars (M$) for generation technologies, fuel and import supply, transmission and distribution, storage, carbon-management/offset activities, and hydrogen or other system components. The figure distinguishes changes in generation-related expenditure from broader system costs and shows how the ITC and RPS affect the overall optimized cost structure.**

The emissions results show that higher renewable activity does not automatically produce the lowest net emissions. Fig. 31 reports CO2e emissions after excluding import-tax emissions and the erroneous CO2 rows in the Mix scenario. ITC has the lowest net CO2e emissions at 3,212, followed by the Base scenario at 3,455 and Mix at 3,625. RPS has the highest net CO2e emissions at 7,600, despite producing the highest wind activity. This occurs because RPS has a much smaller negative-emissions contribution than the other scenarios.

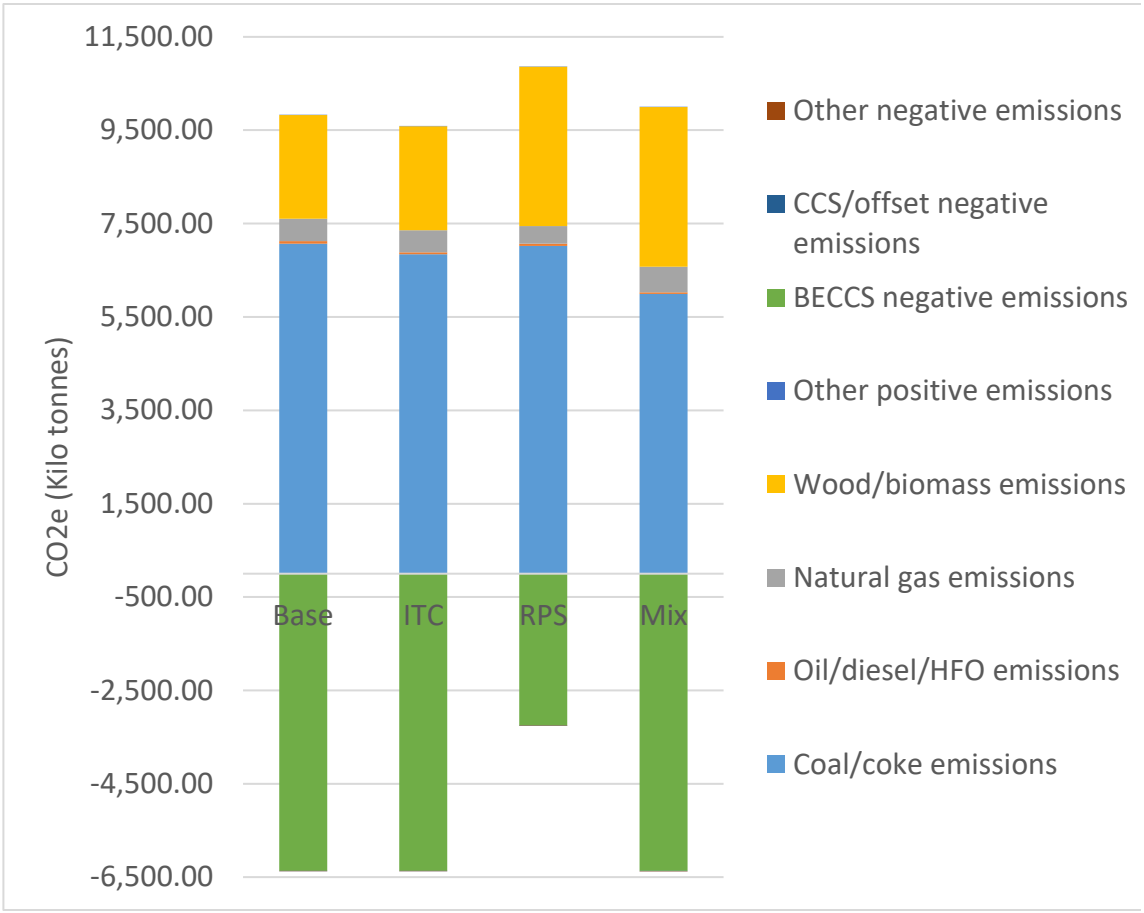


Fig. 31. Atlantic Canada net CO_2-equivalent emissions in 2050 under the Base, ITC, RPS, and combined Mix policy scenarios, disaggregated by major positive and negative emission sources. Positive contributions include coal/coke, oil/diesel/HFO, natural gas, wood/biomass, and other emissions, while negative contributions include BECCS, CCS/offsets, and other negative emission processes. Emissions are reported in kt CO_2e

The technology-level emissions results help explain this outcome. Fig. 31 shows that positive emissions are dominated by imported coal, imported wood, and imported natural gas, while BECCS provides the largest negative-emissions contribution. Wood-related emissions are higher under RPS and Mix than under the Base and ITC scenarios, while the RPS scenario develops substantially less BECCS-related negative emissions. As a result, RPS increases renewable electricity production but does not achieve the lowest net emissions in the modeled system.

The Mix scenario demonstrates that combining ITC and RPS does not produce a simple additive effect. It substantially increases wind deployment relative to the Base and

ITC scenarios, but its total cost remains higher than ITC and Base. Its net CO2e emissions are also higher than both ITC and Base, although lower than RPS. This indicates a countervailing policy interaction: the RPS component pushes the system toward greater wind activity and land-use pressure, while the ITC component partially reduces the cost burden but does not fully offset the emissions and land-use implications of the RPS-driven technology mix.

Overall, all policy scenarios involve substantial onshore wind expansion by 2050, and therefore increase the relevance of land use in electricity-system planning. The Base and ITC scenarios produce more moderate wind growth, while the RPS and Mix scenarios produce the highest wind capacity and activity. These results show that clean-energy policy design affects not only costs and emissions, but also the spatial requirements of the electricity system. For PEI, where land availability and cropland overlap are central concerns, this finding motivates the following colocation analysis.

3.2.3. Colocation Results

Two site-selection approaches are used to compare scenarios that include or exclude cropland from candidate wind sites. Fig. 32 (a) shows the base scenario, which includes 19 sites, while Fig. 32 (c) shows the spacing scenario, where the four cropland sites are excluded. In the absence of cropland sites, the optimal capacity shifts to other sites with lower capacity factors.

The effect of excluding cropland depends on the capacity factors of the affected cells. High-wind locations are not always aligned with agricultural suitability, and compatibility depends on crop type and site conditions. However, in some cases, wind development and agriculture can be colocated. Fig. 32 (e) shows the colocated cells for PEI with red circles; one of the four cropland sites has colocation potential and is selected among the optimal options.

Under no land constraint, the base and colocation model results are similar. However, once land limits are introduced, the model behaves differently across scenarios. The major observations are discussed in the following paragraphs.

One model response to land limits is to reduce overall onshore wind development. Fig. 32 (f) is associated with 0.32 GW less onshore wind capacity compared with the base

value of 0.46 GW in Fig. 32 (e), while almost all of this decrease occurs in noncolocated cells. The resulting shifts in energy production types are discussed in the next subsection.

Another strategy under land-constrained scenarios is to favor colocated cells to minimize land occupation. Fig. 32 (f) shows the results under a 50 km^2 land limit. Based on these results, two of the optimal sites are replaced with a colocated cell to meet demand with lower land use. The colocated cell that is already optimal in Fig. 32 (e) is developed up to its capacity, and much of the capacity from another cell is replaced by the newly selected colocated cell in Fig. 32 (f).

Different modeling approaches yield substantially different results under the same land limit. Fig. 32 (d) and f show these differences clearly. In addition to the choice of cells and capacities for each cell, system costs and emissions also vary between scenarios.

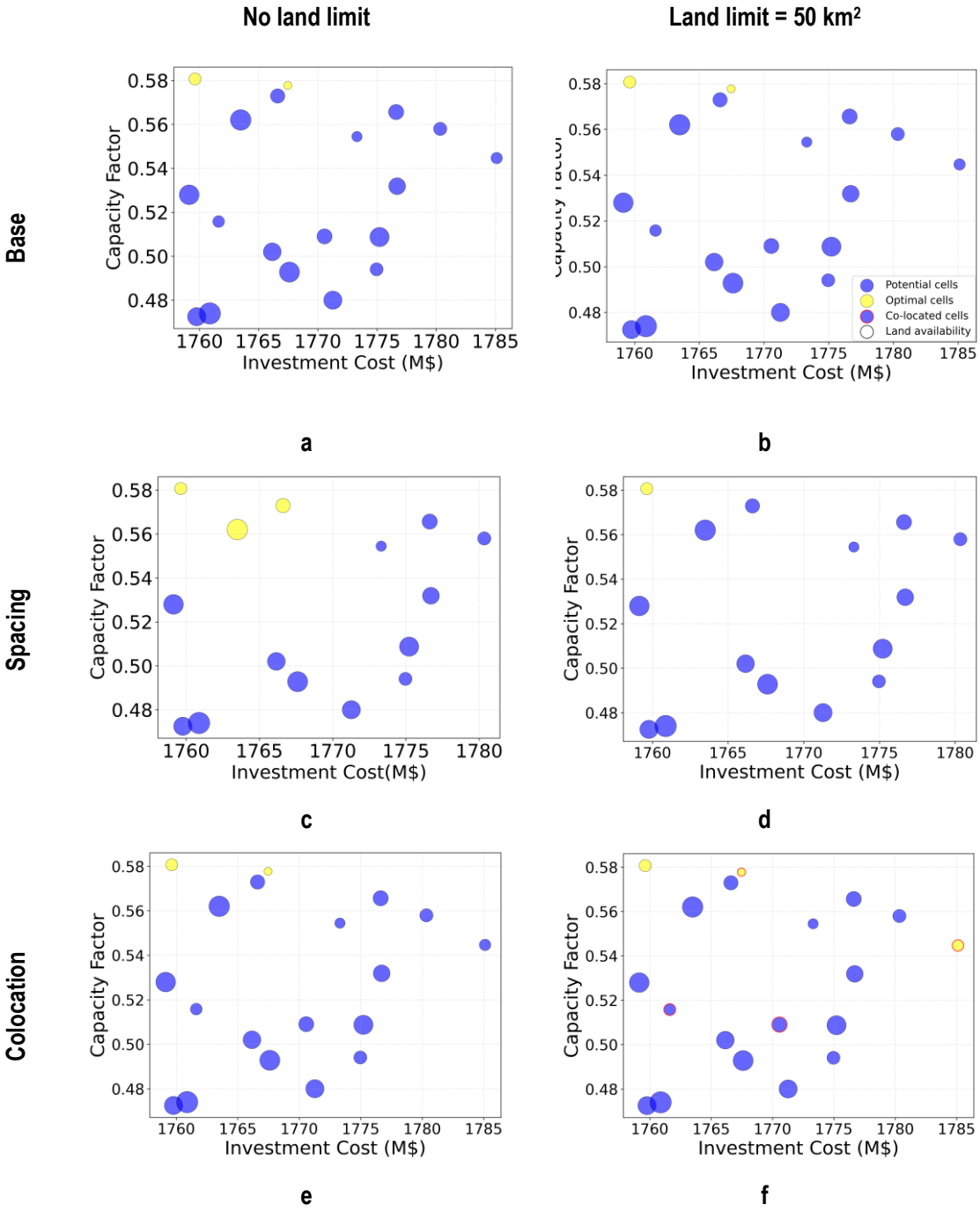

No land limit
Land limit = 50 km²
Base
Spacing
Colocation
Capacity Factor
Investment Cost (M$)
Investment Cost(M$)
Potential cells
Optimal cells
Co-located cells
Land availability
a
b
c
d
e
f

Fig. 32. Candidate and optimized onshore wind cells in Prince Edward Island as a function of capacity factor, investment cost, land availability, cropland treatment, and land constraint. Rows represent the Base/conventional, spacing, and wind–cropland colocation scenarios; columns compare the no-land-limit case with a 50 km² land limit. Each point represents a candidate GIS-derived wind cell positioned according to investment cost and capacity factor; point size represents available land, while colours/symbols distinguish regular candidate cells, cropland-overlapping or colocated cells, and cells selected by the optimization. The panels show how excluding cropland or imposing a land limit changes the set and capacity of wind sites selected by TEMOA-ACES.

Impacts of colocation on energy production

Under the spacing scenario, land limits reduce the ability of onshore wind to meet demand, so the model adds more offshore wind and system costs increase. Fig. 33 shows offshore wind growth under the 50 km2 limit and more extreme land limits. Because offshore wind requires higher investment than other production sources, this shift helps explain the cost increase in the spacing scenario shown in Fig. 34.

Even with additional diesel and offshore wind in the spacing scenario, the model still develops solar to satisfy demand under tighter land limits. Fig. 33 shows the increase in solar capacity in the spacing scenarios. Solar colocation is not defined in the model, so its land use is higher than that of colocated onshore wind. Although solar development has higher land-use and cost implications than colocated onshore wind, it becomes a replacement option in the spacing scenario when onshore wind is constrained.

Cost and emissions also spike in the colocation scenario under the extreme land limit (Fig. 34 and Fig. 35). Despite onshore wind development in the colocation scenario, the model still develops solar and offshore wind when the most restrictive land constraint is applied (Fig. 33). The land cap in this scenario is 1 km^2, which is significantly lower than the current energy system's land occupation. This result suggests that near-zero additional land use is not a practical planning assumption for PEI's electricity transition; instead, the objective is to manage land-use growth while maintaining feasible energy-system outcomes.

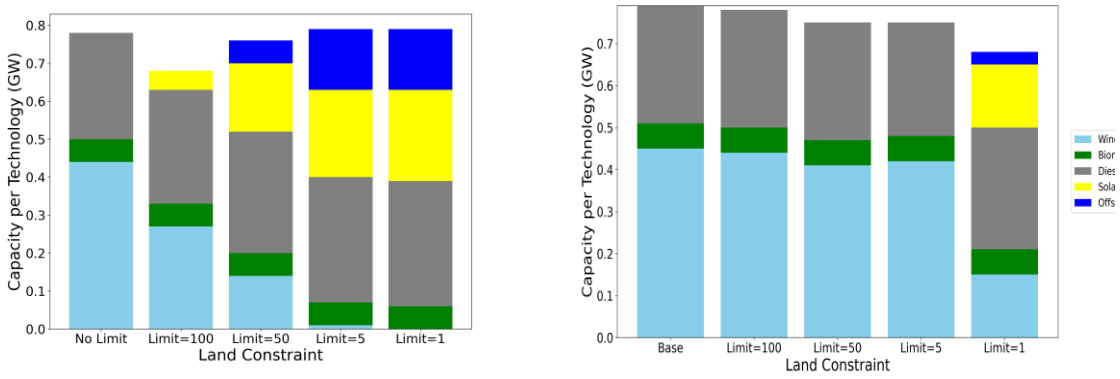


Fig. 33. **Prince Edward Island installed electricity generation capacity by technology in 2050 under progressively more restrictive land limits. (a) conventional scenario** and **(b) wind–cropland colocation scenario. Land-limit cases include no limit and limits of 100, 50, 5, and 1 km². Stacked bars report capacity in GW by technology, including onshore wind, offshore wind, solar, diesel,** and bio**. The comparison shows how tighter land constraints shift capacity away from onshore wind when colocation is not available and how explicit colocation preserves a larger role for onshore wind under constrained land availability.**

Impacts of colocation on cost

Contrary to expectations, colocation scenarios under land limits show lower costs compared with the spacing scenarios. Fig. 34 shows the electricity-system cost change for both spacing and colocation scenarios under a range of land limits. According to these results, the cost of colocation scenarios under land constraints is always lower than that of the spacing scenarios. The cost gap between the two scenarios grows as the land constraint becomes more restrictive. Therefore, colocating wind and agriculture not only decreases land use but also reduces electricity-system costs.

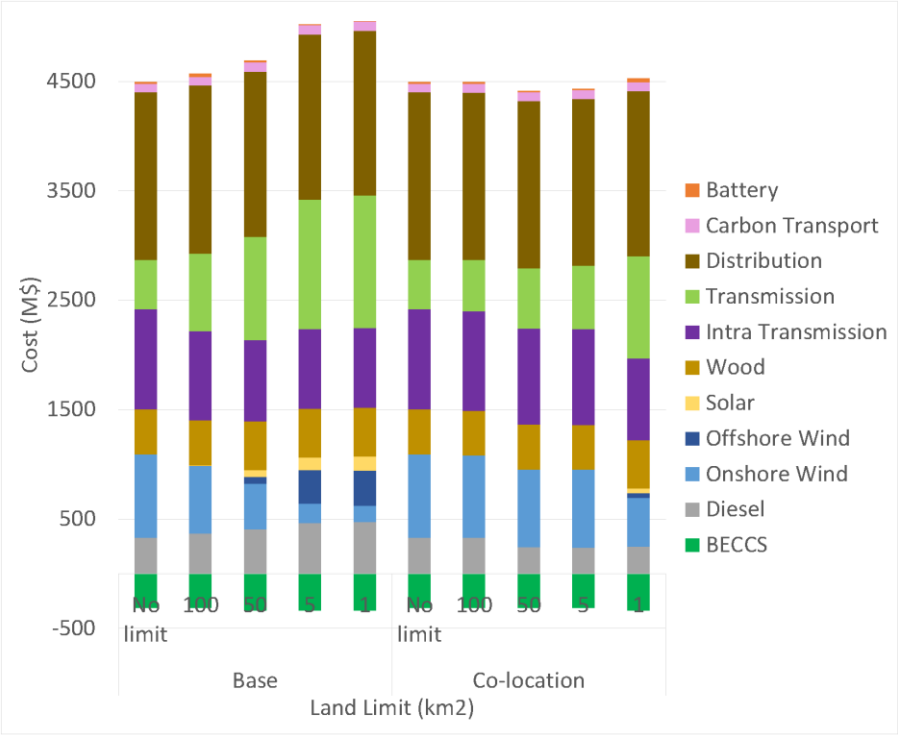


Fig. 34. **Prince Edward Island electricity system cost composition under alternative land constraints for the base and wind–cropland colocation cases. Scenarios are evaluated with no land limit and land limits of 100, 50, 5, and 1 km². Stacked bars report costs in M$ for BECCS, diesel, onshore wind, offshore wind, solar, wood, wind transmission, other transmission, distribution, carbon transport, and battery storage. The figure illustrates how the economic penalty of land scarcity increases when wind–cropland colocation is unavailable and how colocation reduces reliance on more costly substitute technologies.**

Impacts of colocation on emission reduction

Results show that colocation contributes to reducing emissions as well as costs and land use. While land and cost are important decision factors, emissions remain central to clean-energy planning and net-zero targets. Fig. 35 shows that colocation scenarios maintain lower emissions than their spacing counterparts across all land-limits.

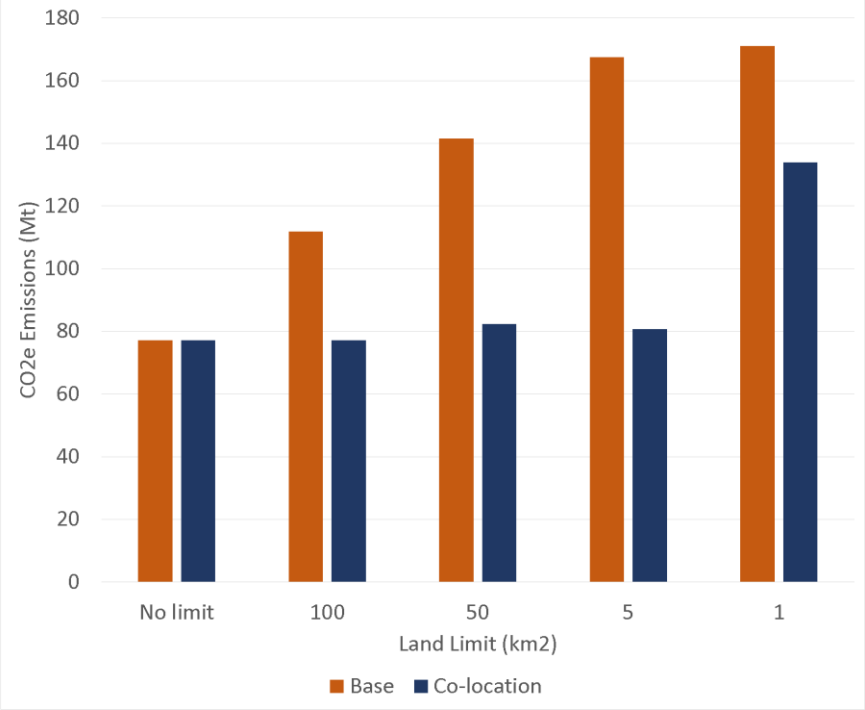


Fig. 35. **Prince Edward Island CO_2-equivalent emissions in 2050 under the base and wind–cropland colocation cases across land limits of no limit, 100, 50, 5, and 1 km². Paired bars compare emissions from the two land use representations at each constraint level. Colocation maintains lower emissions under increasingly restrictive land limits because it preserves greater onshore wind deployment and reduces substitution toward fossil fuel generation.**

Under land limitations, the spacing scenario meets demand with more diesel capacity, whereas the colocation scenario develops more onshore wind. Fig. 33 (a) and (b) show the capacity mix for spacing and colocation scenarios under different land limits. The growth in diesel capacity in the spacing scenario explains the increase in emissions observed in Fig. 35.

Chapter 4. Conclusions

This chapter synthesizes the main conclusions of the thesis in relation to the research questions. The first research question examined which land and wind trade-offs are most important for onshore wind deployment in Canada. The second focused on how a spatially enhanced energy-system modelling tool can be developed to study those trade-offs further. The third considered what this tool shows about the future role of spatial considerations and land use in Prince Edward Island's electricity system. The fourth examined how modelling wind and agricultural land colocation can improve energy-system decision-making under land constraints. The chapter also summarizes the thesis contributions and identifies limitations and future research directions.

4.1. Land-Cover Constraints and Onshore Wind Potential in Canada

The Canada-wide spatial analysis shows that most of the studied land-cover constraints do not materially limit onshore wind deployment at the aggregate provincial scale. Protected and conserved areas, built environments, and Indigenous Reserve Lands occupy relatively small shares of land in most provinces, even when buffers are included. In Atlantic Canada, these layers generally constrain less than 10% of land, while in many other provinces their constrained share is closer to 1-2%. Their exclusion remains important for environmental protection, social sensitivity, and local siting decisions, but at the national or provincial screening scale they do not substantially reduce aggregate onshore wind development potential.

Some high-coverage layers reduce gross land availability without substantially reducing access to high-quality wind resources. In British Columbia, steep slopes are widespread, but high-slope areas largely overlap with lower-wind locations; excluding these areas therefore removes technically difficult sites without strongly changing wind-potential outcomes. A similar pattern occurs for wetlands and water bodies in the provinces where those layers have the greatest coverage. In Ontario and Manitoba, wetland-affected cells have lower average capacity factors than the broader candidate pool, and in Quebec and Manitoba, water-body-affected cells also tend to have below-average capacity factors. These results show that a land layer's importance depends not

only on how much land it covers, but also on whether it overlaps with strong wind resources.

Cropland is the main exception among the studied land-cover layers. In Alberta, Saskatchewan, and Prince Edward Island, cropland overlaps with relatively high-capacity-factor wind locations. In Alberta, cells affected by cropland exclusion have an average capacity factor of 0.22 compared with 0.166 when cropland is included in the candidate pool. In Saskatchewan, the corresponding values are 0.31 and 0.27, while in PEI they are 0.52 and 0.49. This pattern indicates that excluding cropland removes some of the better wind locations in these provinces. The effect is especially important in PEI because the province combines strong wind resources, limited total land area, and a high cropland share. Therefore, the central land-use trade-off identified by the Canada-wide analysis is not simply whether land layers reduce available area, but whether they remove wind-rich land that may also support other important land uses.

Overall, onshore wind development in Canada is not strongly constrained by most of the studied land layers when evaluated at a broad provincial scale. The more consequential constraint is cropland, particularly where agricultural land overlaps with high wind potential. This finding supports the thesis focus on PEI and motivates the later analysis of wind-cropland colocation. It also suggests that land-aware wind planning should distinguish between layers that are true exclusions, layers that remove technically unsuitable land, and layers that may require land-sharing or colocation strategies rather than simple exclusion.

4.2. Spatial Enhancement of TEMOA-ACES

The spatially enhanced TEMOA-ACES model demonstrates that representing onshore wind through spatially specific wind cells changes the model's cost, capacity, and land-use outcomes. Compared with the original ACES representation, the spatially enhanced Base model reduces total electricity system costs by 5.9%. This reduction occurs even after additional grid-connection costs are included, with added transmission-related costs accounting for approximately 2% of the total cost effect. The resulting decline in generation capacity costs is 6.1%, indicating that the original non-spatial representation overestimates the cost of meeting demand because it does not fully capture the availability of higher-quality wind sites.

Spatial enhancement also changes how much capacity is needed to provide the same level of electricity service. The average capacity factor of newly built onshore wind increases from 0.47 in the original ACES model to 0.60 in the spatially enhanced model. Because higher-capacity-factor wind sites generate more electricity per unit of installed capacity, the model requires 1.4 GW less installed capacity to meet the same demand. This result confirms that spatially explicit wind data improve the representation of resource quality and can alter the optimal technology portfolio in a capacity expansion model.

However, spatial enhancement alone does not represent a land-constrained energy system. Although higher-capacity-factor wind sites reduce the installed capacity needed per unit of generation, they also make onshore wind more attractive to the optimization model. As a result, the model shifts toward more onshore wind development, which increases total land use because onshore wind is the dominant land-consuming technology in the model. This finding is important because it shows that better spatial representation is not equivalent to land-use governance. Spatial enhancement improves the realism of the wind resource, but explicit land tracking and land limits are still required if land scarcity is to influence investment decisions directly.

4.3. PEI's Future Electricity Transition and Land-Use Pressure

The Base scenario results show that PEI's future electricity system requires substantial capacity expansion while also reducing emissions. Total installed capacity increases from 0.487 GW in 2022 to 1.080 GW in 2050, which is an increase of approximately 122%. Onshore wind is the main source of this expansion and becomes the dominant renewable technology in the future system. At the same time, emissions fall from 20.57 $MtCO_2e$ in 2022 to 6.78 $MtCO_2e$ in 2050, a reduction of about 67%. The Base scenario therefore shows meaningful decarbonization progress, although it does not eliminate residual emissions by 2050.

The same scenario also shows that PEI's electricity transition creates a significant land-use challenge. Accumulated electricity-sector land use increases from 75.0 km^2 in 2020 to 243.0 km^2 in 2050, an increase of approximately 224%. This growth is mainly driven by onshore wind because diesel and other conventional generation technologies occupy much less direct land per unit of capacity. The result is a mismatch between

capacity shares and land-use shares: technologies such as diesel can remain visible in the capacity mix while contributing very little to land occupation, whereas wind dominates the land-use profile. Consequently, PEI's clean electricity transition cannot be evaluated only through cost, capacity, and emissions; it must also be evaluated through land availability and land-use compatibility.

These results establish the importance of studying PEI as a land-constrained energy-system case. PEI has strong wind resources, but it also has a limited land base and a high share of cropland. Expanding onshore wind helps the province meet future electricity demand and reduce emissions, but it also increases the spatial footprint of the electricity system. This means that PEI's energy transition involves a trade-off between renewable electricity development and competing land uses, especially agriculture. The following policy and colocation conclusions build directly on this baseline land-use challenge.

4.4. Policy Design, Technology Choice, and Land Use

The policy scenarios show that the Investment Tax Credit (ITC) and Renewable Portfolio Standard (RPS) affect the electricity system through different mechanisms. The ITC is primarily an investment-cost policy, so it improves system economics and supports additional clean-capacity development without directly requiring a specific renewable generation share. Under the ITC scenario, total discounted system cost is 64,643, which is lower than the Base cost of 65,247. ITC also produces the lowest net emissions among the policy cases considered, with net emissions of 3,212 after excluding import-tax emissions. These results indicate that investment incentives can reduce system cost while supporting a cleaner technology portfolio.

The RPS produces a stronger renewable-generation response, but also increases cost, emissions, and land-use pressure relative to the ITC scenario. By 2050, the RPS scenario develops 4.93 GW of onshore wind capacity and 90.91 PJ of onshore wind activity, compared with 3.49 GW and 63.90 PJ in the Base scenario. Because land use in the model is strongly linked to onshore wind, the RPS creates the greatest land-use pressure among the policy scenarios. It also has the highest total discounted system cost, at 66,961, and the highest net emissions, at 7,600 after import-tax emissions are excluded.

This outcome reflects the fact that the RPS increases renewable electricity generation but develops less negative-emission output than the other scenarios.

The Mix scenario demonstrates that combining policies does not automatically produce the best result across all metrics. The combined policy reinforces the renewable-generation effect of the RPS, leading to 4.82 GW of onshore wind capacity and 88.89 PJ of wind activity by 2050. However, its total cost of 65,511 and net emissions of 3,625 are both higher than the ITC scenario. The Mix scenario therefore produces an interaction effect rather than a simple additive benefit: the RPS component increases renewable activity and associated land-use pressure, while the ITC component moderates the cost burden of that shift. This finding shows that clean-energy policy design must consider technology choice, land use, emissions, and cost together rather than evaluating renewable generation targets in isolation.

Across all scenarios, onshore wind expands substantially by 2050. This common result shows that wind remains central to the region's future electricity system, regardless of whether the policy instrument is cost-based, generation-based, or combined. However, the amount of wind development and its associated land requirements vary by policy design. For PEI, where land availability and cropland overlap are major planning concerns, policy decisions therefore have spatial consequences. Policies that increase renewable activity can also intensify land-use competition unless land constraints and colocation options are represented in the model.

4.5. Wind-Cropland Colocation as a Land-Use Strategy

The colocation results show that conventional land-energy modelling can underestimate onshore wind potential under land constraints. A conventional model with land limits develops approximately 400 MW less onshore wind than a spatially integrated model that allows wind-cropland colocation. This difference occurs because conventional approaches often treat cropland either as fully available or fully unavailable, rather than distinguishing between the turbine footprint and the surrounding spacing area. When colocation is represented explicitly, the model can use high-quality wind sites on cropland while preserving the possibility of continued agricultural activity between turbines.

A similar underestimation occurs when spatial models exclude cropland entirely from candidate wind sites. Since PEI's cropland overlaps with some of the province's strongest wind locations, excluding agricultural land removes high-capacity-factor sites from the development pool. This can shift development toward lower-quality wind sites or toward other technologies that may be more costly, more emissions-intensive, or still land-consuming. The conclusion is not that all cropland should be available for wind development, but rather that cropland should not be treated as a simple hard exclusion without considering colocation potential, landowner participation, and agricultural compatibility.

Colocation also changes how land-use results should be interpreted. A conventional land-use metric may count the full wind-farm spacing area as land-use change, which can make wind development appear to displace agricultural land at a scale close to the cropland area affected by the project. In PEI, this could be interpreted as approximately 30 km² of land-use change under some conventional assumptions. However, if turbine spacing remains agriculturally usable, then the physical land transformation is much smaller than the full spacing area. This distinction is central to the thesis because it shows that land-use accounting can influence not only model outputs, but also how decision-makers interpret the social and agricultural impacts of wind development.

Under land constraints, colocated energy systems are more cost-effective than non-colocated systems. The model results show that wind-cropland colocation can reduce PEI electricity system costs by up to 22% relative to scenarios where land constraints restrict wind development more severely. This cost reduction occurs because colocation allows the model to retain more onshore wind, which is a relatively attractive resource in PEI, rather than substituting toward more expensive alternatives such as offshore wind, solar, or fossil-based backup generation. Colocation therefore reduces the economic penalty of land scarcity.

Colocation also supports emissions reduction under land constraints. The results show that allowing wind-cropland colocation can reduce emissions by up to 35% by enabling greater onshore wind development and reducing reliance on fossil generation. This does not mean that colocation alone guarantees net-zero emissions, but it does show that land-sharing assumptions can affect emissions pathways by changing the feasible

technology mix. In land-constrained systems such as PEI, the treatment of agricultural land is therefore directly connected to both system cost and emissions performance.

4.6. Limitations and Future Work

The main limitation of the colocation analysis is that it does not include the social and economic costs required to implement colocation in practice. Wind-cropland colocation depends on landowner and farmer participation, compensation agreements, access-road design, turbine placement, and operational coordination. This is especially important in PEI because much of the agricultural land is privately owned. Since these costs and institutional conditions are not included in the model objective function, the optimization may favor colocation more strongly than would occur under real project-development conditions. Future work should include landowner incentives, lease payments, compensation mechanisms, and transaction costs to better represent the practical economics of colocation.

The model is also limited to onshore wind colocation and does not evaluate solar colocation. This is important because solar is one of the next most land-consuming technologies in the model after onshore wind. Agrovoltaics, solar siting on marginal land, and hybrid solar-agricultural systems could change land-use outcomes in ways that are not captured here. Future research should therefore extend the framework to represent solar colocation and compare its land, cost, and emissions implications with those of wind-cropland colocation.

Another limitation is that crop type and crop yield are not explicitly represented. The model treats cropland as a spatial layer, but it does not distinguish between different crops, farm practices, crop values, or compatibility with turbine infrastructure. In reality, the effects of wind development may differ depending on crop type, machinery access, drainage, soil conditions, and field layout. Future work should include crop-specific compatibility and yield effects so that colocation scenarios can account for both electricity-system benefits and agricultural productivity impacts.

The PEI analysis also suggests a possible regional energy-planning opportunity that is not fully tested in this thesis. PEI's onshore wind potential is strong relative to some neighboring areas, which raises the possibility that the province could play a larger role in

regional renewable electricity supply if transmission capacity were expanded. However, the current analysis focuses on the PEI case and does not fully optimize regional transmission expansion or evaluate PEI as a renewable generation hub. Future work should therefore assess transmission development, interprovincial electricity trade, and the feasibility of using PEI wind resources to support broader Atlantic Canadian decarbonization.

Finally, the treatment of Indigenous Reserve Lands and related buffers should be expanded in future spatial energy analysis. In this thesis, these layers are represented as socially sensitive constraints for wind siting, but transmission and grid-connection routes may also intersect Indigenous lands or buffers. Since transmission infrastructure is essential for renewable energy deployment, future research should evaluate not only where generation technologies are located, but also how grid connections interact with Indigenous territories, consultation requirements, and social acceptance. This extension would improve the social relevance of spatial energy-system modelling.

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Appendix A.

Prince Edward Island's electricity network diagram

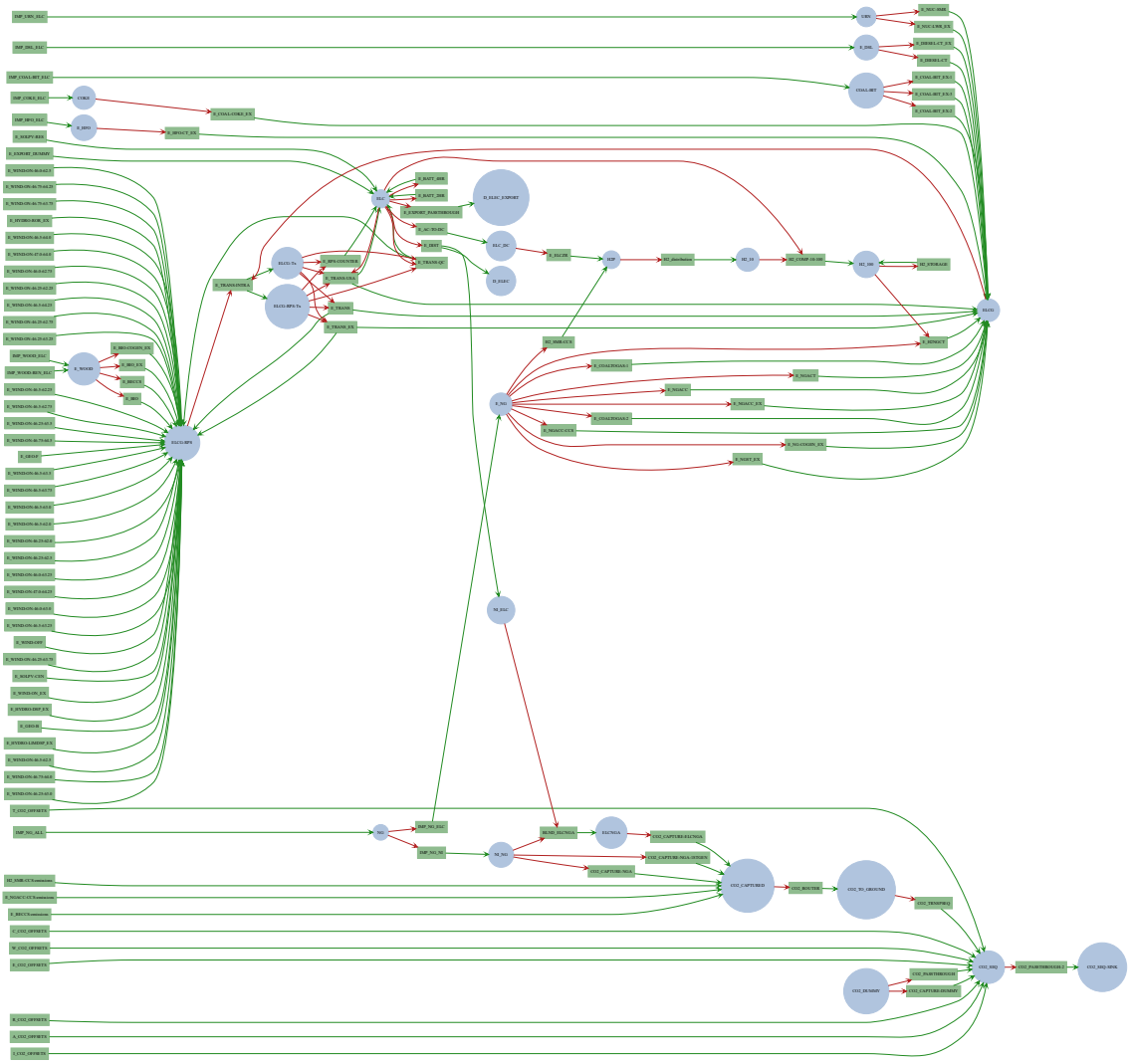
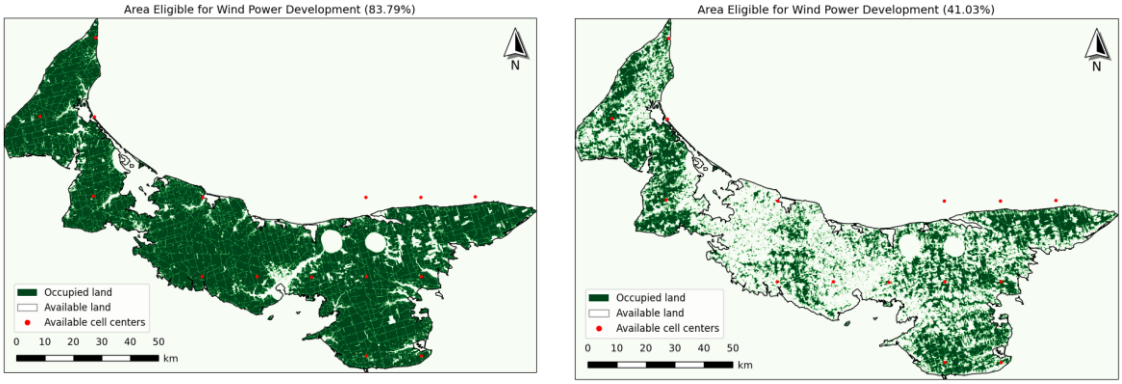


Fig. A.1. **Reference Energy System (RES) representation of Prince Edward Island in the TEMOA-ACES model. The network diagram shows the modeled relationships among imported energy carriers, electricity generation technologies, transmission and distribution processes, storage, hydrogen pathways, CO_2 capture and sequestration processes, renewable energy accounting, electricity exports, and final electricity demand. Nodes represent model commodities and technologies, while connecting links represent permitted energy or material flows within the optimization framework.**

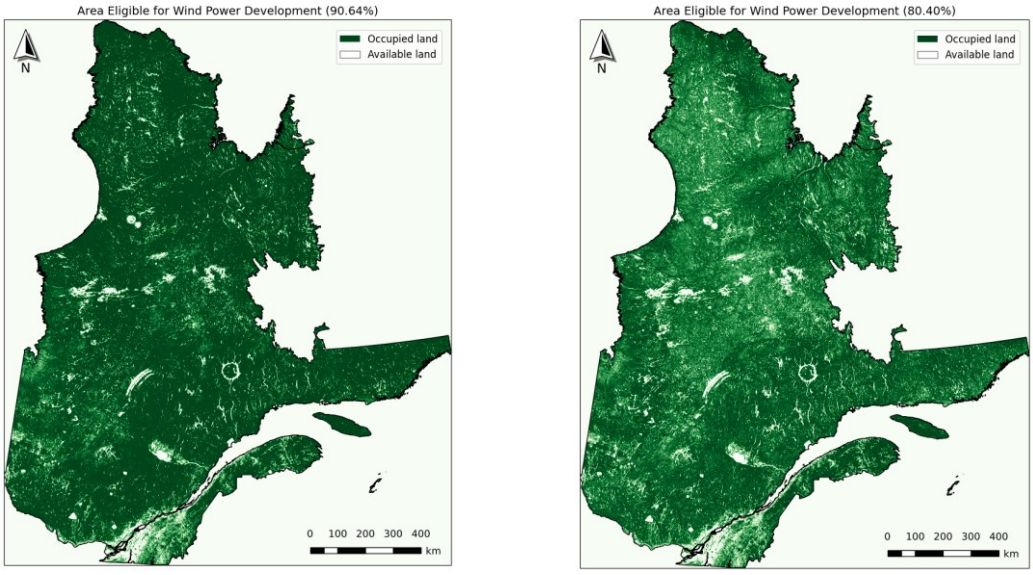
Appendix B.

Maps of different provinces and their available land under different land cover assumptions



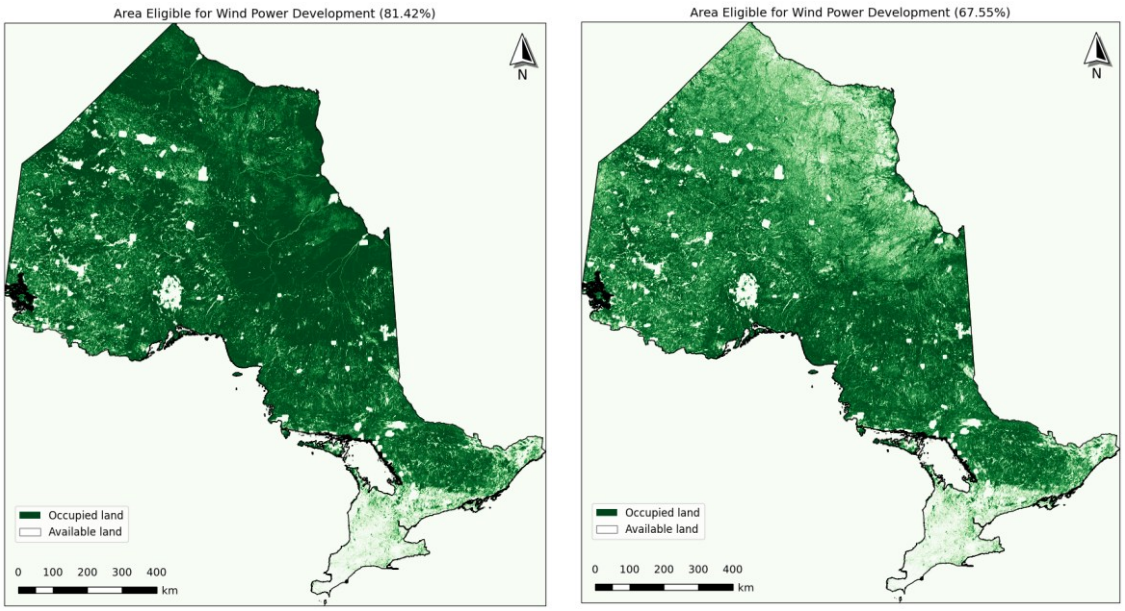
a b

Fig. B.1. **Spatial distribution of land eligible for onshore wind development inPrince Edward Island under two cropland assumptions: (a) included and (b) excluded cropland.**



a b

Fig. B.2. **Spatial distribution of land eligible for onshore wind development in Quebeca)before and b)after excluding wetlands**



a b

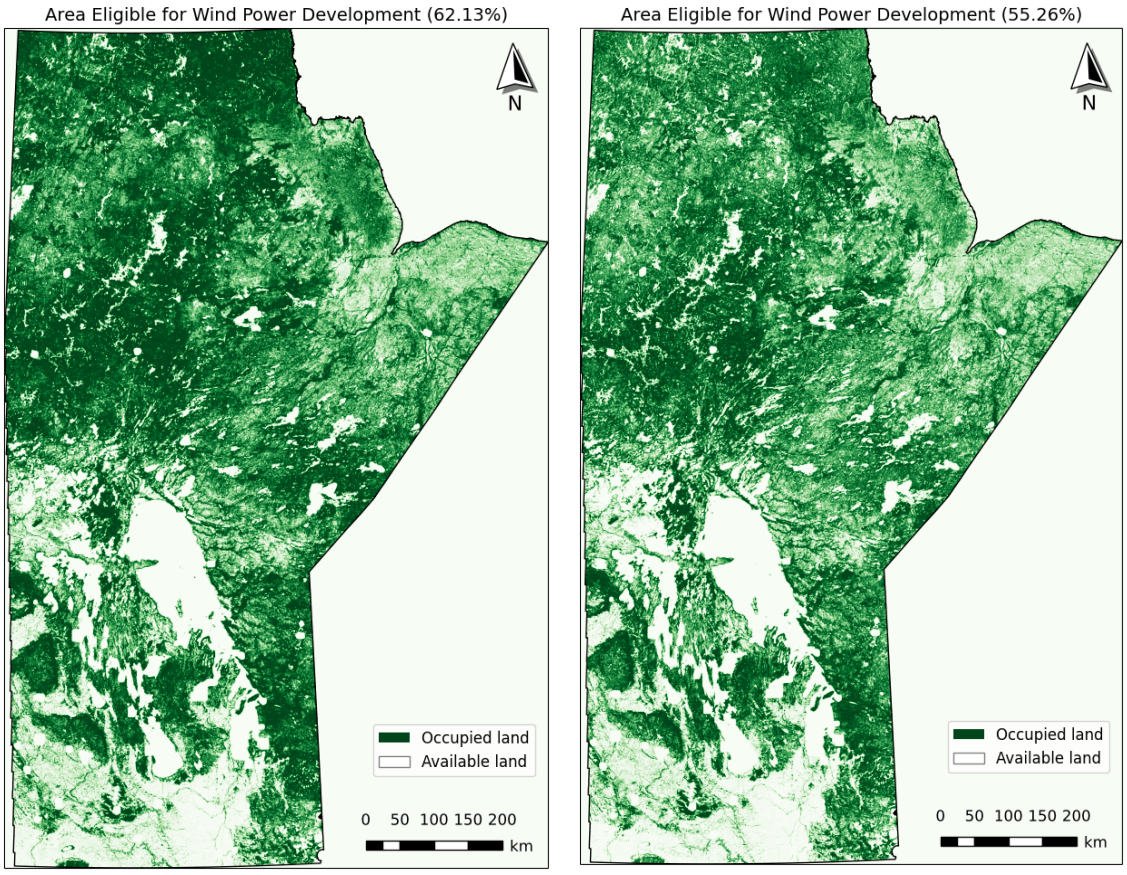
Fig. B.3. Spatial distribution of land eligible for onshore wind development in Ontarioa)before and b) after excluding waterbodies


Area Eligible for Wind Power Development (79.05%)
N
Occupied land
Available land
0 50 100 150 200
km

a

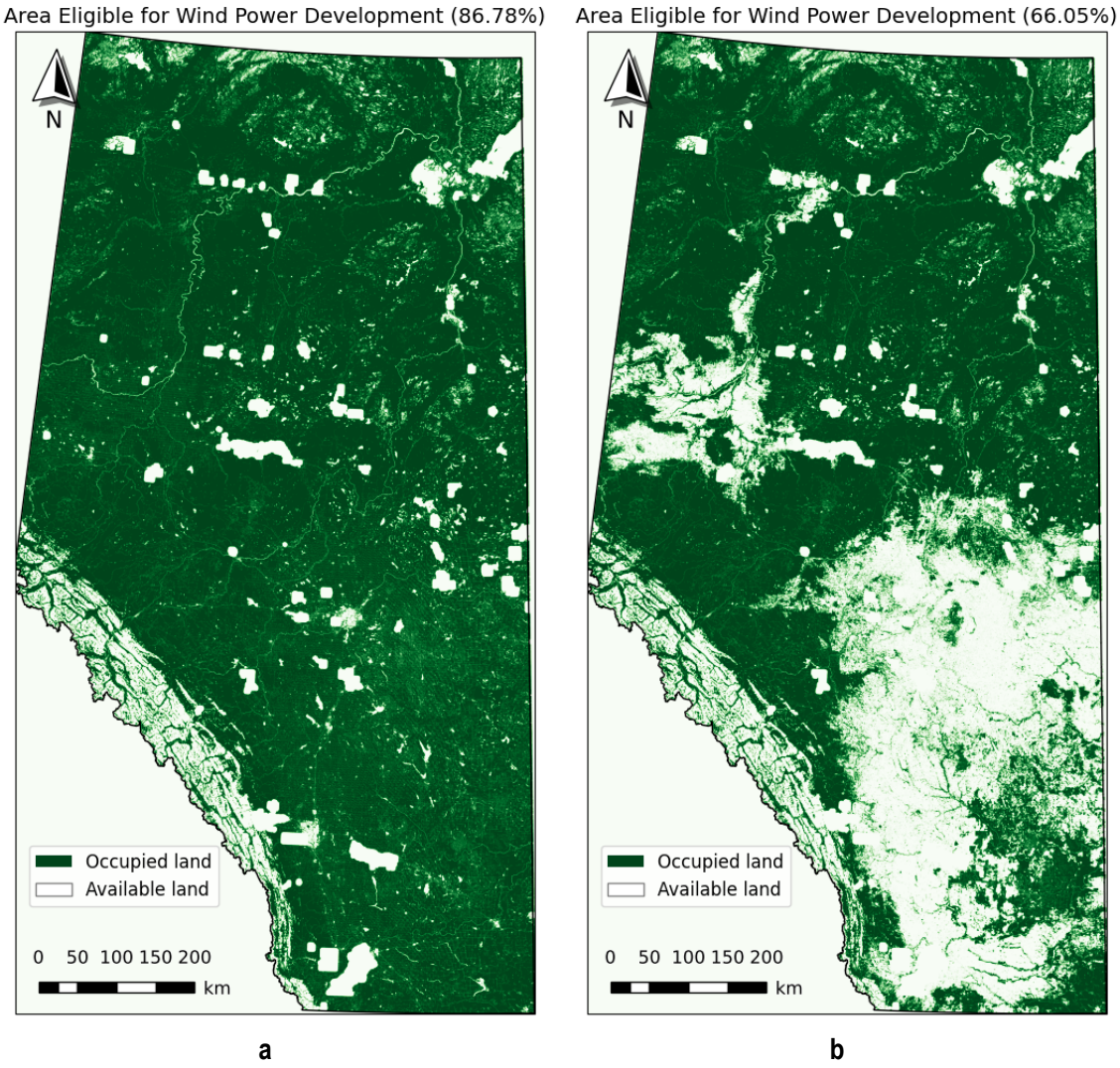

Area Eligible for Wind Power Development (72.18%)
N
Occupied land
Available land
0 50 100 150 200
km

b



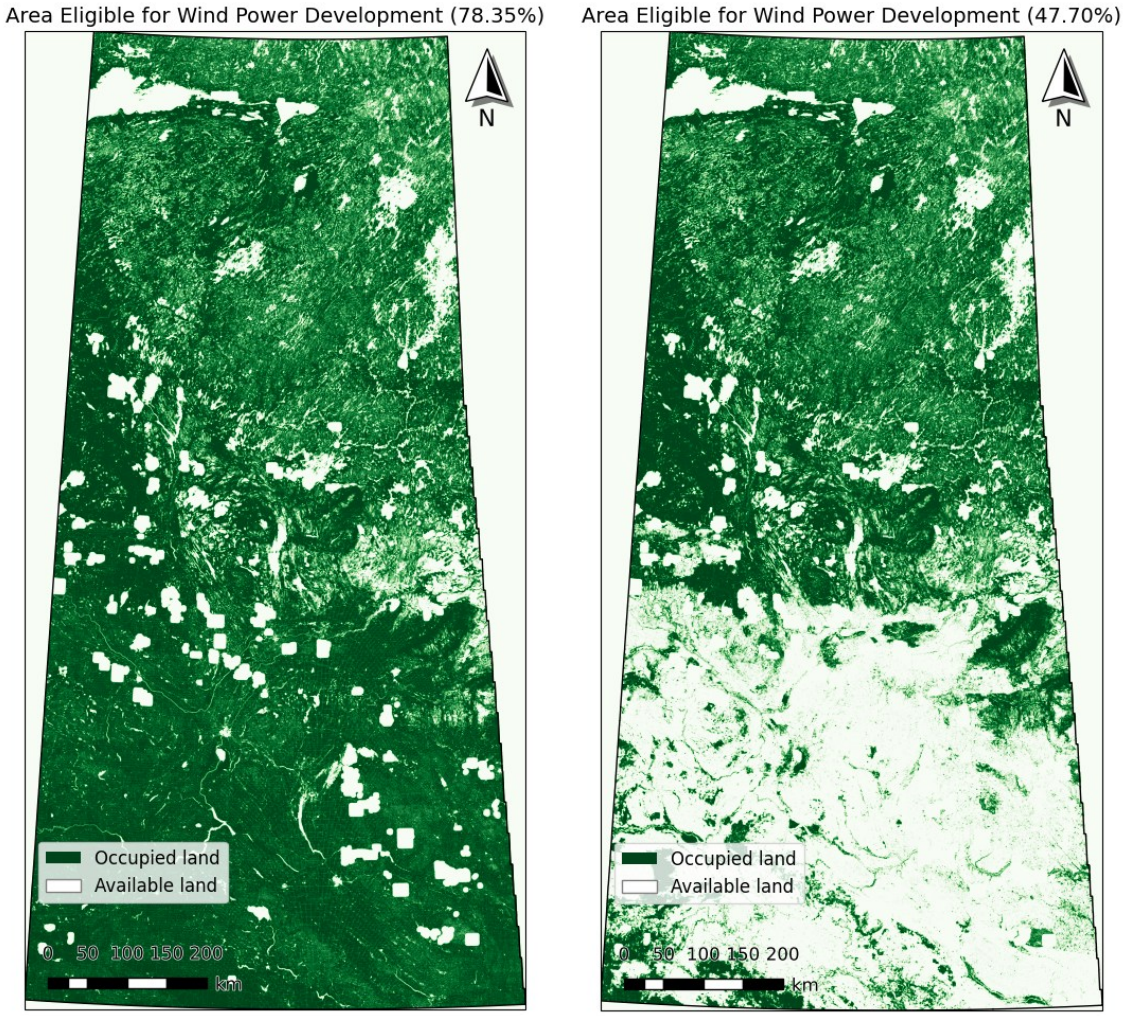
c d

Fig. B.4. **Spatial distribution of land eligible for onshore wind development in Manitobaa)before and after excluding b)wetlands, c)waterbodies and d)both**



a b

Fig. B.5. Spatial distribution of land eligible for onshore wind development in Alberta a)before and b)after excluding cropland



a b

Fig. B.6. Spatial distribution of land eligible for onshore wind development in Saskatchewana)before and b)after excluding cropland

Appendix C. ACES model's data sources

Table. C.1. Data sources used to develop the ACES model

	Data	Source
1	[A1]	Best Farm Animals, "How long do domesticated cows live? Cows lifespan according to us." [Online]. Available: https://bestfarmanimals.com/how-long-do-domesticated-cows-live-cows-lifespan-according-use/
2	[A2]	Frank, Stefan, et al. "Structural change as a key component for agricultural non-CO 2 mitigation efforts." Nature communications 9.1 (2018): 1-8.
3	[A3]	Statistics Canada. (2021). Table: 32-10-0139-01 Cattle statistics, supply and disposition of cattle (x 1,000) [Data table]. https://doi.org/10.25318/3210013901-eng
4	[A4]	Statistics Canada. (2021). Table: 32-10-0200-01 Hogs statistics, supply and disposition of hogs, semi-annual (x 1,000) [Data table]. https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=3210020001
5	[A5]	Canada's Official Greenhouse Gas Inventory - GHG_IPCC_Can_Prov_Terr.csv. [Online]. Available: http://donnees.ec.gc.ca/data/substances/monitor/canada-s-official-greenhouse-gas-inventory/GHG_IPCC_Can_Prov_Terr.csv
6	[A6]	Canada's Official Greenhouse Gas Inventory - EN_GHG_ECON_Prov_Terr.csv. [Online]. Available: https://donnees.ec.gc.ca/data/substances/monitor/canada-s-official-greenhouse-gas-inventory/D-Tables-Canadian-Economic-Sector-Provinces-Territories/?lang=en
7	[A7]	IPCC's Fourth Assessment Report - Errata. [Online]. Available: https://www.ipcc.ch/report/ar4/wg1/
8	[A8]	Statistics Canada. (2021). Table: 25-10-0029-01 Supply and demand of primary and secondary energy in terajoules, annual [Data table]. https://doi.org/10.25318/2510002901-eng
9	[A9]	Government of Nova Scotia. (2018). Standards for Quantification, Reporting, and Verification of Greenhouse Gas Emissions. https://climatechange.novascotia.ca/sites/default/files/uploads/Nova-Scotia-Standards-for-QRV-of-Greenhouse-Gas-Emissions.pdf
10	[A10]	Statistics Canada. 2017. Newfoundland and Labrador (table). Census Profile. 2016 Census. Statistics Canada Catalogue no. 98-316-X2016001. Ottawa. Released November 29, 2017. https://www12.statcan.gc.ca/census-recensement/2016/dp-pd/prof/index.cfm?Lang=E (accessed October 28, 2021).https://www12.statcan.gc.ca/census-recensement/2016/dp-pd/prof/index.cfm?Lang=E (accessed May 11, 2021).
11	[A11]	Labrador. (2021). Retrieved May 11, 2021, from https://en.wikipedia.org/wiki/Labrador
12	[A12]	Canada Energy Regulator. Canada's Energy Future 2020 Data Appendices (Reference Case). DOI: https://doi.org/10.35002/zjr8-8x75

	Data	Source
13	[B1]	NRCan, "Comprehensive Energy Use Database. Residential Sector. Atlantic Provinces. Table 2: Secondary Energy Use and GHG Emissions by End-Use". [Online]. Available: https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/showTable.cfm?type=CP&sector=res&juris=atl&rn=2&page=0
14	[B2]	NRCan, "Comprehensive Energy Use Database. Commercial/Institutional Sector. Atlantic Provinces. Table 2: Secondary Energy Use and GHG Emissions by End-Use". [Online]. Available: https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/showTable.cfm?type=CP&sector=com&juris=atl&rn=2&page=0
15	[B3]	Natural Resources Canada (NRCan) Heat Pump Study (Confidential)
16	[B4]	RSMeans Mechanical Data 2018, MF2016 Weighted Average, p. 703
17	[B5]	RSMeans Residential Costs (Gordian), 2019
18	[B6]	Updated Buildings Sector Appliance and Equipment Costs and Efficiencies, EIA, US DOE, June 2018. Available at: https://www.eia.gov/analysis/studies/buildings/equipcosts/pdf/full.pdf
19	[B7]	The Home Depot, Canada. [Online]. https://www.homedepot.ca/en/home/categories/building-materials/plumbing/water-heaters/tank-water-heaters.html
20	[B8]	Canada Energy Regulator. Canada's Energy Future 2020 Data Appendices (Reference Case). DOI: https://doi.org/10.35002/zjr8-8x75
21	[B9]	U.S. Energy Information Administration. Annual Energy Outlook 2021. Table 3: Energy Prices by Sector and Source. [Online]. Available: https://www.eia.gov/outlooks/aeo/data/browser/#/?id=3-AEO2021&region=1-1&cases=ref2021&start=2019&end=2050&f=A&linechart=ref2021-d113020a.3-3-AEO2021.1-1&map=ref2021-d113020a.4-3-AEO2021.1-1&sourcekey=0
22	[B10]	Open Energy Outlook for the United States (2021), GitHub Repository, https://github.com/TemoaProject/oeo
23	[B11]	Natural Resources Canada (NRCan), Heat Pump Study, "Commercial_bldg_electrification_results_v2_Nov 3.xls" (Confidential)
24	[B12]	Arkansas TRM Version 8.1 Vol. 2, 2.3.1 Water Heater Replacement, p125-138
25	[B13]	ENERGY STAR Certfiied Water Heater Product Finder [Online]. https://www.energystar.gov/productfinder/download/certified-water-heaters/
26	[B14]	Jadun, Paige, et al. Electrification futures study: end-use electric technology cost and performance projections through 2050. No. NREL/TP-6A20-70485. National Renewable Energy Lab.(NREL), Golden, CO (United States), 2017.
27	[B15]	Natural Resources Canada (NRCan), 2015 Survey of Household Energy Use (SHEU-2015) Data Tables". [On-Line]. Available at: https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/menus/sheu/2015/tables.cfm .
28	[B16]	Natural Resources Canada (NRCan), "Comprehensive Energy Use Database, Residential Sector." [On-line]. Available at: https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/menus/trends/comprehensive_tables/list.cfm

	Data	Source
29	[B17]	New York State Joint Utilities, "New York Standard Approach for Estimating Energy Savings from Energy Efficiency Programs," V7.0, 2019. Available at: https://www3.dps.ny.gov/W/PSCWeb.nsf/All/72C23DECFF52920A85257F1100671BDD?OpenDocument .
30	[B18]	Illinois Statewide TRM v8.0 Vol.3, 2019, P-108
31	[B19]	Custom Measure Life Review, Ontario Energy Board, May 2018
32	[B20]	State of Minnesota Techncial Reference Manual, Version 3.0, 2019. P-143
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